

v7 cloning campaign: subset generation, pre-training, in-sample optimization and held-out generalization

2026-09-09

State of the results on 2026-09-09 (revised report; the first version was dated 2026-09-08). Every number in this report is read from a file of the pulled runs (`~/Documents/Research/xcquinox-results/runs/dfs_step7/`) or from a CSV the figure suite wrote beside the figure it belongs to; the file is named at each table. The figures are the tracked sets under `notebooks/analysis/figures_dfs_step7_*`; the validation-best checkpoint channel is used for every optimization figure unless a caption says otherwise. Each figure is described panel by panel with the values it shows, so that the text can be read without magnifying the figure.

The sections follow the pipeline: the model and its descriptors (Sec. 1), the generation of the training subsets, which is the step before any network is fitted (Sec. 2), the pre-training of each architecture to its parent functional (Sec. 3), the optimization on the training subsets read on the training data itself (Sec. 4), and the held-out generalization (Sec. 5); the pending items close the document (Sec. 6).

Scope and inventory

The v7 campaign trains one network pair (an exchange network and a correlation network) per architecture and per training-subset size, under one protocol, on the cluster as separate runs that differ in their architecture axis. Each run carries its own pre-training (the cloning of the parent functional), a fidelity certificate that gates its training array, and the array itself, one cell per (architecture, subset size). The figures and tables of this report are drawn once over every finished cell of every run: the runs are merged into one view (`family_runs.yaml`, `merge_family_runs.py`; the runbook's v7 section) and the architecture axis is the one of Sec. 1.5, with the run of each cell recorded in the tables' first column. The architecture names are the shown names of Sec. 1.5 (the registry keys the runs are filed under are given there beside them).

Run	Configuration file	Architectures (registry key)	Cells done / planned	Run directory
size	dfs_step7.dfs6311_grid3_v7g1_size.yaml	deep_2x8 (shallow), deep_attn_2x8 (shallow_attn), deep_3x16 (medium), deep_attn_3x16 (medium_attn)	deep_3x16 11/11, deep_attn_3x16 9/11, deep_2x8 0/11, deep_attn_2x8 0/11	run_20260902T145245Z
families	dfs_step7.dfs6311_grid3_v7g2a_families_core.yaml	deep0_3x16 (deep_3x16), deep0_attn_3x16 (deep_attn_3x16), deep0_cusp_3x16 (deep_cusp_3x16)	deep0_3x16 11/11, deep0_attn_3x16 2/11, deep0_cusp_3x16 0/11	run_20260902T145247Z
meta-GGA families	dfs_step7.dfs6311_grid3_v7g2_families_mgga.yaml	deep0_cusp_mgga_3x16, deep0_mgga_3x16, deep0_mgga_attn_3x16, deep0_rung35_mgga_3x16, deep0_rung35ms_mgga_3x16	0/55 (the first certificate reads FAIL, Sec. 3.6; the other four are pending)	run_20260902T145250Z
25-cycle arm	dfs_step7.dfs6311_grid3_v7g1_c25.yaml	deep_3x16 (medium), tag [25 cycles]	0/5 (pre-training done, training queued)	run_20260908T153856Z
dpyscf-parity arm	dfs_step7.dfs6311_grid3_v7g1_dfsparity.yaml	deep_3x16 (medium), tag [dpyscf parity]	0/5 (pre-training done, training queued)	run_20260908T153908Z

The subset-size axis is 1, 2, 3, 4, 5, 6, 7, 12, 15, 18 and 26 training points, drawn from the 26-point DFS training pool by the Jensen-Shannon subset selection of Sec. 2; the membership of every subset is in the table of Sec. 2.4. Basis 6-311++G(3df,2pd), grid level 3, density fitting on, for every stage (fidelity_certificate.json, keys basis, grid_level, density_fit). The cells that have not finished are the attention and cusp architectures at the larger subsets and the whole meta-GGA group; they appear as "pending" in the tables below and are absent from the figures. Two arms of a follow-up (the 25-cycle SCF arm and the dpyscf-parity arm, both on the deep_3x16 architecture) were submitted on 2026-09-08 and carry no results yet.

1. Model, descriptors and the functional form

Sources: xcquinox/alec/networks.py (the two networks), xcquinox/alec/descriptors.py and xcquinox/alec/metagga.py (the extra descriptors), xcquinox/net.py (the attention block), xcquinox/alec/config.py (the architecture registry). The coordinate convention is the DFS one (model.descriptor_coordinates: dfs in every v7 file), following Dick and Fernandez-Serra, PRB 104, L161109 (2021), whose equation numbers are quoted as "DFS Eq. n".

1.1 Density variables

With $n = n_\alpha + n_\beta$ the total density and $\sigma = |\nabla n|^2$,

$$k_F = (3\pi^2 n)^{1/3}, \quad s = \frac{|\nabla n|}{2k_F n}, \quad r_s = \left(\frac{3}{4\pi n}\right)^{1/3}, \quad \zeta = \frac{n_\alpha - n_\beta}{n}, \quad \phi(\zeta) = \frac{1}{2} \left[(1 + \zeta)^{4/3} + (1 - \zeta)^{4/3} \right].$$

s is the reduced density gradient (DFS Eq. 5), ϕ the spin-scaling factor of DFS Eq. 4 ($\phi = 1$ at $\zeta = 0$, $2^{1/3}$ at full polarization). The density is floored at the network's lower_rho_cutoff before any of these is formed.

1.2 Network coordinates (DFS coordinates)

The exchange network reads one coordinate at the GGA level and two at the meta-GGA level; the correlation network reads three and four. They are logarithmic compressions of the variables above (DFS Eqs. 7 to 10):

$$x_0 = \ln(n^{1/3} + 10^{-5}), \quad x_1 = \ln \phi(\zeta), \quad x_s = (1 - e^{-s^2}) \ln(s + 1), \quad x_\alpha = \ln \frac{\alpha + 1}{2}.$$

Exchange input: $[x_s, \dots]$, where s is that of the doubled spin channel $2n_\sigma$ (the exchange network is evaluated per spin channel, Sec. 1.4). Correlation input: $[x_0, x_1, x_s, \dots]$ on the total density. The ellipsis holds the extra descriptors of the architecture, in the registry's list order; on the meta-GGA rung the raw α column among them is replaced by x_α .

1.3 Extra descriptors

Iso-orbital indicator (meta-GGA rung, metagga), SCAN's α (Sun, Ruzsinszky and Perdew, PRL 115, 036402 (2015), Eq. 2; DFS Eq. 6):

$$\alpha = \frac{\tau - \tau_W}{\tau_{\text{unif}}}, \quad \tau = \frac{1}{2} \sum_i |\nabla \psi_i|^2 = \frac{1}{2} \sum_{\mu\nu} P_{\mu\nu} \nabla \chi_\mu \cdot \nabla \chi_\nu, \quad \tau_W = \frac{|\nabla n|^2}{8n}, \quad \tau_{\text{unif}} = \frac{3}{10} (3\pi^2)^{2/3} n^{5/3}.$$

τ is a contraction of the live one-particle density matrix P against the AO gradients, so it is self-consistent and differentiable through the SCF without second derivatives. The bound $\alpha \geq 0$ (the von Weizsacker inequality) is imposed by a smooth positive part rather than a hard clip, so the descriptor and its derivative stay continuous where a spin channel holds one orbital.

Nuclear cusp pair (cusp), two bounded features per grid point:

$$x_4 = \exp(-2Z_{\text{nearest}} r_{\text{min}}), \quad x_5 = \tanh\left(\frac{1}{5} \ln \sum_A \frac{Z_A}{r_A}\right).$$

x_4 is a proximity feature following the density-form cusp envelope e^{-2Zr} (Steiner, J. Chem. Phys. 39, 2365 (1963), the density counterpart of Kato’s wavefunction cusp); x_5 is the log-compressed bare-nuclear potential $-V_{\text{ext}}$, the log and the 1/5 scale being the conditioning of this work. Both are geometric, so the synthetic pre-training mesh (Sec. 3.1) carries no values for them.

The remaining registry descriptors (density-matrix statistics, the rung-3.5 occupancies) are not on any v7 axis and are not described here.

1.4 The functional form

Each network is a multilayer perceptron with GELU activations, depth hidden layers of nodes units, a final linear layer (initialized at zero in the `deep_*` family, `zero_init_final_layer`), and an output squash. With g the MLP output at a grid point,

$$F_x = 1 + \mathcal{L}_{1.804}(G_x \cdot g_x), \quad F_c = 1 + \mathcal{L}_2(G_c \cdot g_c), \quad \mathcal{L}_\lambda(z) = \lambda \text{sigmoid}(z - \ln(\lambda - 1)) - 1.$$

$\mathcal{L}_\lambda(0) = 0$, so a zero MLP output returns $F = 1$ (the local density approximation), and F stays in $(0, \lambda)$ for every z . $\mathcal{L}_\lambda$ is the DFS I_a map (DFS Eq. 11) to machine precision. For exchange $\lambda = 1.804 = 1 + \kappa_{\text{PBE}}$, the PBE Lieb-Oxford-motivated ceiling (PBE, PRL 77, 3865 (1996), Eq. 14); DFS uses 1.174 (the tighter bound of Perdew, Ruzsinszky, Sun and Burke, J. Chem. Phys. 140, 18A533 (2014)), and the v7 meta-GGA architectures keep 1.804 as well (`config.resolved_xnet_lob_lim`), a recorded deviation. For correlation $\lambda = 2$ is the symmetric sigmoid of DFS Eq. 13; its purpose is $F_c \geq 0$, the upper value 2 is incidental.

G is the uniform-electron-gas gate that pins $F = 1$ where the gradient vanishes:

$$G = \tanh^2(s) \text{ (GGA rung)}, \quad G = x_s + \tanh^2(x_\alpha) \text{ (meta-GGA rung, DFS Eq. 12)},$$

evaluated on the raw s (and α), never on the compressed coordinate. On the meta-GGA rung the same gate form is used for both networks (DFS’s correlation gate applies $\tanh$ to x_s ; recorded deviation).

The exchange-correlation energy is assembled with the exchange spin-scaling relation of Oliver and Perdew (PRA 20, 397 (1979)) and the PW92 correlation baseline:

$$E_{xc} = \frac{1}{2} (E_x[2n_\alpha] + E_x[2n_\beta]) + E_c[n_\alpha, n_\beta], \quad E_x[n] = \int n \epsilon_x^{\text{LDA}}(n) F_x d^3r, \quad E_c = \int n \epsilon_c^{\text{PW92}}(r_s, \zeta) F_c d^3r,$$

so the exchange network sees each spin channel at its doubled density (its x_s is that of $2n_\sigma$), and the correlation network sees the total density with ζ through x_1 .

Attention variants. In the `_attn` architectures a self-attention block (Vaswani et al., 2017, Eqs. 1 and 2, pre-layer-norm as in Xiong et al., 2020) sits after the first hidden layer. The hidden vector of H units at a grid point is treated as h tokens of dimension $d_k = H/h$ and attention runs across the token axis,

$$\text{Attn}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V, \quad x \leftarrow x + W^O \text{Concat}(\text{head}_1, \dots, \text{head}_h)(\text{LN}(x)).$$

It is a per-point re-mixing of hidden units, not a spatial or non-local operation.

1.5 Architectures on the v7 axes

The architectures lie on four axes: the size (hidden layers x units), the initialization of the last layer, attention, and the extra descriptors (which set the rung and the parent). The name states

them: the family word is the initialization (deep for the Glorot start of every layer; deep0 for a last layer whose weight and bias are zeroed at construction, so that the untrained network is the LDA, $F = 1$, and pre-training starts from it), then the descriptor and attention tokens, then the size. The registry keys the runs are filed under are given beside the names; deep_3x16 and deep0_3x16 are the same network (the registry's medium and deep_3x16: depth 3, width 16, the same inputs, the same parameter count), separated by the initialization alone, so the two columns are replicates of one architecture under two starts. Nothing is frozen and nothing is appended in either; after pre-training both are PBE clones that differ in their fitted weights.

Architecture	Registry key	Start	Rung, parent	Layers x units	Heads	Extra descriptors
deep_2x8	shallow	Glorot	GGA, PBE	2 x 8	none	none
deep_attn_2x8	shallow_attn	Glorot	GGA, PBE	2 x 8	2	none
deep_3x16	medium	Glorot	GGA, PBE	3 x 16	none	none
deep_attn_3x16	medium_attn	Glorot	GGA, PBE	3 x 16	4	none
deep0_3x16	deep_3x16	last layer zeroed	GGA, PBE	3 x 16	none	none
deep0_attn_3x16	deep_attn_3x16	last layer zeroed	GGA, PBE	3 x 16	4	none
deep0_cusp_3x16	deep_cusp_3x16	last layer zeroed	GGA, PBE	3 x 16	none	cusp
deep0_cusp_mgga_3x16	deep_cusp_mgga_3x16	last layer zeroed	meta-GGA, SCAN	3 x 16	none	cusp, α

The network inputs follow from the rung and the extra descriptors:

Architecture	Exchange input	Correlation input
deep_2x8, deep_attn_2x8, deep_3x16, deep_attn_3x16, deep0_3x16, deep0_attn_3x16	x_s	x_0, x_1, x_s
deep0_cusp_3x16	x_s, x_4, x_5	x_0, x_1, x_s, x_4, x_5
deep0_cusp_mgga_3x16	x_s, x_4, x_5, x_α	$x_0, x_1, x_s, x_4, x_5, x_\alpha$

The deep0 entries of the registry also carry `dm_entropy_intensive` and `descriptor_log_transform`, both inert on this axis (no density-matrix descriptor; the DFS coordinates already compress s , and only the cusp column-1 logarithm acts). All v7 networks are unanchored (`model.parent_anchor: false`): the network output is the whole enhancement factor, and the parent functional enters only as the pre-training target. A cell of a run trained under another protocol carries the run’s tag on its name (`deep_3x16 [25 cycles]`, `deep_3x16 [dpyscf parity]`; Sec. 6), so that the same architecture under two protocols sits on one axis.

2. Training-subset generation (step 0 of the pipeline)

Sources: `xcquinox/alec/dfs_pool.py` (the pool), `xcquinox/alec/training_points.py` (build `dfs_pool_points`), `xcquinox/alec/subset_selection.py` (the descriptors, the histograms, the metric, the search), the selection ledger `notebooks/checkpoints_step7/alpha_on/subset_index_log.json`, the cached per-species descriptors `notebooks/checkpoints_step7/subset_descriptors/` and the cached reference histogram `notebooks/checkpoints_step7/alpha_on/dfs_pool_full_hist/reference.npz`. Every value quoted below was recomputed from those caches with the package’s own functions and agrees with the ledger to the last digit.

2.1 The pool

The training pool is the DFS training set (Dick and Fernandez-Serra, PRB 104, L161109 (2021), SI Sec. II): 21 atomization energies of G2/97 molecules, 3 BH76 reactions and 2 IP13 ionization potentials, 26 selectable points, with the H and Li atomic densities of the DFS set entering as the anchors of the points that contain those elements. The atomization references are the non-relativistic, zero-point-exclusive values of Haunschild and Klopper (J. Chem. Phys. 136, 164102 (2012)) with H₂O and C₂H₂ from W4-11 (Karton, Daon and Martin, CPL 510, 165 (2011)); the reaction energies are the GMTKN55 BH76RC values; the ionization potentials the NIST values. Each point carries the species its loss term needs: an atomization point its molecule plus the H and Li anchors it contains, a reaction point every reactant and product, an ionization point the neutral and the cation. The 26 points, in the index order of the ledger:

Index	Point	Kind	Species (name, charge, spin)
0	H2	atomization energy	H2 (0, 0), H (0, 1)
1	N2	atomization energy	N2 (0, 0)
2	FLi	atomization energy	FLi (0, 0), Li (0, 1)
3	CHN	atomization energy	CHN (0, 0), H (0, 1)
4	CO2	atomization energy	CO2 (0, 0)
5	F2	atomization energy	F2 (0, 0)
6	C2H2	atomization energy	C2H2 (0, 0), H (0, 1)
7	CO	atomization energy	CO (0, 0)
8	HLi	atomization energy	HLi (0, 0), H (0, 1), Li (0, 1)
9	Na2	atomization energy	Na2 (0, 0)
10	NO	atomization energy	NO (0, 1)
11	CH	atomization energy	CH (0, 1), H (0, 1)
12	HO	atomization energy	HO (0, 1), H (0, 1)
13	NO2	atomization energy	NO2 (0, 1)
14	HN	atomization energy	HN (0, 2), H (0, 1)
15	O3	atomization energy	O3 (0, 0)
16	N2O	atomization energy	N2O (0, 0)

Index	Point	Kind	Species (name, charge, spin)
17	CH3	atomization energy	CH3 (0, 1), H (0, 1)
18	CH2	atomization energy	CH2 (0, 2), H (0, 1)
19	H2O	atomization energy	H2O (0, 0), H (0, 1)
20	H3N	atomization energy	H3N (0, 0), H (0, 1)
21	OH+N2_to_H+N2O	reaction energy	HO (0, 1), N2 (0, 0), H (0, 1), N2O (0, 0)
22	OH+CH3_to_O+CH4	reaction energy	HO (0, 1), CH3 (0, 1), O (0, 2), CH4 (0, 0), H (0, 1)
23	HF+F_to_H+F2	reaction energy	HF (0, 0), F (0, 1), H (0, 1), F2 (0, 0)
24	Li_IP	ionization potential	Li (0, 1), Li+ (1, 0)
25	C_IP	ionization potential	C (0, 2), C+ (1, 1)

2.2 The descriptor distributions

For every unique species of the pool (30 with the anchors and the cations) one self-consistent PBE calculation at def2-SVP on a level-1 quadrature grid gives, at every grid point g with quadrature weight w_g , the three descriptors

$$\rho^{1/3}, \quad s = \frac{|\nabla\rho|}{2(3\pi^2)^{1/3}\rho^{4/3}}, \quad \alpha = \frac{\tau - \tau_W}{\tau_{\text{unif}}} \quad (\text{clipped to } [0, 100]),$$

with $\tau_W = |\nabla\rho|^2/8\rho$ and $\tau_{\text{unif}} = \frac{3}{10}(3\pi^2)^{2/3}\rho^{5/3}$ as in Sec. 1.3. A point’s descriptor sample is the concatenation of its species’ grid points, so a reaction contributes in proportion to the number of grid points of its species. The reference distribution is the concatenation over all 26 points. For each descriptor $x \in \{\rho^{1/3}, s, \alpha\}$ the reference sample is binned into 200 bins of equal width between its 0.1 and 99.9 percentiles, weighted by the quadrature weights, and every candidate subset is binned on the same edges:

$$h_x^{(b)} = \sum_{g: x_g \in \text{bin } b} w_g, \quad P_x^{(b)} = \frac{h_x^{(b)}}{\sum_{b'} h_x^{(b')}}.$$

The normalization to a probability mass function removes the bin width and the sample size, so the comparison is between the shapes of the weighted distributions.

Two properties of these histograms follow from the linear edges and the percentile range and are stated here because they determine what the metric measures. The ranges are set by the low-density tail of the grids, where s and α grow without bound: the s edges run from 8×10^{-9} to 2.0×10^{10} and the α edges from 0 to 2.6×10^{10} , while the $\rho^{1/3}$ edges run from 10^{-10} to 9.04. With 200 bins of equal width, the quadrature weights (which grow into the tail) place 98.1 percent of the reference mass in the first $\rho^{1/3}$ bin, 88.9 percent in the first s bin and 97.2 percent in the first α bin; 99 percent of the mass lies within the first 2, 126 and 12 bins. What the first bin holds differs between the descriptors. For s and α it is the whole chemically relevant range ($s = 5$ and $\alpha = 10$ both fall in bin 0, whose widths are 10^8), so the metric compares those two distributions over the remaining 11 and 3 percent of the weight, which sits in the low-density tails of the molecular grids. For $\rho^{1/3}$ the first bin is the vacuum tail itself ($\rho < 9.2 \times 10^{-5}$ per bohr³), and the remaining 1.9 percent of the weight is the valence and core region ($\rho = 0.01$ falls in bin 4, $\rho = 1$ in bin 22). Samples outside the edges enter no histogram: 1.92 percent of the pool weight for s and 0.20 percent for α . The s marginal carries most of the divergence (Sec. 2.4).

2.3 The metric and the search

The distance between a candidate subset S and the pool is the Jensen-Shannon divergence (Lin, IEEE Trans. Inf. Theory 37, 145 (1991)) summed over the three marginals with equal weights:

$$\text{JSD}(S) = \sum_{x \in \{\rho^{1/3}, s, \alpha\}} \frac{1}{2} [\text{KL}(P_x \parallel M_x) + \text{KL}(Q_x \parallel M_x)], \quad M_x = \frac{1}{2}(P_x + Q_x),$$

$$\text{KL}(P \parallel Q) = \sum_b P^{(b)} \ln \frac{P^{(b)}}{Q^{(b)}},$$

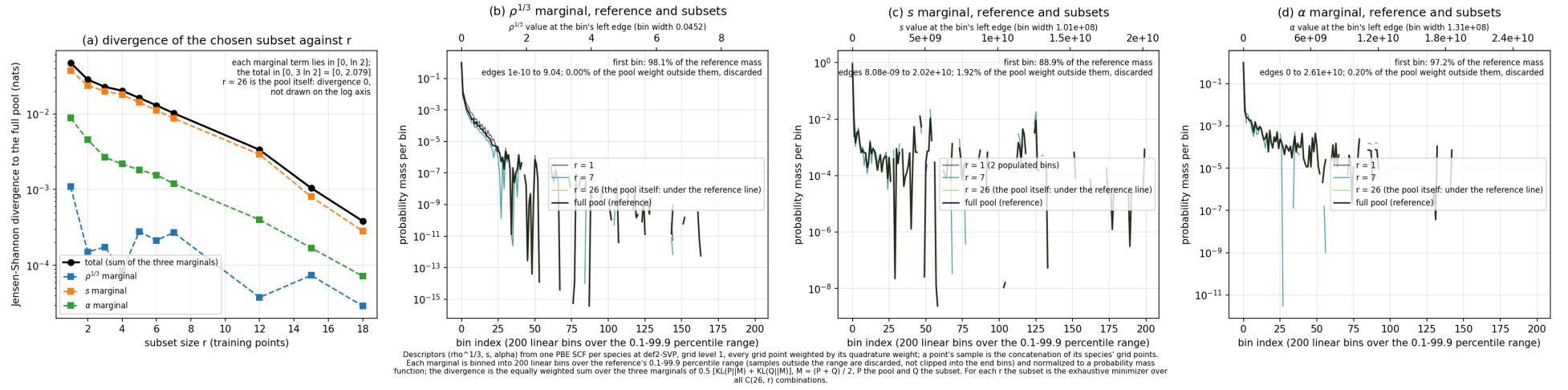
with P_x the reference and Q_x the subset mass function of descriptor x , probabilities floored at 10^{-12} inside the logarithm, in nats. Each marginal term lies in $[0, \ln 2]$ and the sum in $[0, 3 \ln 2]$; a subset whose grid points all fall outside a marginal’s range is assigned an infinite divergence and never selected. For each subset size $r \in \{1, \dots, 7, 12, 15, 18, 26\}$ the subset is the exhaustive minimizer of $\text{JSD}(S)$ over all $\binom{26}{r}$ combinations (9.7 million at $r = 12$), computed with the pre-binned per-point histograms; the L2 distance between the same mass functions was run as a second metric and is recorded in the ledger but not used by any v7 run. The subsets are nested only where the minimizer happens to nest: the 7-molecule subset does not contain the 6-molecule one.

2.4 The chosen subsets

r	Points (ledger index order)	AE / BH76 / IP	JSD to the pool	$\rho^{1/3}$ term	s term	α term	s share
1	H2O	1 / 0 / 0	0.0475	0.00109	0.03764	0.00881	79%
2	HLi, OH+N2_to_H+N2O	1 / 1 / 0	0.0284	0.00015	0.02372	0.00455	83%
3	OH+N2_to_H+N2O, OH+CH3_to_O+CH4, Li_IP	0 / 2 / 1	0.0228	0.00017	0.01995	0.00267	88%
4	CH2, OH+N2_to_H+N2O, OH+CH3_to_O+CH4, Li_IP	1 / 2 / 1	0.0203	0.00008	0.01801	0.00218	89%
5	HLi, OH+N2_to_H+N2O, OH+CH3_to_O+CH4, Li_IP, C_IP	1 / 2 / 2	0.0163	0.00028	0.01418	0.00183	87%
6	HLi, CH, OH+N2_to_H+N2O, OH+CH3_to_O+CH4, Li_IP, C_IP	2 / 2 / 2	0.0130	0.00021	0.01120	0.00156	86%
7	CHN, CO2, HLi, CH, OH+N2_to_H+N2O, Li_IP, C_IP	4 / 1 / 2	0.0102	0.00027	0.00877	0.00120	86%
12	CO2, C2H2, CO, HLi, CH, NO2, CH2, OH+N2_to_H+N2O, OH+CH3_to_O+CH4, HF+F_to_H+F2, Li_IP, C_IP	7 / 3 / 2	0.0034	0.00004	0.00292	0.00040	87%
15	FLi, CHN, CO2, C2H2, CO, HLi, NO, CH, NO2, CH2, OH+N2_to_H+N2O, OH+CH3_to_O+CH4, HF+F_to_H+F2, Li_IP, C_IP	10 / 3 / 2	0.0010	0.00007	0.00081	0.00017	77%
18	N2, FLi, CHN, CO2, C2H2, CO, HLi, Na2, NO, CH, NO2, N2O, CH2, OH+N2_to_H+N2O, OH+CH3_to_O+CH4, HF+F_to_H+F2, Li_IP, C_IP	13 / 3 / 2	0.0004	0.00003	0.00028	0.00007	74%
26	H2, N2, FLi, CHN, CO2, F2, C2H2, CO, HLi, Na2, NO, CH, HO, NO2, HN, O3, N2O, CH3, CH2, H2O, H3N, OH+N2_to_H+N2O, OH+CH3_to_O+CH4, HF+F_to_H+F2, Li_IP, C_IP	21 / 3 / 2	0.0000	0.00000	0.00000	0.00000	-

Fig. 2.1 Left: the Jensen-Shannon divergence of the chosen subset to the full pool against the subset size, the total and its three marginal terms (log scale; the bound $3\ln 2 = 2.08$ lies far above the axis). The divergence falls from 0.0475 at one molecule to 0.0102 at seven, 0.0034 at twelve, 0.0010 at fifteen and 0.0004 at eighteen, and to zero at the full pool; the s term is 74 to 89 percent of the total at every size (0.0376 of 0.0475 at one molecule, 0.00028 of 0.00038 at eighteen). Right, three panels: the mass functions of the reference (black) and of the subsets at $r = 1, 7$ and 26 over the bin index, log scale, with the first-bin mass printed in each panel.

Training-subset selection: Jensen-Shannon divergence to the full DFS pool



3. Pre-training (functional cloning)

Sources: `xcquinox/alec/pretrain.py`, `xcquinox/alec/pretrain_data_gen.py`, the `pretrain:` and `fidelity:` blocks of the resolved configurations, `pretrain/<arch>/pretrain_metadata.json` and `pretrain/<arch>/fidelity_certificate.json` of each run.

3.1 Set, targets and objective

Each network is fitted to reproduce its parent functional’s enhancement factor pointwise on the parent’s own self-consistent densities: PBE for the GGA rung, SCAN for the meta-GGA rung. The set is the DFS pre-training inventory (8 free atoms and 22 G2/97 molecules at the GGA level; 20 molecules at the meta-GGA level, which drops H2 and N2) plus every single-atom species of the BH76 and W4-11 pools at its production charge and spin (the 12 neutral elements and the F- and Cl- anions), 39 systems for the GGA groups and 38 for the meta-GGA group (`n_systems` of the certificates). The targets are stored per grid point as

$$t_x = F_x^{\text{parent}} - 1 = \frac{\epsilon_x^{\text{parent}}}{\epsilon_x^{\text{LDA}}} - 1, \quad t_c = F_c^{\text{parent}} - 1 = \frac{\epsilon_c^{\text{parent}}}{\epsilon_c^{\text{PW92}}} - 1,$$

from spin-resolved libxc evaluations, the exchange rows posed per spin channel at the doubled density ($2n_\sigma, 4\sigma_{\sigma\sigma}$), the footing the production exchange evaluates (`exchange_footing: spin_channel`). A synthetic (r_s, s, α) mesh (the DFS SI’s (s, α) grid extended to three dimensions) is added as a regularizer at a fixed 30 percent share of the integration weight (`mesh_fraction: 0.3`); its rows belong to no system.

The objective per network is the integration-weighted pointwise residual plus a per-system energy term (`_PretrainLoss`):

$$\begin{aligned} \mathcal{L}_{\text{pre}} = & \frac{\sum_i w_i (F_i^{\text{NN}} - 1 - t_i)^2}{\sum_i w_i} \\ & + w_E \frac{1}{N_{\text{sys}}} \sum_s \left(\sum_{i \in s} w_i^{\text{grid}} n_i \epsilon_i^{\text{LDA}} F_i^{\text{NN}} - E_{xc,s}^{\text{parent}} \right)^2, \\ & w_i = |n_i \epsilon_i^{\text{LDA}}| w_i^{\text{grid}}, \quad w_E = 0.1. \end{aligned}$$

The energy term is present because the pointwise residual alone does not bound a system’s energy: in the earlier campaign the architecture with the lowest exchange residual carried the largest atomization-energy offset from its parent. Schedule: Adam, 20000 steps, learning rate 10^{-3} decaying linearly to 10^{-5} from the midpoint, gradient clip 1.0, a 20 percent held-out validation slice scored every 50 steps with patience 300 checks (`pretrain: block`).

3.2 The fidelity certificate

After the fit, the saved networks are re-evaluated on every system of the set at the production basis and grid and compared with the parent: the per-system exchange-correlation energy error $\Delta E_{xc,s} = E_{xc,s}^{\text{NN}} - E_{xc,s}^{\text{parent}}$, and, for every molecule, the atomization-energy error $\Delta AE = \Delta E_{xc,\text{mol}} - \sum_A \Delta E_{xc,A}$ against the parent’s own atomization energy. The gate (`fidelity: block`, two-tier since 2026-09-02) is

$$\max_{\text{atoms}} |\Delta E_{xc,A}| \leq 1.0 \text{ mHa}, \quad \text{mean}_{\text{mol}} |\Delta AE| \leq 1.0 \text{ kcal/mol}, \quad \max_{\text{mol}} |\Delta AE| \leq 2.0 \text{ kcal/mol},$$

and a failing certificate makes the pre-training task exit without success, which holds the group’s preflight and training array. The certificate is what gives the training-stage comparison against PBE (or SCAN) its meaning: every trained cell starts from a network that is its parent to within these tolerances.

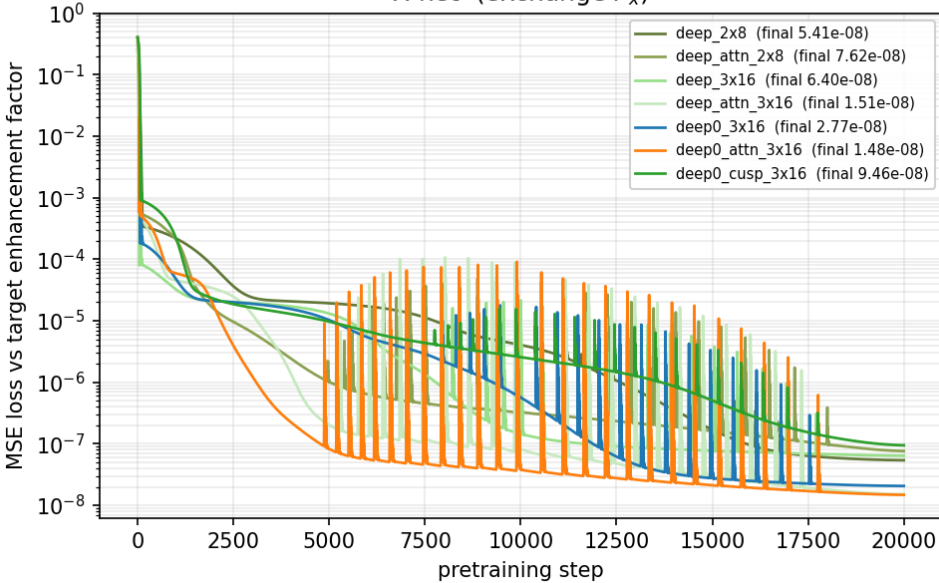
3.3 Pre-training loss curves

Fig. 3.1 The pointwise term of the objective per step taken, for the exchange network (left) and the correlation network (right), one curve per architecture, log scale over the 20000 steps; the final value of each curve is printed in the legend. The regular spikes from step 5000 onward are the validation checks: every 50 steps the objective is evaluated on the held-out slice and the best network is snapshotted, and the training step that follows a snapshot reload sits above the trend line for one step. The first figure holds the seven certified GGA clones of the size and families runs; the second the meta-GGA clone (its certificate failed, so it is not in the merged view and is drawn from its run).

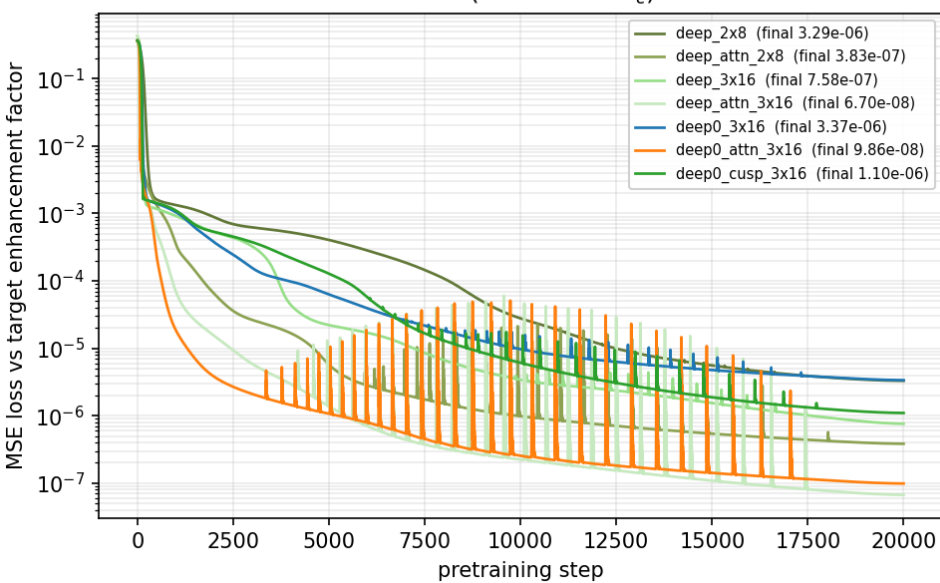
The exchange curves of the four Glorot-start architectures reach 5×10^{-8} to 8×10^{-8} (deep_attn_3x16 1.5×10^{-8}), the correlation curves 7×10^{-8} (deep_attn_3x16) to 3×10^{-6} (deep_2x8); deep_attn_3x16 descends earliest and ends lowest on both networks, the deep_2x8 correlation network is the slowest. Of the zero-init architectures deep0_attn_3x16 is lowest on both networks (1.5×10^{-8} , 1.0×10^{-7}), deep0_3x16 and deep0_cusp_3x16 within a factor 3.5 of each other (2.8×10^{-8} and 9.5×10^{-8} ; 3.4×10^{-6} and 1.1×10^{-6}). The meta-GGA clone stops at 3.7×10^{-6} (exchange) and 5.2×10^{-6} (correlation), fifty to a hundred times above the GGA clones after the same 20000 steps, with a smooth trajectory and no divergence.

Pretraining loss curves -- 7 archs, 20000 steps
run_20260902T145247Z

X-net (exchange F_x)

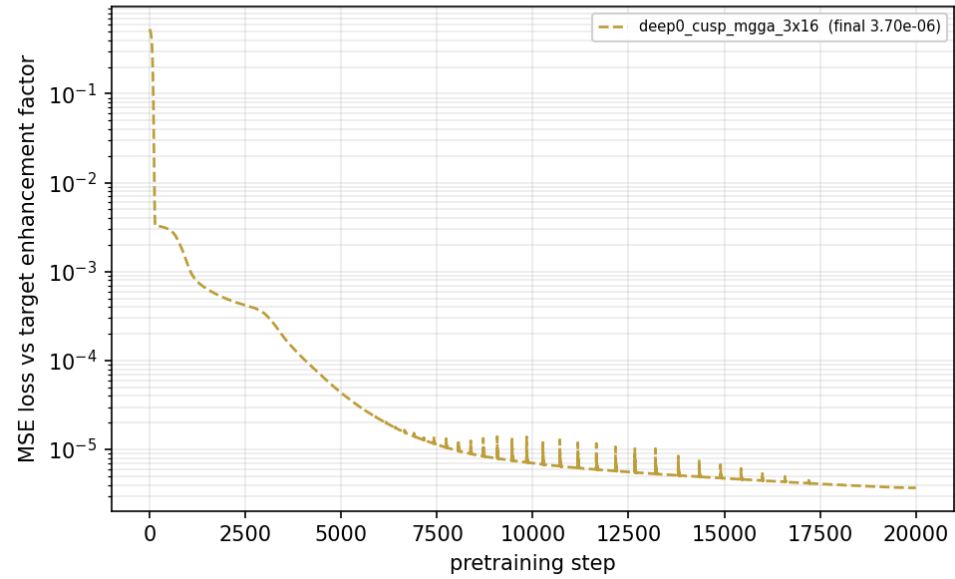


C-net (correlation F_c)

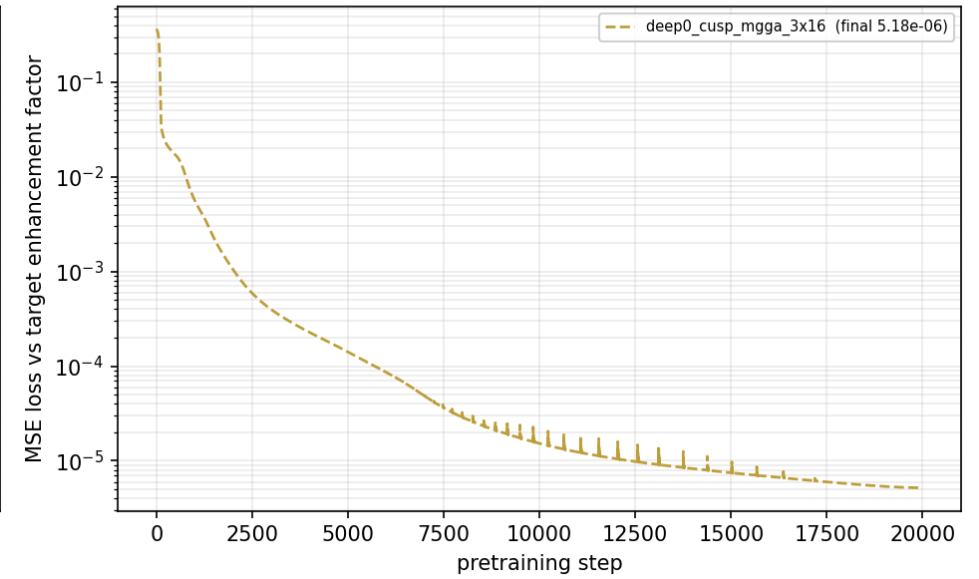


Pretraining loss curves -- 1 archs, 20000 steps
run_20260902T145250Z

X-net (exchange F_x)

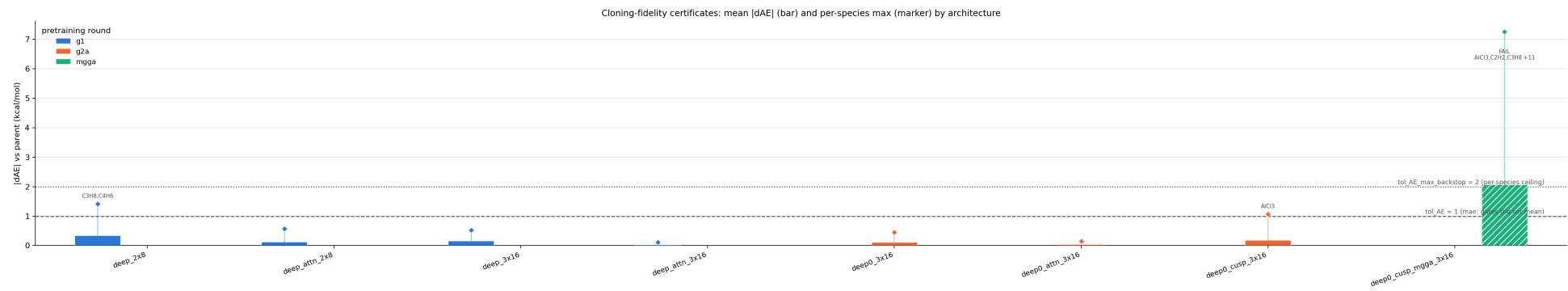


C-net (correlation F_c)



3.4 Certificate summary

Fig. 3.2 One bar per architecture, the mean $|\Delta AE|$ against the parent over the set's molecules; the marker above each bar is the largest single-species $|\Delta AE|$, with the species above 1 kcal/mol printed beside it. Blue bars are the v7 round (this campaign, the two-tier gate: the dashed line at 1.0 kcal/mol gates the mean, the dotted line at 2.0 kcal/mol every species), one colour per run (the size run, the families run, the meta-GGA run); the meta-GGA bar is the FAILED certificate, hatched, with its flagged species printed. The numbers are in the table of Sec. 3.6. Reading: every v7 bar sits below 0.35 kcal/mol and every v7 marker below 1.5 kcal/mol; the attention architectures are the tightest (deep_attn_3x16 0.031 mean, 0.115 max; deep0_attn_3x16 0.042 and 0.149), the deep_2x8 plain network the loosest (0.343 mean, 1.421 max on $\bar{\text{C}}\text{H}_8$ and C_4H_6).

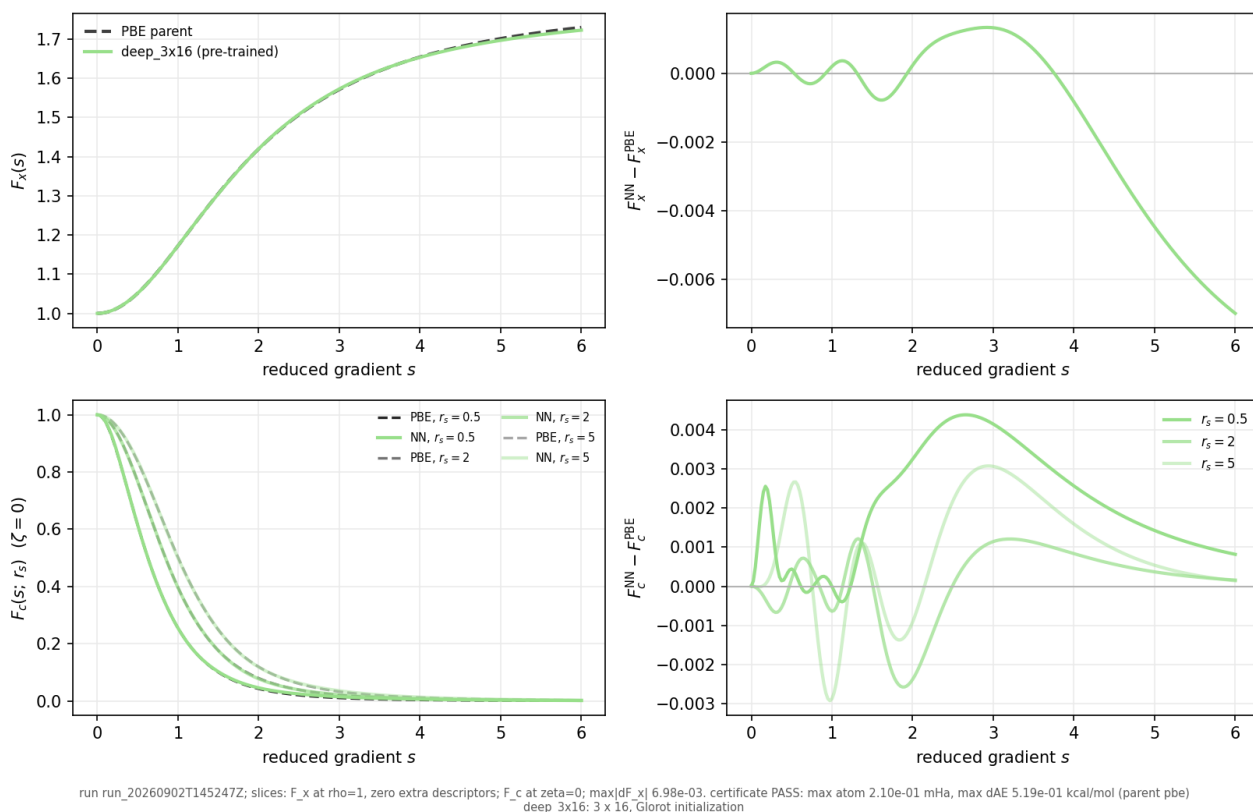


3.5 Pre-trained enhancement factors

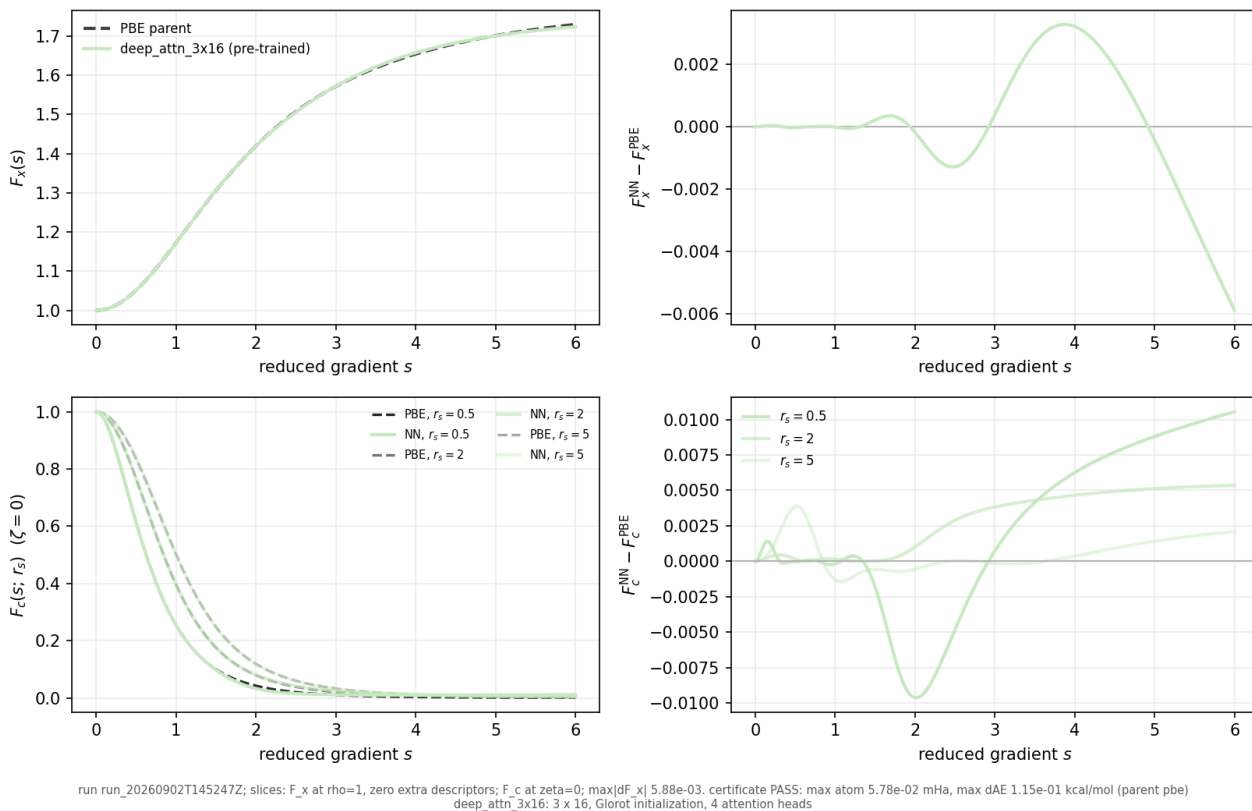
Fig. 3.3 For one architecture, the pre-trained network against its parent along two slices of descriptor space. Top left: the exchange enhancement factor $F_x(s)$ at density $n = 1$ and zero extra descriptors, the parent dashed, the network solid, from $s = 0$ to 6. Top right: the difference $F_x^{\text{NN}} - F_x^{\text{parent}}$ on the same slice, which is where the fit is visible at all. Bottom left: the correlation factor $F_c(s; r_s)$ at $\zeta = 0$ for $r_s = 0.5, 2$ and 5, parent dashed and network solid; bottom right: the three differences. For the meta-GGA clone the layout is three columns: $F_x(s; \alpha)$ at $\alpha = 0$ and $\alpha = 1$ (the SCAN Fig. 1 convention, α the exact iso-orbital indicator of the slice) and F_c at the same two α values, each with its difference panel. The footer of each figure names the run, the slices, the largest $|\Delta F_x|$ on the slice and the certificate verdict.

deep_3x16: $|\Delta F_x| \leq 7 \times 10^{-3}$ over $s \leq 6$, a smooth negative deviation growing beyond $s = 4$ (the region the pre-training set barely samples), $|\Delta F_c| \leq 4.5 \times 10^{-3}$ with oscillations below $s = 1.5$ at every r_s . deep_attn_3x16: $|\Delta F_x| \leq 6 \times 10^{-3}$, $|\Delta F_c| \leq 10^{-2}$ with one dip at $s = 2$, $r_s = 0.5$. deep0_3x16: $|\Delta F_x| \leq 2.6 \times 10^{-3}$ (the tightest exchange clone of the three shown), $|\Delta F_c| \leq 1.2 \times 10^{-2}$ at $r_s = 0.5$ and 5. deep0_cusp_mgga_3x16 against SCAN: F_x at $\alpha = 0$ runs 0.005 below SCAN up to $s = 1$ and 0.01 to 0.027 above it for $s > 2.5$ (the maximum at $s = 4.3$), and F_x at $\alpha = 1$ 0.005 to 0.013 below it between $s = 1$ and 4; F_c at $\alpha = 1$, $r_s = 0.5$ has a spike of +0.04 near $s = 0.2$, a dip of -0.022 at $s = 0.8$ and sits 0.005 to 0.033 below SCAN from $s = 2$ onward; at $\alpha = 0$ the correlation differences stay within ± 0.02 . These are systematic, same-sign offsets over wide ranges of s , which is the shape of error that does not cancel between a molecule and its atoms (Sec. 3.6).

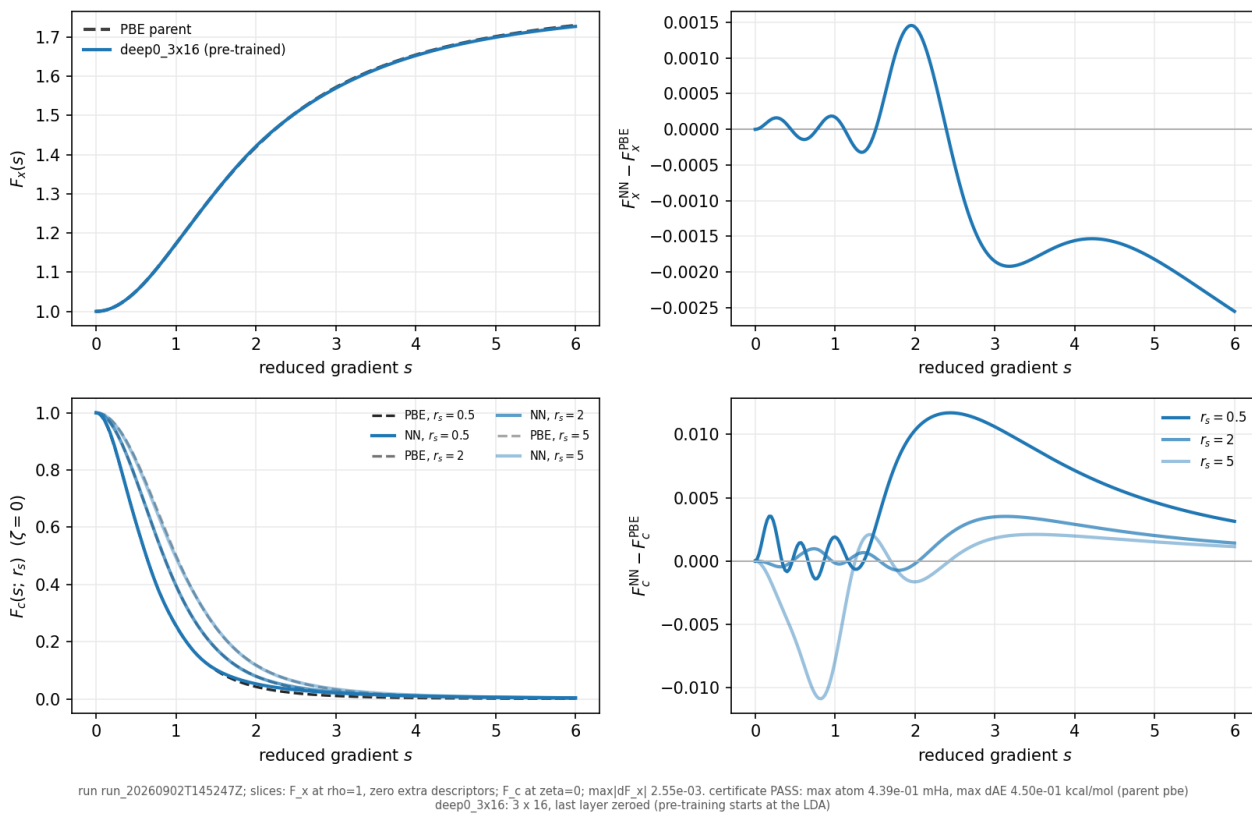
deep_3x16: pre-trained network against the PBE parent

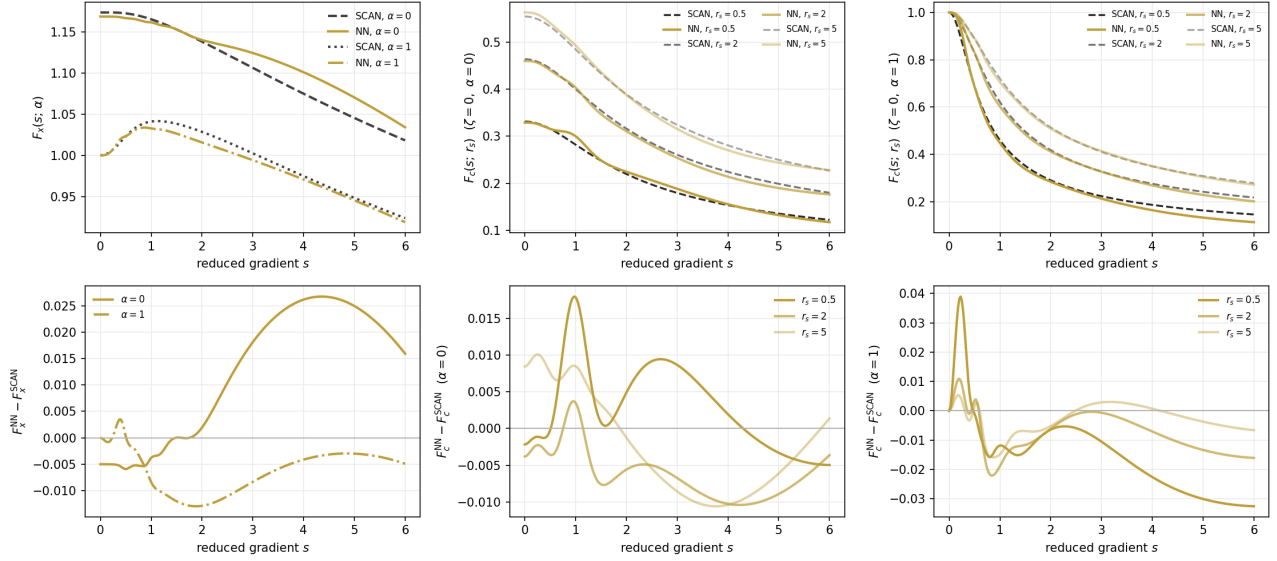


deep_attn_3x16: pre-trained network against the PBE parent



deep0_3x16: pre-trained network against the PBE parent

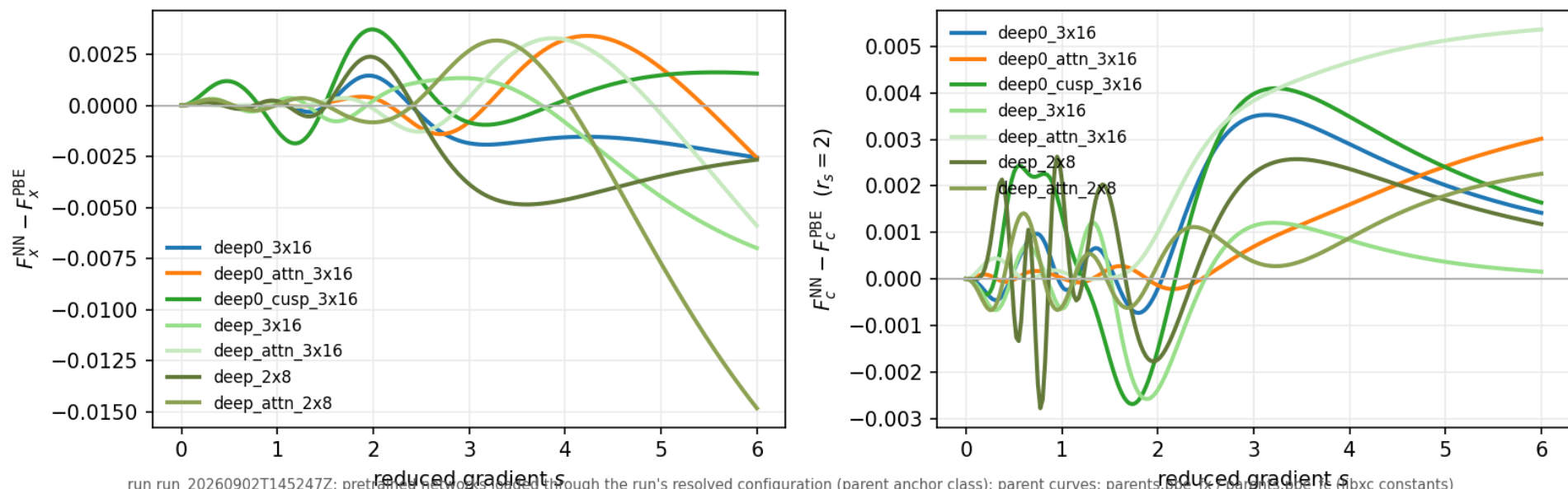




run run_20260902T145250Z; parent SCAN; slices: F_x at $\rho=1$ at the exact iso-orbital indicator α in $\{0, 1\}$, zero other descriptors; F_c at $\zeta=0$, same α slices; $\max(\text{dF}_x)$ 2.67e-02, certificate FAIL: max atom 3.13e+00 mHa, max dAE 7.26e+00 kcal/mol (parent scan)
 deep0_cusp_mgga_3x16: 3 x 16, last layer zeroed (pre-training starts at the LDA), cusp descriptors, meta-GGA (SCAN parent)

Fig. 3.4 The exchange differences (left) and the correlation differences at $r_s = 2$ (right) of every certified GGA clone on one pair of axes. The four Glorot-start exchange clones agree with PBE to 1×10^{-3} below $s = 2.5$ and fan out beyond it (deep_2x8 to -0.005 at $s = 3.3$, deep_attn_3x16 to $+0.003$ at $s = 3.9$, and at $s = 6$ deep_attn_2x8 -0.015 , deep_3x16 -0.007 , deep_attn_3x16 -0.006 , deep_2x8 -0.003); their correlation clones stay within 3×10^{-3} below $s = 3$, deep_attn_3x16 drifting to $+5 \times 10^{-3}$ by $s = 6$. The three zero-init clones stay within 4.2×10^{-3} everywhere on both slices (deep0_attn_3x16 exchange $+0.0034$ at $s = 4.2$; deep0_cusp_3x16 correlation $+0.0041$ at $s = 3.2$).

Pre-trained networks against the PBE parent: differences, all architectures



run run_20260902T145247Z; pretrained networks loaded through the run's resolved configuration (parent anchor class); parent curves: parents_pbe_1x16 (parent pbe_1x16), parents_pbe_1x16 (parent pbe_1x16), parents_pbe_1x16 (parent pbe_1x16), parents_pbe_1x16 (parent pbe_1x16), parents_pbe_1x16 (parent pbe_1x16), parents_pbe_1x16 (parent pbe_1x16), parents_pbe_1x16 (parent pbe_1x16), parents_pbe_1x16 (parent pbe_1x16)

deep0_3x16: 3 x 16, last layer zeroed (pre-training starts at the LDA); deep0_attn_3x16: 3 x 16, last layer zeroed (pre-training starts at the LDA), 4 attention heads; deep0_cusp_3x16: 3 x 16, last layer zeroed (pre-training starts at the LDA), cusp descriptors; deep_3x16: 3 x 16, Glorot initialization; deep_attn_3x16: 3 x 16, Glorot initialization, 4 attention heads; deep_2x8: 2 x 8, Glorot initialization; deep_attn_2x8: 2 x 8, Glorot initialization, 2 attention heads

The deep_2x8 pair and the deep0_attn_3x16 and deep0_cusp_3x16 clones are drawn in the same directories (pretrain_fx_fc_<arch>.png).

3.6 Certificate results

Read from pretrain/<arch>/fidelity_certificate.json and pretrain_metadata.json; the pointwise losses are the saved network's, in units of the squared enhancement-factor residual.

Every GGA clone passes on all three criteria; the two with a species above 1 kcal/mol (deep_2x8: C3H8, C4H6; deep0_cusp_3x16: AlCl3) pass on the mean and under the 2.0 kcal/mol backstop. The attention variants are the tightest clones by every measure (deep_attn_3x16: max atom 0.058 mHa, mean 0.031 kcal/mol; deep0_attn_3x16: 0.123 mHa, 0.042 kcal/mol), at the price of the longest fits (about 28 h against 3.6 to 3.8 h for their plain twins, duration_seconds).

The meta-GGA clone deep0_cusp_mgga_3x16 fails all three criteria after a converged 20000-step fit (pointwise residuals $3.7\text{e-}6$ and $5.2\text{e-}6$, an order of magnitude above the GGA clones'): the largest free-atom error is 3.13 mHa (O; B 2.91, Be 2.61), the mean atomization error 2.06 kcal/mol and the largest 7.26 kcal/mol (C4H6; C3H8 7.19, O2 4.77, SiCH6 3.52, CH4 3.47), 14 of 22 molecules above 1 kcal/mol, every one of the eight largest errors of the same sign. Fig. 3.3 (bottom) shows where the clone misses SCAN: the $\alpha = 0$ exchange slice is 0.015 to 0.027 above SCAN for $s > 2$ and the $\alpha = 1$ correlation slices sit 0.01 to 0.03 below it at $r_s = 0.5$, systematic offsets that do not cancel between a molecule and its atoms. The v6 meta-GGA certificates, all PASS, were fitted in the anchored model class (the network learns a correction on top of SCAN); this is the first unanchored meta-GGA clone the two-tier gate has judged. The four remaining meta-GGA architectures are pending; whether the rung or this architecture is the cause is decided by their certificates. Until then the meta-GGA training array is held.

Run	Architecture	Parent	Steps	Pointwise loss X / C	max atom $ \Delta E_{xc} $ (mHa)	mean $ \Delta AE $ (kcal/mol)	max $ \Delta AE $ (kcal/mol)	Species over 1 kcal/mol	Verdict
size	deep_2x8	PBE	20000	5.41e-08 / 3.29e-06	0.369	0.343	1.421	C3H8, C4H6	PASS
size	deep_attn_2x8	PBE	20000	7.62e-08 / 3.83e-07	0.201	0.131	0.573	none	PASS
size	deep_3x16	PBE	20000	6.40e-08 / 7.58e-07	0.210	0.162	0.519	none	PASS
size	deep_attn_3x16	PBE	20000	1.51e-08 / 6.70e-08	0.058	0.031	0.115	none	PASS
families	deep0_3x16	PBE	20000	2.77e-08 / 3.37e-06	0.439	0.114	0.450	none	PASS
families	deep0_attn_3x16	PBE	20000	1.48e-08 / 9.86e-08	0.123	0.042	0.149	none	PASS
families	deep0_cusp_3x16	PBE	20000	9.46e-08 / 1.10e-06	0.439	0.188	1.071	AlCl3	PASS
meta-GGA	deep0_cusp_mgga_3x16	SCAN	20000	3.70e-06 / 5.18e-06	3.132	2.062	7.257	AlCl3, C2H2, C3H8, C4H6, CH2, CH4, CO, CO2, F2, H2O, HCN, O2, PH3, SiCH6	FAIL

4. Optimization, in-sample

Sources: `xcquinox/alec/losses.py` (L5_gradnorm_vxc_step7, the channel terms), `xcquinox/alec/train.py` (the per-molecule loop, the optimizer), `xcquinox/alec/oneshot.py` (the SCF tail window), `xcquinox/alec/solver_manual.py` (the differentiable SCF), the hyperparams: and solvers: blocks of the resolved configurations, and per cell `train_metadata.json`, `losses.npy`, `aux_log.pkl`, `eval/per_molecule.json`.

4.1 Training protocol

Each cell trains the certified clone of its architecture on a subset of the 26-point DFS pool. The per-molecule update scheme takes one optimizer step per training group (a reaction, or a molecule with its density and potential references) per epoch, so an epoch is one pass over the groups. The loss of a group is the fixed weighted sum

$$\mathcal{L} = 1 \cdot \mathcal{L}_{AE} + 1 \cdot \mathcal{L}_{\text{rxn}} + 1 \cdot \mathcal{L}_{IP} + 1 \cdot \mathcal{L}_{v_{xc}} + 20 \cdot \mathcal{L}_{\rho},$$

the Letter’s $\lambda_{RE} = 1$, $\lambda_n = 20$ structure (effective_channel_weights; the GradNorm balancer is configured but dormant under this scheme, and the vxc_weight 0.01 and density_weight 0.1 pre-scales are forced to 1). The channels, with E^{NN} the self-consistent total energy of the 3-cycle SCF (Sec. 4.2) and $t = 0, \dots, N - 1$ the cycle index:

- Reaction energies (ae_as_reactions: true routes the W4-11 atomizations here as molecule-to-atoms reactions with the network’s own atoms, beside the BH76 barriers):

$$\mathcal{L}_{\text{rxn}} = \frac{1}{|T|} \sum_{t \in T} w_t^2 r_t^2, \quad r_t = \sum_i c_i E_{i,t}^{\text{NN}} - E^{\text{ref}}, \quad w_t = \left(\frac{t}{N-1} \right)^2,$$

T the kept SCF tail (the last $\min(N, 10)$ cycles; at $N = 3$ all three, $w_t^2 = 0, 1/16, 1$), the DFS convergence-tail scoring (DFS Eqs. 15 and 16) that penalizes a non-converging SCF instead of one arbitrary final cycle. Absolute Ha² (loss_metric: absolute). References: W4-11 atomization energies and the W2-F12 BH76 barrier heights of GMTKN55.

- Atom anchors (the residue of the AE channel once the atomizations are reactions): the free atoms of the subset are held near their exact non-relativistic energies E_Z^{anchor} (Chakravorty et al., PRA 47, 3649 (1993)), $\mathcal{L}_{AE} = 0.01 \cdot \text{mean}_Z (E_Z^{\text{NN}} - E_Z^{\text{anchor}})^2 / (E_Z^{\text{anchor}})^2$.
- Ionization potentials, $\mathcal{L}_{IP} = \text{mean} (E_+^{\text{NN}} - E_0^{\text{NN}} - IP^{\text{ref}})^2$, present in the form but empty on every v7 subset (no IP13 point is selected).
- Exchange-correlation potential, on the reference density (frozen features): $\mathcal{L}_{v_{xc}} = \|V_{xc}^{\text{NN}} - V_{xc}^{\text{ref}}\|_F^2 / n_{\text{AO}}^2$, summed over spin channels for open shells.
- Density, the final mixed density of the SCF against the CCSD reference density on the same grid, per electron squared (the Letter’s Eq. 17 normalization; density_per_electron: true):

$$\mathcal{L}_{\rho} = \text{mean}_{\text{mol}} \frac{\sum_g w_g (n_g^{\text{NN}} - n_g^{\text{ref}})^2}{N_e^2}, \quad N_e = \sum_g w_g n_g^{\text{ref}}.$$

Optimizer: AdamW, learning rate 10^{-3} held for the first half of the run and decayed linearly to 10^{-5} over the second (lr_decay_start: 0.5), weight decay 10^{-4} , gradient clip 1.0. 200 epochs (n_steps), a validation reaction-energy MAE on the 47-reaction validation slice every 25 epochs, early stop after 5 checks without a 0.1 kcal/mol improvement; three checkpoints are kept (final, lowest training loss, lowest validation MAE), and the figures of Secs. 4 and 5 read the validation-best one. Over the 33 cells the runs took 150 to 200 epochs (early-stopped in 17) and 1.2 to 34.9 h of wall each (train_metadata.json).

4.2 The differentiable SCF

The energies and densities above are self-consistent: the manual backend runs a fixed $N = 3$ cycles from the converged PBE density (solver: full_3), diagonalizing the Kohn-Sham matrix built from the network’s V_{xc} each cycle, with the DFS decaying linear mixing $D_{t+1} = (1 - a_t)D_t + a_tD_t^{\text{out}}$, $a_t = 0.3^t + 0.3$, a freeze once $|\Delta E| < 10^{-6}$ Ha, and gradients through every cycle. The density scored by $\mathcal{L}_\rho$ and by the density metrics is the last mixed density; three cycles do not converge the SCF for a functional far from a fixed point, which is the subject of the two arms submitted on 2026-09-08 (25 cycles; dpyscf parity) and of the converged-SCF re-evaluation in flight. The density record of 2026-09-07 (notebooks/analysis/AUDIT_2026-09-01.md, the density section) records the full parity table against dpyscf.

4.3 Training losses

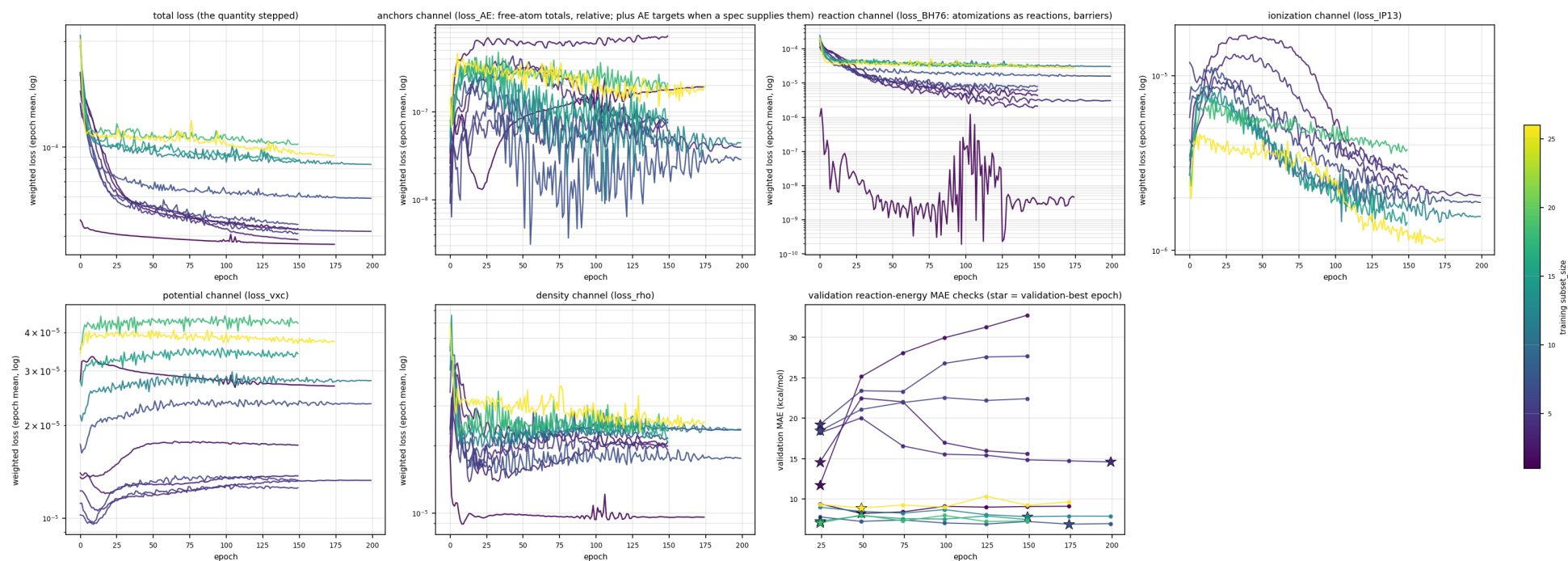
Fig. 4.1 Per-channel training losses, one figure per architecture, eight panels. Every loss panel is the weighted per-epoch mean over the epoch’s group updates read from the cell’s `aux_log.pkl`, on a log axis against the epoch, one curve per subset size (the color bar: dark one molecule, yellow twenty-six). Panel 1 is the total loss, the quantity the optimizer steps; panels 2 to 6 are the five channels of Sec. 4.1 at their trained weights, and they add up to panel 1 at every epoch. Panel 7 is the validation reaction-energy MAE at each 25-epoch check, the star marking the validation-best epoch whose checkpoint every later figure reads. The footer states the weights and their source.

deep_3x16 (the first figure): the total falls from $1.3\text{e-}4$ to $2.1\text{e-}4$ at the first epoch to a plateau reached by epoch 25 in every cell, the plateau ordered by subset size from $3.7\text{e-}5$ (one molecule) to $1.0\text{e-}4$ (eighteen), because the potential and density channels grow with the number of supervised molecules. The reaction channel is the one that falls by orders of magnitude: from $1\text{e-}4$ at the first epoch to $4\text{e-}9$ (one molecule), $2\text{e-}6$ to $8\text{e-}6$ (two to six) and $1.6\text{e-}5$ to $3.1\text{e-}5$ (seven to twenty-six) at the end. The anchors channel is $1\text{e-}8$ to $1\text{e-}6$ throughout and drifts upward in the large-subset cells (the free atoms are released as the molecules pull the functional). The ionization channel (present from three molecules, where Li^+ enters the subset) peaks at $1\text{e-}5$ within the first 50 epochs and falls to $1\text{e-}6$ to $4\text{e-}6$. The potential channel is the one that rises: from $1\text{e-}5$ to $1.3\text{e-}5$ at the start to $1.4\text{e-}5$ (one to six molecules) and $2.4\text{e-}5$ to $4.3\text{e-}5$ (seven to twenty-six) at the end, the rise steepest in the first 25 epochs. The density channel drops from $2.6\text{e-}5$ to $4.8\text{e-}5$ to $1.0\text{e-}5$ (one molecule) and $1.8\text{e-}5$ to $2.5\text{e-}5$ (the rest) inside the first 10 epochs and is flat afterwards. The validation panel separates the cells into two families: the two-to-six-molecule cells start at 12 to 19 kcal/mol and rise to 15 to 33 by the last check (the star at epoch 25 for two, three, five and six; at 200 for four, whose MAE falls back from 20 to 14.6), while the one-molecule cell and the cells from seven upward hold 7 to 10 kcal/mol across every check.

deep_attn_3x16 (the second figure): the same shapes with a reaction channel that falls further ($5\text{e-}7$ to $2\text{e-}6$ at two to six molecules, $8\text{e-}7$ at seven), a density channel that ends lower ($6\text{e-}6$ to $1.3\text{e-}5$), and a validation panel where the two-to-six-molecule cells fall rather than rise: from 22 to 37 kcal/mol at the first check to 7 to 13 by epoch 100, with the validation-best epochs at 75 to 200. deep0_3x16 (the third figure) reproduces the deep_3x16 picture, validation included (two molecules: 11.2 to 32.4 kcal/mol; seven to eighteen: 6.2 to 8.8).

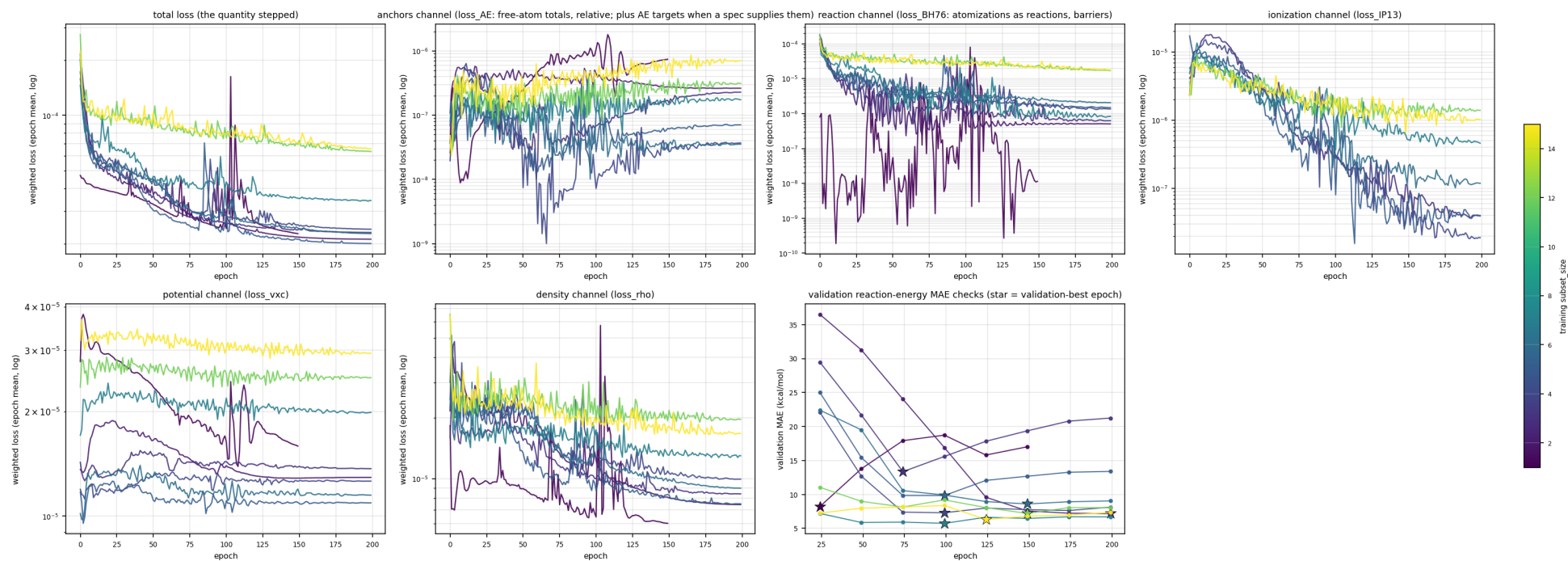
The first and last five-epoch means of every channel and every validation check are in the table of Sec. 4.5.

Per-channel training losses -- deep_3x16 (weighted epoch means; panels sum to the total) · run_20260902T145247Z · val-best



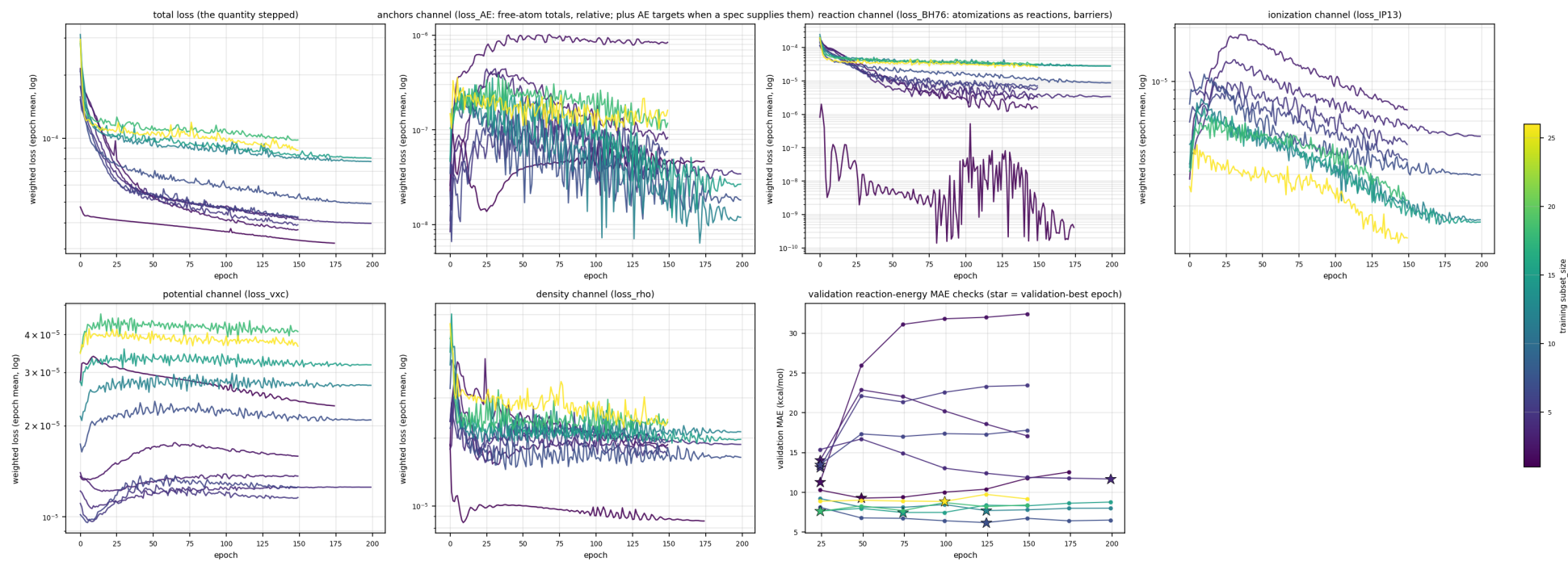
Held-out reactions: 4/29 archs (deep_3x16, deep_attn_3x16, deep0_3x16, deep0_attn_3x16). NOT TRAINED in this run: deep_2x8, deep_attn_2x8, deep0_cusp_3x16.
Channel weights as trained: loss_AE 1, loss_BH76 1, loss_IP13 1, loss_vxc 1, loss_rho 20 (source: train_metadata); every panel is the weighted epoch mean over the epoch's group updates of aux_log.pkl, so the five channel panels add up to the total.

Per-channel training losses -- deep_attn_3x16 (weighted epoch means; panels sum to the total) · run_20260902T145247Z · val-best



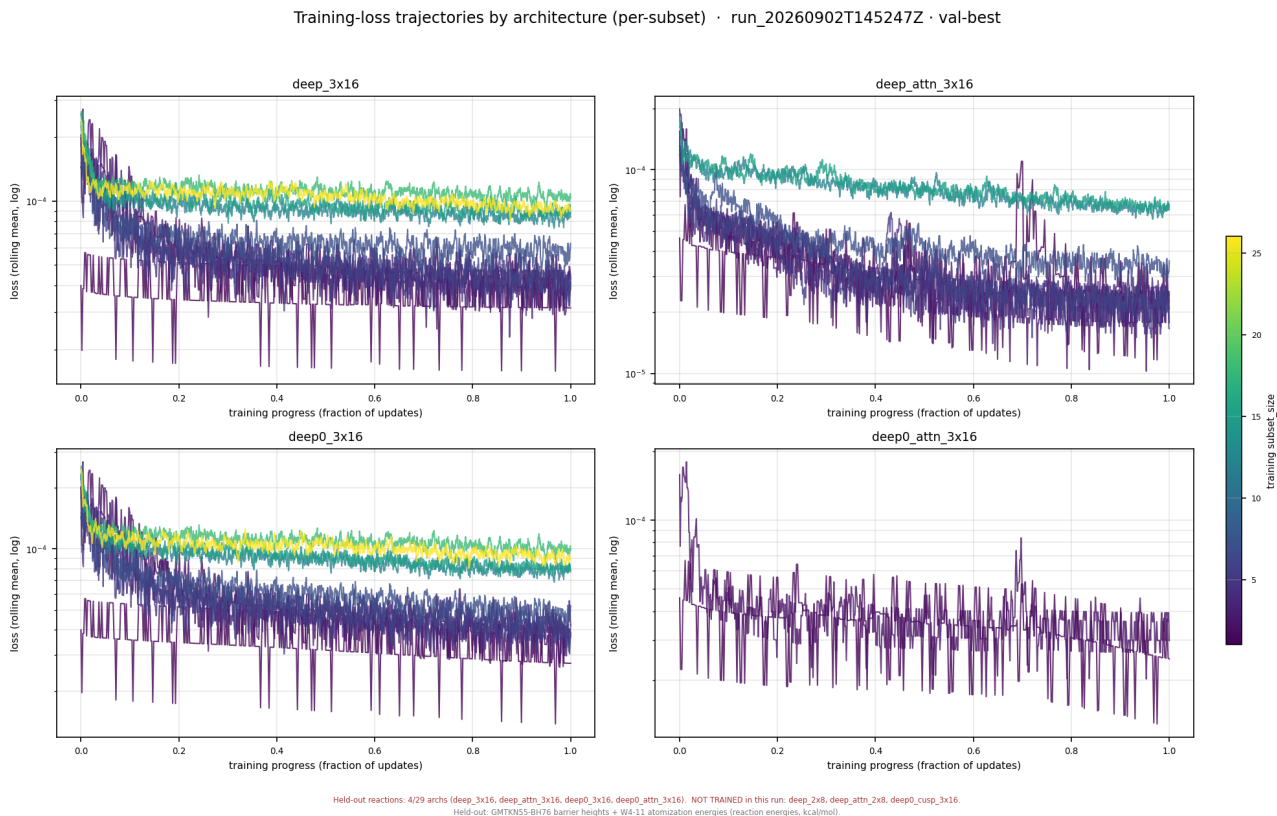
Held-out reactions: 4/29 archs (deep_3x16, deep_attn_3x16, deep0_3x16, deep0_attn_3x16). NOT TRAINED in this run: deep_2x8, deep_attn_2x8, deep0_cusp_3x16.
Channel weights as trained: loss_AE 1, loss_BH76 1, loss_IP13 1, loss_vxc 1, loss_rho 20 (source: train_metadata); every panel is the weighted epoch mean over the epoch's group updates of aux_log.pkl, so the five channel panels add up to the total.

Per-channel training losses -- deep0_3x16 (weighted epoch means; panels sum to the total) · run_20260902T145247Z · val-best



Held-out reactions: 4/29 archs (deep_3x16, deep_attn_3x16, deep0_3x16, deep0_attn_3x16). NOT TRAINED in this run: deep_2x8, deep_attn_2x8, deep0_cusp_3x16.
Channel weights as trained: loss_AE 1, loss_BH76 1, loss_IP13 1, loss_vxc 1, loss_rho 20 (source: train_metadata); every panel is the weighted epoch mean over the epoch's group updates of aux_log.pkl, so the five channel panels add up to the total.

Fig. 4.2 The total training loss per cell as a rolling mean over the group updates (log axis against the fraction of the run’s updates; the first panel of Fig. 4.1 on the update rather than the epoch axis), one panel per architecture, one curve per subset size. The curves are the quantity stepped; the fall between the first and the last five percent of the updates is the “loss fall” column of the table in Sec. 4.5 (1.2 to 5.1 over the 33 cells). The deep_3x16 panel separates into the two plateaus of Fig. 4.1 (1e-4 for the cells from twelve molecules upward, 3e-5 to 5e-5 below); the one-molecule curve alternates between two levels from update to update. The deep_attn_3x16 panel shows transient rises near 45 and 70 percent of the run in the one- and two-molecule cells, the spikes of the potential and density channels around epochs 85 to 105 in Fig. 4.1.



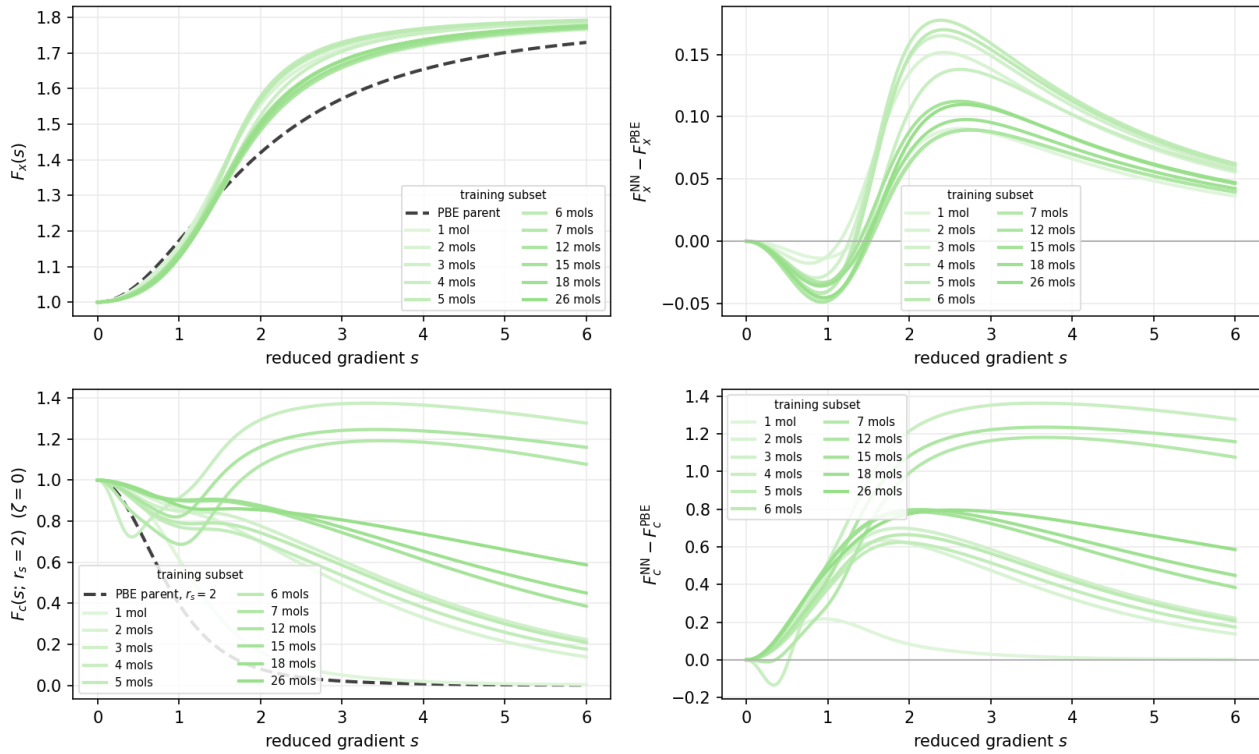
4.4 Trained enhancement factors

Fig. 4.3 The trained networks against PBE, one figure per architecture, four panels, one curve per subset size at the validation-best checkpoint. Top left: $F_x(s)$ at $n = 1$ and zero extra descriptors, PBE dashed. Top right: $F_x^{\text{NN}} - F_x^{\text{PBE}}$ on the same slice. Bottom left: $F_c(s; r_s = 2)$ at $\zeta = 0$, PBE dashed. Bottom right: the correlation difference. The pre-trained curves of Fig. 3.3 are the starting point of every curve here, and the footer gives the largest $|\Delta F_x|$ over the slice.

Exchange, all three architectures: every trained network lowers F_x below PBE between $s = 0.3$ and 1.3 (by 0.02 to 0.06 at $s \approx 0.8$) and raises it above PBE from $s \approx 1.4$ onward, with the maximum of the raise at $s = 2.5$ to 3 (deep_3x16 up to 0.18; deep_attn_3x16 0.07 to 0.13; deep0_3x16 0.05 to 0.14) and a slow decay to +0.03 to +0.06 at $s = 6$, so that the trained F_x crosses 1.7 near $s = 3.5$ where PBE crosses it near $s = 5$. The shape is the same at every subset size; the one-molecule cell moves least. Correlation at $r_s = 2$: PBE’s F_c falls from 1 at $s = 0$ to 0.02 at $s = 3$; every trained network holds F_c between 0.6 and 1.0 up to $s = 2$, and beyond that the cells separate. deep_3x16: the one-molecule cell returns to PBE (a difference of 0.2 at $s = 1$ and 0 beyond 3), the two- and three-molecule cells rise to 1.1 to 1.4 at $s = 6$, the four-to-six-molecule cells decay to 0.15 to 0.25 and the cells from seven to twenty-six molecules to 0.4 to 0.6. deep0_3x16: the one-molecule cell as for deep_3x16, the two-molecule cell 1.5 at $s = 6$, the fifteen-molecule cell flat at 0.8 from $s = 2$, the twelve- and eighteen-molecule cells 1.1 to 1.2, the others 0.1 to 0.25. deep_attn_3x16: the one-molecule cell returns to PBE, every other cell rises above 1 from $s = 2$ to 3 onward, to 1.2 (two molecules), 1.4 (three and seven), 1.7 (four) and 1.9 (five and six) at $s = 6$. The correlation difference therefore reaches 0.2 to 1.9 at large s depending on the cell.

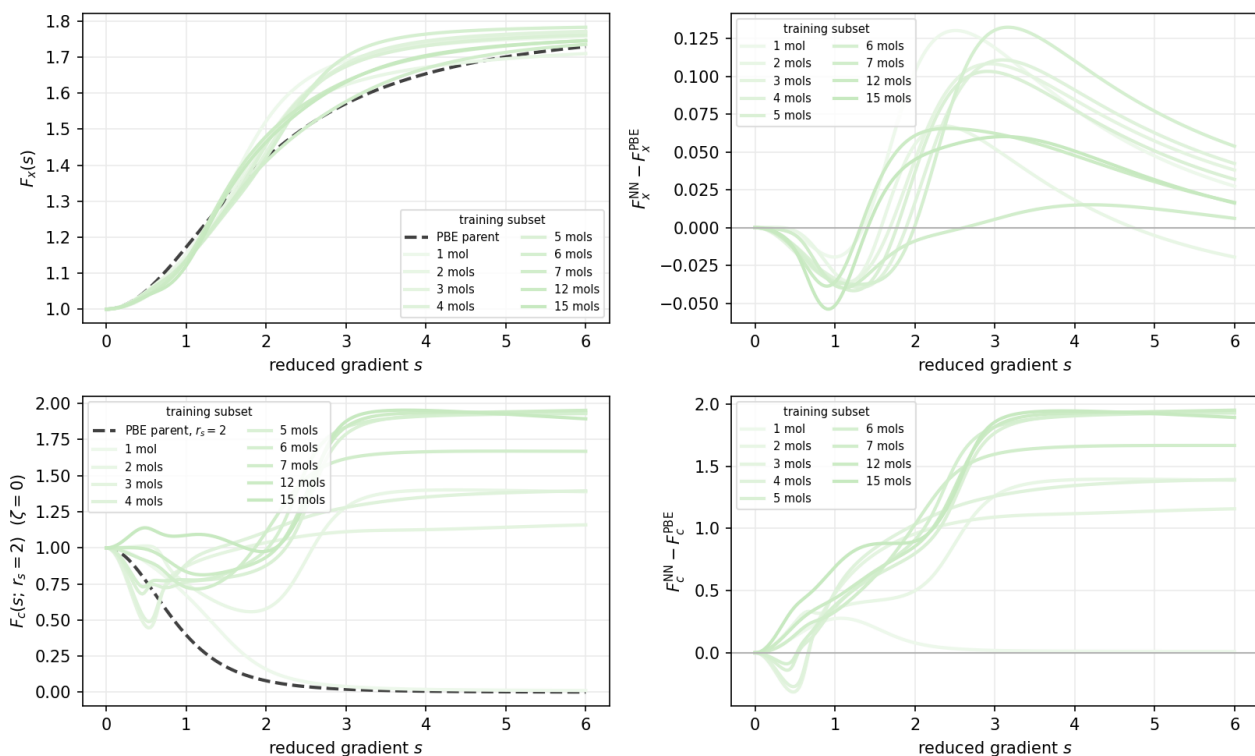
The corrections are therefore not small perturbations of PBE: the density-gradient dependence of both channels is reshaped in the region $s > 1.5$, which is populated by the valence tails and the bonding regions of molecules, and what Sec. 5 measures is how that reshaping transfers.

deep_3x16: trained networks against the PBE parent



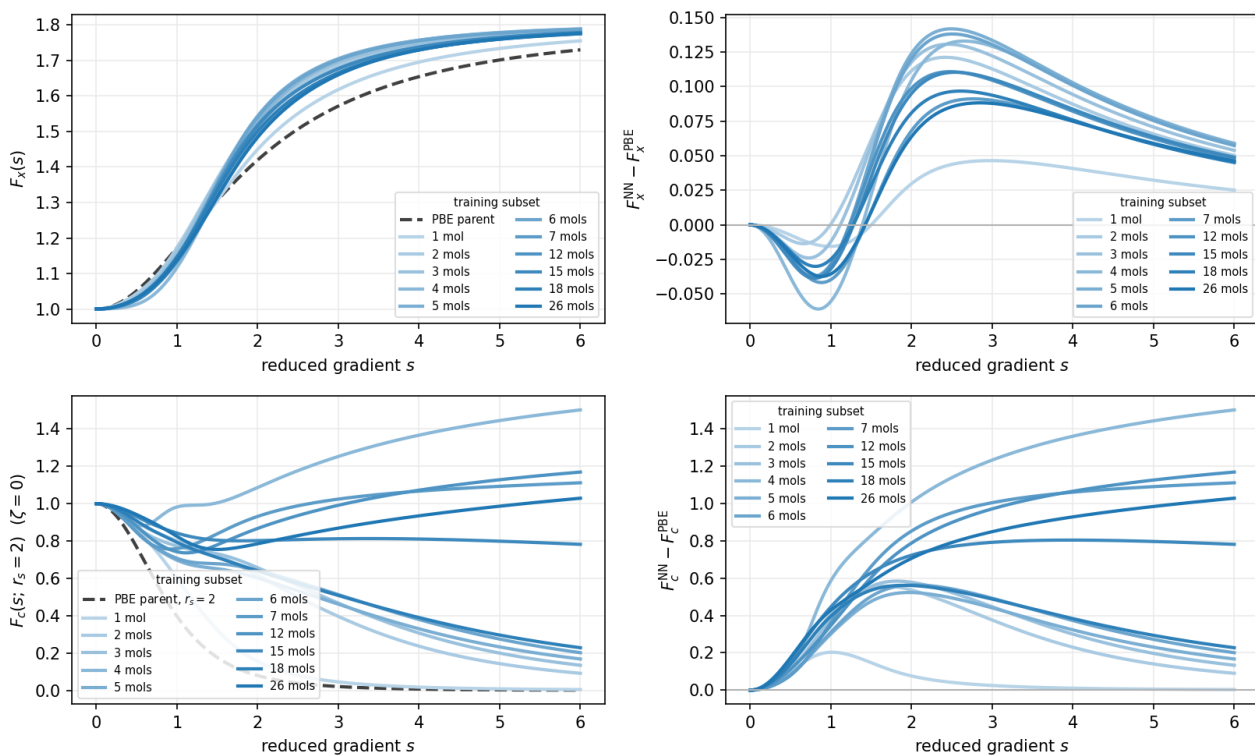
run_run_20260902T145247Z; checkpoint channel val_best (model_val_best.eqx); slices: F_x at rho=1, zero extra descriptors; F_c at zeta=0, r_s=2; 11 completed cell(s); max|dF_x| 1.78e-01; parent curves parents.pbe_fx / parents.pbe_fc (libxc constants).
deep_3x16: 3 x 16, Glorot initialization

deep_attn_3x16: trained networks against the PBE parent



run run_20260902T145247Z; checkpoint channel val_best (model_val_best.eqx); slices: F_x at $\rho=1$, zero extra descriptors; F_c at $\zeta=0$, $r_s=2$; 9 completed cell(s); max|dF_x| 1.32e-01; parent curves parents.pbe_fx / parents.pbe_fc (libxc constants).
deep_attn_3x16: 3 x 16, Glorot initialization, 4 attention heads

deep0_3x16: trained networks against the PBE parent



run run_20260902T145247Z; checkpoint channel val_best (model_val_best.eqx); slices: F_x at $\rho=1$, zero extra descriptors; F_c at $\zeta=0$, $r_s=2$; 11 completed cell(s); max|dF_x| 1.42e-01; parent curves parents.pbe_fx / parents.pbe_fc (libxc constants).
deep0_3x16: 3 x 16, last layer zeroed (pre-training starts at the LDA)

4.5 In-sample fit

Fig. 4.4 Training-set density against CCSD in grid-RMSE units, final checkpoint (the in-sample evaluation exists for the final checkpoint only; its validation-best twin is identical because the in-sample directory is shared). Left: the cell mean over the trained species against the subset size (the species count printed at each point), one line per architecture, with PBE on the same species as the dashed line with crosses. Middle: every trained species on the abscissa (22, the union of the subsets, the same for every run), the network’s RMSE at every cell that trained it as a point and PBE’s as a cross, log scale: the network’s points sit below the cross for most species, FLi and HLi ($0.7\text{e-}5$ to $2\text{e-}5$ against $1.5\text{e-}5$ and $4\text{e-}5$) and the second-row hydrides among them, and above it for a few (C_2H_2 in the small-subset cells). Right: the NN/PBE ratio of the cell means against the subset size, PBE at 1: deep_3x16 0.72 at one molecule, 0.81 to 0.97 at two to six, 0.73 to 0.79 from seven; deep_attn_3x16 0.57 to 0.76; deep0_3x16 0.68 to 0.93; deep0_attn_3x16 0.64 and 0.91 at one and two molecules. The table below gives the ratios and the count of improved species per cell (198 of the 233 (cell, species) pairs over the 33 cells, 50 of deep_attn_3x16’s 51).

IN-SAMPLE density fit on TRAINING molecules (atoms excluded; weighted-mean grid RMSE vs CCSD, NOT N e-normalized). Training-set fit, NOT generalization; not comparable to the held-out energy panels.

IN-SAMPLE density fit on TRAINING molecules (atoms excluded; weighted-mean grid RMSE vs CCSD, NOT N e-normalized). Training-set fit, NOT generalization; not comparable to the held-out energy panels.

Fig. 4.5 In-sample energy and density in DFS units, the 3x3 layout of Fig. 5.5 on the training set: rows WTMAD-2 / ε_n / $\mathcal{ED}$, columns BH76 / W4-11 / combined, each cell against PBE on its own training reactions and molecules. The energy leg is the training reactions themselves (the W4-11 atomizations as molecule-to-atoms reactions and the BH76 barriers of `train_metadata.json`), each formed from the final checkpoint’s self-consistent species energies in `eval/per_molecule.json` against its reference; PBE on the same reactions from the same file’s PBE energies. The dashed line is PBE on the union of the run’s training reactions (18.6 kcal/mol for the BH76 barriers, 3.83 for the W4-11 atomizations, 5.67 combined, the same union for every run since the runs train the same subsets); the capped spans and the verdict triangles use each cell’s own reactions. A cell whose subset has no reaction of a column (the one-molecule cell in the BH76 column; the three-molecule cell, whose subset is H₂O, Li⁺ and a barrier, in the W4-11 column) has no bar there.

Panel A (BH76 energy): `deep_3x16`’s barrier residual grows with the subset from 2.7 kcal/mol at two molecules to 9.5 at twenty-six, against 18.6 for PBE on the union and 12 to 25 on the cells’ own barriers; `deep_attn_3x16` 0.8 to 3.2. Panel B (W4-11): `deep_3x16` 0.02 (one molecule, H₂O fitted to 0.02 kcal/mol) to 1.6, `deep_attn_3x16` 0.05 to 0.36, against PBE’s 3.83. Panel C: the two-subset combination, 0.02 to 4.3 against 5.67 on the union of the training reactions of every cell. Panels D to F (density): the trained molecules’ ε_n is 0.0073 (one molecule) to 0.0135 (two) for `deep_3x16` and 0.0049 to 0.0073 for `deep_attn_3x16`, against PBE’s 0.0089 on the union of the trained molecules; the `deep_3x16` cells at two to seven molecules sit above PBE on this per-electron L1 while their grid RMSE (Fig. 4.4) sits below it: the two norms rank the small-subset fits differently. Panels G to I: every cell’s $\mathcal{ED}$ is below PBE’s on its own slice (0.002 to 8.5 against 0.8 to 13.3), the energy leg deciding the combination at every subset.

In-sample per-channel story (DF5 units): WTMAD-2 | c_{DH} | c_{DH} (BH76, W4-11, combined) -- training fit · run_20260902T14524Z · val-best

$\langle D_{\text{cell}} \rangle = 25.1 \cdot E^* = 30 \text{ gamma } c_{\text{cell}} \rangle \cdot c_{\text{cell}} = \frac{1}{N} \sum_{i=1}^N \langle D_{\text{cell}} \rangle = \langle D_{\text{cell}} \rangle$ (Eq. 20 per species). ONE gamma shared by all channels, the letters published 2024.87 kspiral (near the non-axis RL) stamped in each panel -- $\langle D_{\text{cell}} \rangle$ compares across channels, $\langle D_{\text{cell}} \rangle$ of PRE is $\frac{1}{N} \sum_{i=1}^N \langle D_{\text{cell}} \rangle$. Density row = cell-mean c_{cell} near the trained molecule. The dashed PRE (pink) line reduces the UNDO of the cell's training reaction's molecules, not any one cell's; the dotted green and the dash-dot red use each cell's own reaction. CCSD reference predictions carry no doubt mark here (no CCSD computer on the training reactions). Final checkpoint only (final) has no val best variant: identical in the final step and val best direction, only the title's checkpoint stamp differs.

Warning: final checkpoint: each cell's own training reaction (the last) is: BH76 3 + W4-11, alternates as reaction and barrier. P-pairs included: reaction energies, full-cell, density. CCSD species in CCSD cell values excluded.



4.6 Results per cell

The training-set density panels (Fig. 4.4) are the direct check that the density channel trains: on the supervised molecules the cell-mean density RMSE against CCSD is below PBE’s in all 33 cells, the NN/PBE ratio 0.72 to 0.97 for deep_3x16, 0.57 to 0.76 for deep_attn_3x16, 0.68 to 0.93 for deep0_3x16 and 0.64 and 0.91 for deep0_attn_3x16 at one and two molecules, with the weakest fits at two to six molecules and 198 of the 233 trained species-cells improved (the second table below, computed from eval/per_molecule.json with case twins collapsed). The total loss (Fig. 4.2) falls by a factor 1.2 to 5.1 between the first and the last 5 percent of the updates and is flat by the end.

Per channel (Fig. 4.1; the first table below gives the first and last five-epoch means of every weighted channel, computed from the logs, and the third the validation checks): the reaction channel falls in every cell, by 3x (deep0_3x16 at 18 molecules) to 190x (deep_attn_3x16 at 2), and the density channel by 1.3x to 4.6x, both settling within the first 25 epochs; the potential channel, the term the DFS objective does not carry (Sec. 4.2), rises over training from seven to eighteen molecules in the deep_3x16 and deep0_3x16 cells (deep_3x16 at 12: $2.2\text{e-}5$ to $2.8\text{e-}5$; at 18: $3.9\text{e-}5$ to $4.3\text{e-}5$), is flat at 26 ($3.7\text{e-}5$ in both) and falls in the attention cells at 12 and 15 (deep_attn_3x16: $2.6\text{e-}5$ to $2.5\text{e-}5$, $3.3\text{e-}5$ to $2.9\text{e-}5$); the anchors channel stays at $1\text{e-}7$ to $1\text{e-}6$ and the ionization channel (populated from three molecules, where the Li ionization enters) peaks in the first 50 epochs and falls. The validation MAE of the 2-to-6-molecule cells of deep_3x16 and deep0_3x16 rises after the first check (deep_3x16 at 2: 11.7 to 32.7 kcal/mol, validation-best at epoch 25), while the same cells of deep_attn_3x16 fall over the run, and deep0_attn_3x16 splits (at 1: 7.8 to 15.1, best at 25; at 2: 33.8 to 23.7, best at 150); the cells from seven molecules upward hold 6 to 11 kcal/mol across their checks (deep_attn_3x16 at 12 opens at 11.0 and ends at 8.0; deep_3x16 at 26 peaks at 10.4). The held-out rise at the same subset sizes is recorded in Sec. 5.8.

Training losses per cell (aux_log.pkl of each cell; the weighted per-epoch means of the first and last five epochs; the channel weights 1, 1, 1, 1, 20 of Sec. 4.1).

Run	Architecture	Subset	Epochs	total first/last	AE first/last	reactions first/last	IP13 first/last	V_xc first/last	rho first/last
size	deep_3x16	1	175	4.47e-5 / 3.66e-5	7.77e-8 / 1.93e-7	7.25e-7 / 4.16e-9	0.00e+0 / 0.00e+0	3.13e-5 / 2.68e-5	1.26e-5 / 9.62e-6
size	deep_3x16	2	150	1.79e-4 / 4.29e-5	1.66e-7 / 7.16e-7	1.31e-4 / 4.49e-6	0.00e+0 / 0.00e+0	1.35e-5 / 1.73e-5	3.43e-5 / 2.05e-5
size	deep_3x16	3	150	1.83e-4 / 3.85e-5	4.91e-8 / 7.93e-8	1.36e-4 / 2.04e-6	5.12e-6 / 2.63e-6	1.39e-5 / 1.37e-5	2.82e-5 / 2.01e-5
size	deep_3x16	4	200	1.57e-4 / 4.19e-5	4.50e-8 / 3.98e-8	1.10e-4 / 3.05e-6	3.94e-6 / 2.06e-6	1.21e-5 / 1.33e-5	3.12e-5 / 2.35e-5
size	deep_3x16	5	150	1.39e-4 / 4.10e-5	3.75e-8 / 6.83e-8	9.16e-5 / 5.96e-6	1.08e-5 / 2.89e-6	1.07e-5 / 1.25e-5	2.62e-5 / 1.96e-5
size	deep_3x16	6	150	1.34e-4 / 4.51e-5	2.52e-8 / 7.90e-8	8.98e-5 / 8.13e-6	7.05e-6 / 2.11e-6	1.01e-5 / 1.33e-5	2.70e-5 / 2.15e-5
size	deep_3x16	7	200	1.58e-4 / 5.90e-5	1.29e-7 / 2.96e-8	9.01e-5 / 1.60e-5	8.12e-6 / 1.89e-6	1.69e-5 / 2.35e-5	4.29e-5 / 1.76e-5
size	deep_3x16	12	200	2.14e-4 / 8.38e-5	1.03e-7 / 4.38e-8	1.40e-4 / 3.07e-5	4.65e-6 / 1.56e-6	2.17e-5 / 2.79e-5	4.78e-5 / 2.36e-5
size	deep_3x16	15	150	1.96e-4 / 8.79e-5	1.03e-7 / 8.94e-8	1.15e-4 / 3.02e-5	4.00e-6 / 1.50e-6	2.92e-5 / 3.39e-5	4.74e-5 / 2.22e-5
size	deep_3x16	18	150	1.94e-4 / 1.03e-4	1.05e-7 / 2.08e-7	1.08e-4 / 3.13e-5	3.45e-6 / 3.78e-6	3.92e-5 / 4.31e-5	4.38e-5 / 2.46e-5
size	deep_3x16	26	175	1.79e-4 / 9.16e-5	1.98e-7 / 1.84e-7	9.34e-5 / 2.76e-5	3.08e-6 / 1.15e-6	3.73e-5 / 3.73e-5	4.54e-5 / 2.53e-5
size	deep_attn_3x16	1	150	4.49e-5 / 2.28e-5	1.48e-7 / 7.30e-7	3.63e-7 / 1.39e-8	0.00e+0 / 0.00e+0	3.49e-5 / 1.60e-5	9.52e-6 / 6.03e-6
size	deep_attn_3x16	2	200	1.44e-4 / 2.11e-5	3.02e-7 / 2.63e-7	9.54e-5 / 5.08e-7	0.00e+0 / 0.00e+0	1.36e-5 / 1.29e-5	3.43e-5 / 7.44e-6
size	deep_attn_3x16	3	200	1.51e-4 / 2.30e-5	8.53e-8 / 2.28e-7	9.82e-5 / 6.22e-7	7.06e-6 / 3.99e-8	1.32e-5 / 1.37e-5	3.28e-5 / 8.41e-6
size	deep_attn_3x16	4	200	1.25e-4 / 2.40e-5	4.86e-8 / 3.67e-8	8.16e-5 / 1.36e-6	4.99e-6 / 1.87e-8	1.16e-5 / 1.26e-5	2.66e-5 / 9.94e-6
size	deep_attn_3x16	5	200	1.12e-4 / 2.00e-5	7.20e-8 / 7.07e-8	6.49e-5 / 1.49e-6	1.12e-5 / 4.01e-8	1.06e-5 / 1.09e-5	2.51e-5 / 7.51e-6
size	deep_attn_3x16	6	200	1.04e-4 / 2.27e-5	4.82e-8 / 3.52e-8	6.12e-5 / 2.06e-6	6.95e-6 / 1.19e-7	1.05e-5 / 1.15e-5	2.57e-5 / 8.98e-6
size	deep_attn_3x16	7	200	1.18e-4 / 3.43e-5	1.18e-7 / 1.77e-7	5.79e-5 / 8.12e-7	7.39e-6 / 4.76e-7	1.98e-5 / 1.98e-5	3.24e-5 / 1.30e-5
size	deep_attn_3x16	12	200	1.64e-4 / 6.37e-5	1.05e-7 / 3.10e-7	9.69e-5 / 1.72e-5	4.93e-6 / 1.39e-6	2.63e-5 / 2.51e-5	3.61e-5 / 1.97e-5
size	deep_attn_3x16	15	200	1.43e-4 / 6.59e-5	1.56e-7 / 6.95e-7	7.06e-5 / 1.80e-5	5.18e-6 / 1.02e-6	3.26e-5 / 2.94e-5	3.43e-5 / 1.67e-5
families	deep0_3x16	1	175	4.46e-5 / 3.20e-5	5.86e-8 / 4.65e-8	8.47e-7 / 3.81e-10	0.00e+0 / 0.00e+0	3.12e-5 / 2.33e-5	1.25e-5 / 8.61e-6
families	deep0_3x16	2	150	1.80e-4 / 3.69e-5	1.37e-7 / 8.33e-7	1.35e-4 / 1.59e-6	0.00e+0 / 0.00e+0	1.33e-5 / 1.59e-5	3.09e-5 / 1.86e-5
families	deep0_3x16	3	150	1.82e-4 / 4.23e-5	3.82e-8 / 8.71e-8	1.37e-4 / 3.21e-6	4.91e-6 / 7.03e-6	1.35e-5 / 1.36e-5	2.60e-5 / 1.83e-5
families	deep0_3x16	4	200	1.55e-4 / 3.96e-5	4.02e-8 / 3.46e-8	1.10e-4 / 3.40e-6	3.92e-6 / 4.92e-6	1.18e-5 / 1.25e-5	2.93e-5 / 1.87e-5
families	deep0_3x16	5	150	1.38e-4 / 3.91e-5	3.06e-8 / 3.63e-8	9.35e-5 / 5.75e-6	1.00e-5 / 4.56e-6	1.04e-5 / 1.15e-5	2.42e-5 / 1.72e-5
families	deep0_3x16	6	150	1.33e-4 / 4.26e-5	2.23e-8 / 5.93e-8	9.04e-5 / 7.17e-6	6.70e-6 / 3.65e-6	9.91e-6 / 1.25e-5	2.57e-5 / 1.92e-5
families	deep0_3x16	7	200	1.53e-4 / 4.92e-5	1.42e-7 / 1.90e-8	8.72e-5 / 8.77e-6	8.61e-6 / 3.00e-6	1.71e-5 / 2.09e-5	4.03e-5 / 1.65e-5
families	deep0_3x16	12	200	2.11e-4 / 7.79e-5	1.22e-7 / 1.17e-8	1.36e-4 / 2.77e-5	4.97e-6 / 1.66e-6	2.21e-5 / 2.72e-5	4.71e-5 / 2.13e-5
families	deep0_3x16	15	200	1.93e-4 / 8.08e-5	1.30e-7 / 2.62e-8	1.13e-4 / 2.78e-5	4.52e-6 / 1.62e-6	2.93e-5 / 3.18e-5	4.60e-5 / 1.97e-5
families	deep0_3x16	18	150	1.86e-4 / 9.83e-5	1.42e-7 / 1.21e-7	1.01e-4 / 3.13e-5	4.22e-6 / 2.23e-6	3.84e-5 / 4.10e-5	4.26e-5 / 2.36e-5
families	deep0_3x16	26	150	1.76e-4 / 8.92e-5	2.06e-7 / 1.50e-7	9.07e-5 / 2.71e-5	3.24e-6 / 1.35e-6	3.76e-5 / 3.74e-5	4.47e-5 / 2.33e-5
families	deep0_attn_3x16	1	150	4.48e-5 / 2.50e-5	1.32e-7 / 8.15e-7	3.07e-7 / 1.30e-8	0.00e+0 / 0.00e+0	3.48e-5 / 1.67e-5	9.49e-6 / 7.50e-6
families	deep0_attn_3x16	2	200	1.52e-4 / 3.29e-5	2.43e-7 / 2.68e-7	1.04e-4 / 6.97e-7	0.00e+0 / 0.00e+0	1.33e-5 / 1.49e-5	3.51e-5 / 1.70e-5

Validation checks per cell (the validation reaction-energy MAE at every 25-epoch check and the epoch whose checkpoint the figures read).

Run	Architecture	Subset	Epochs	Validation MAE at each 25-epoch check (kcal/mol)	Validation-best epoch
size	deep_3x16	1	175	9.4, 8.2, 8.4, 9.1, 9.0, 9.1, 9.1	50
size	deep_3x16	2	150	11.7, 25.2, 28.1, 30.0, 31.3, 32.7	25
size	deep_3x16	3	150	14.6, 22.5, 22.0, 17.0, 16.0, 15.6	25
size	deep_3x16	4	200	18.2, 20.0, 16.6, 15.6, 15.4, 14.9, 14.7, 14.6	200
size	deep_3x16	5	150	19.2, 23.4, 23.3, 26.8, 27.6, 27.7	25
size	deep_3x16	6	150	18.4, 21.1, 21.9, 22.6, 22.2, 22.4	25
size	deep_3x16	7	200	7.8, 7.3, 7.4, 7.1, 6.9, 7.2, 6.9, 7.0	175
size	deep_3x16	12	200	9.0, 8.5, 8.3, 8.7, 8.1, 7.8, 7.9, 7.9	150
size	deep_3x16	15	150	7.3, 8.0, 7.6, 7.5, 7.9, 7.5	25
size	deep_3x16	18	150	7.1, 8.0, 7.3, 8.0, 7.2, 7.3	25
size	deep_3x16	26	175	9.3, 8.9, 9.3, 9.0, 10.4, 9.3, 9.7	50
size	deep_attn_3x16	1	150	8.2, 13.8, 17.9, 18.7, 15.8, 17.0	25
size	deep_attn_3x16	2	200	36.5, 31.3, 24.0, 16.9, 9.6, 7.6, 7.2, 7.2	200
size	deep_attn_3x16	3	200	29.5, 21.7, 13.3, 15.6, 17.8, 19.4, 20.8, 21.2	75
size	deep_attn_3x16	4	200	22.1, 12.7, 7.4, 7.3, 8.0, 7.8, 7.6, 8.1	100
size	deep_attn_3x16	5	200	25.0, 15.4, 9.8, 9.8, 12.1, 12.7, 13.3, 13.4	100
size	deep_attn_3x16	6	200	22.4, 19.5, 10.6, 9.9, 8.9, 8.6, 8.9, 9.0	150
size	deep_attn_3x16	7	200	7.2, 5.9, 5.9, 5.7, 6.6, 6.5, 6.7, 6.7	100
size	deep_attn_3x16	12	200	11.0, 9.0, 8.1, 9.2, 8.1, 7.2, 8.0, 8.0	150
size	deep_attn_3x16	15	200	7.2, 8.0, 8.1, 8.4, 6.3, 6.8, 7.0, 7.3	125
families	deep0_3x16	1	175	10.3, 9.3, 9.4, 10.0, 10.4, 11.8, 12.5	50
families	deep0_3x16	2	150	11.2, 26.0, 31.1, 31.8, 32.0, 32.4	25
families	deep0_3x16	3	150	14.0, 22.9, 22.0, 20.2, 18.6, 17.1	25
families	deep0_3x16	4	200	15.4, 16.7, 14.9, 13.0, 12.4, 11.9, 11.8, 11.7	200
families	deep0_3x16	5	150	13.1, 22.1, 21.4, 22.6, 23.3, 23.4	25
families	deep0_3x16	6	150	13.5, 17.3, 17.0, 17.4, 17.3, 17.8	25
families	deep0_3x16	7	200	8.1, 6.8, 6.7, 6.4, 6.2, 6.7, 6.4, 6.5	125
families	deep0_3x16	12	200	9.2, 8.1, 8.1, 8.5, 7.7, 7.8, 8.0, 8.0	125
families	deep0_3x16	15	200	7.6, 8.0, 7.5, 7.5, 8.4, 8.3, 8.6, 8.8	75
families	deep0_3x16	18	150	7.6, 8.3, 7.7, 8.7, 8.2, 8.4	25
families	deep0_3x16	26	150	8.9, 9.0, 8.9, 8.8, 9.7, 9.2	100
families	deep0_attn_3x16	1	150	7.8, 9.2, 9.5, 9.6, 10.3, 15.1	25
families	deep0_attn_3x16	2	200	33.8, 29.2, 26.6, 26.6, 27.1, 23.7, 25.5, 26.0	150

On the training reactions themselves (Fig. 4.5, the table below): every one of the 33 cells is below PBE on the DFS-units $\mathcal{ED}$ over its own training reactions and molecules; the training-reaction WTMAD-2 is 0.00 to 4.4 kcal/mol against 5.7 for PBE on the union of the training reactions, the attention twins the closest fits (deep_attn_3x16 0.05 to 2.45, deep0_attn_3x16 0.05 and 0.77). The training-molecule ε_n sits above PBE’s pooled value in every deep_3x16 cell but 1 and 26 and every deep0_3x16 cell but 1, 18 and 26, below it in every deep_attn_3x16 cell and in deep0_attn_3x16 at 1 (above at 2), while the grid RMSE of the same molecules is below PBE’s in every cell: the per-electron L1 and the grid RMSE rank the same fits differently, and both are reported.

In-sample per cell in DFS units (insample_by_pool_3x3_eps.csv of the family_val_best set, identical in the final-step set; combined leg; E = 2-subset WTMAD-2 over the cell’s own training reactions, kcal/mol; eps over its training molecules; ED under gamma = 1084.87 kcal/mol; the PBE columns are the union of the run’s training reactions / molecules for E and eps and the cell’s own slice for ED; n = training reactions and trained species with a density).

Architecture	Subset	n rxn	n species	E NN	E PBE (pool)	eps NN	eps PBE (pool)	ED NN	ED PBE (cell)	dED	beats
deep_3x16	1	1	1	0.02	5.67	0.00728	0.00888	0.05	0.79	-0.74	yes
deep_3x16	2	2	3	2.14	5.67	0.01353	0.00888	3.73	11.27	-7.55	yes
deep_3x16	3	2	3	2.96	5.67	0.01106	0.00888	4.75	12.57	-7.82	yes
deep_3x16	4	3	4	2.63	5.67	0.01118	0.00888	4.33	11.58	-7.25	yes
deep_3x16	5	3	4	4.31	5.67	0.01315	0.00888	6.63	13.25	-6.63	yes
deep_3x16	6	4	5	3.92	5.67	0.01234	0.00888	6.06	11.78	-5.72	yes
deep_3x16	7	5	6	2.04	5.67	0.01042	0.00888	3.46	7.89	-4.44	yes
deep_3x16	12	10	11	3.11	5.67	0.01026	0.00888	4.86	8.86	-4.00	yes
deep_3x16	15	13	14	2.78	5.67	0.00900	0.00888	4.32	8.08	-3.76	yes
deep_3x16	18	16	16	2.60	5.67	0.00944	0.00888	4.14	8.01	-3.86	yes
deep_3x16	26	24	22	2.23	5.67	0.00731	0.00888	3.49	7.14	-3.66	yes
deep_attn_3x16	1	1	1	0.05	5.67	0.00494	0.00888	0.09	0.79	-0.70	yes
deep_attn_3x16	2	2	3	0.59	5.67	0.00617	0.00888	1.08	11.27	-10.20	yes
deep_attn_3x16	3	2	3	1.57	5.67	0.00626	0.00888	2.55	12.57	-10.02	yes
deep_attn_3x16	4	3	4	1.68	5.67	0.00648	0.00888	2.71	11.58	-8.87	yes
deep_attn_3x16	5	3	4	1.77	5.67	0.00732	0.00888	2.89	13.25	-10.36	yes
deep_attn_3x16	6	4	5	1.72	5.67	0.00724	0.00888	2.83	11.78	-8.96	yes
deep_attn_3x16	7	5	6	0.53	5.67	0.00640	0.00888	0.98	7.89	-6.92	yes
deep_attn_3x16	12	10	11	2.45	5.67	0.00784	0.00888	3.80	8.86	-5.05	yes
deep_attn_3x16	15	13	14	2.20	5.67	0.00667	0.00888	3.37	8.08	-4.70	yes
deep0_3x16	1	1	1	0.00	5.67	0.00753	0.00888	0.00	0.79	-0.79	yes
deep0_3x16	2	2	3	0.91	5.67	0.01392	0.00888	1.72	11.27	-9.56	yes
deep0_3x16	3	2	3	3.81	5.67	0.01046	0.00888	5.71	12.57	-6.86	yes
deep0_3x16	4	3	4	2.92	5.67	0.00969	0.00888	4.57	11.58	-7.01	yes
deep0_3x16	5	3	4	4.36	5.67	0.01327	0.00888	6.69	13.25	-6.56	yes
deep0_3x16	6	4	5	3.61	5.67	0.01257	0.00888	5.71	11.78	-6.08	yes
deep0_3x16	7	5	6	1.58	5.67	0.01038	0.00888	2.77	7.89	-5.12	yes
deep0_3x16	12	10	11	3.00	5.67	0.01037	0.00888	4.74	8.86	-4.12	yes
deep0_3x16	15	13	14	2.72	5.67	0.00913	0.00888	4.27	8.08	-3.80	yes
deep0_3x16	18	16	16	2.66	5.67	0.00873	0.00888	4.16	8.01	-3.85	yes
deep0_3x16	26	24	22	2.22	5.67	0.00721	0.00888	3.46	7.14	-3.68	yes
deep0_attn_3x16	1	1	1	0.05	5.67	0.00508	0.00888	0.09	0.79	-0.70	yes
deep0_attn_3x16	2	2	3	0.77	5.67	0.01118	0.00888	1.45	11.27	-9.83	yes

In-sample density fit per cell (eval/per_molecule.json; the cell mean of the grid RMSE against CCSD over the trained species, twins collapsed; the loss fall is the ratio of the mean total loss over the first five percent of the updates to that over the last five percent).

Run	Architecture	Subset	Trained species	Density RMSE NN	PBE	NN/PBE	Species improved	Loss fall (first/last 5%)	Updates
size	deep_3x16	1	1	9.52e-05	1.32e-04	0.72	1/1	1.2	525
size	deep_3x16	2	3	8.28e-05	1.03e-04	0.81	2/3	3.7	600
size	deep_3x16	3	3	1.29e-04	1.33e-04	0.97	2/3	4.4	750
size	deep_3x16	4	4	1.16e-04	1.22e-04	0.96	2/4	3.4	1200
size	deep_3x16	5	4	9.34e-05	1.03e-04	0.90	2/4	3.5	1050
size	deep_3x16	6	5	8.99e-05	1.02e-04	0.88	3/5	2.7	1200
size	deep_3x16	7	6	1.03e-04	1.32e-04	0.78	5/6	2.2	1800
size	deep_3x16	12	11	1.17e-04	1.49e-04	0.79	8/11	2.0	2800
size	deep_3x16	15	14	1.10e-04	1.45e-04	0.76	11/14	1.9	2550
size	deep_3x16	18	16	1.21e-04	1.52e-04	0.79	13/16	1.6	3000
size	deep_3x16	26	22	1.18e-04	1.60e-04	0.73	20/22	1.6	4900
size	deep_attn_3x16	1	1	7.52e-05	1.32e-04	0.57	1/1	1.9	450
size	deep_attn_3x16	2	3	7.11e-05	1.03e-04	0.69	3/3	5.1	800
size	deep_attn_3x16	3	3	8.95e-05	1.33e-04	0.67	3/3	4.8	1000
size	deep_attn_3x16	4	4	8.30e-05	1.22e-04	0.68	4/4	4.1	1200
size	deep_attn_3x16	5	4	6.81e-05	1.03e-04	0.66	4/4	4.3	1400
size	deep_attn_3x16	6	5	6.61e-05	1.02e-04	0.65	5/5	3.7	1600
size	deep_attn_3x16	7	6	1.00e-04	1.32e-04	0.76	6/6	2.9	1800
size	deep_attn_3x16	12	11	1.10e-04	1.49e-04	0.74	10/11	2.1	2800
size	deep_attn_3x16	15	14	9.77e-05	1.45e-04	0.67	14/14	1.9	3400
families	deep0_3x16	1	1	9.00e-05	1.32e-04	0.68	1/1	1.4	525
families	deep0_3x16	2	3	8.04e-05	1.03e-04	0.78	2/3	4.3	600
families	deep0_3x16	3	3	1.23e-04	1.33e-04	0.93	2/3	4.0	750
families	deep0_3x16	4	4	1.05e-04	1.22e-04	0.86	3/4	3.5	1200
families	deep0_3x16	5	4	9.19e-05	1.03e-04	0.89	2/4	3.7	1050
families	deep0_3x16	6	5	8.72e-05	1.02e-04	0.86	4/5	2.8	1200
families	deep0_3x16	7	6	1.03e-04	1.32e-04	0.78	5/6	2.6	1800
families	deep0_3x16	12	11	1.14e-04	1.49e-04	0.77	10/11	2.1	2800
families	deep0_3x16	15	14	1.06e-04	1.45e-04	0.73	13/14	1.9	3400
families	deep0_3x16	18	16	1.17e-04	1.52e-04	0.77	13/16	1.7	3000
families	deep0_3x16	26	22	1.12e-04	1.60e-04	0.70	21/22	1.8	4200
families	deep0_attn_3x16	1	1	8.37e-05	1.32e-04	0.64	1/1	1.7	450
families	deep0_attn_3x16	2	3	9.33e-05	1.03e-04	0.91	2/3	3.5	800

The in-sample atomization-energy panels of the suite’s overview figure (insample_overview.png, panels A and B) are not included: that metric subtracts the network’s molecular energy from the fixed exact-atom anchors, so it reports the network’s absolute-energy offset (30 to 110 kcal/mol here) rather than an atomization error; the trained objective compares molecules with the network’s own atoms (Sec. 4.1). Fig. 4.5 carries the residuals the training minimized.

5. Optimization, hold-out

Sources: xcquinox/alec/evaluation.py (the density metrics), notebooks/analysis/HOLDOUT_SET.md (the set), notebooks/analysis/make_ablation_arch_figure.py (the reductions, WTMAD-2 and ED), per cell eval_holdout_val_best/per_reaction.json and per_molecule.json, and the CSVs written beside the figures.

5.1 The held-out set and the comparators

The held-out pool is the union of the GMTKN55 BH76 barrier heights and the W4-11 atomization energies. Its test slice, after the removal of the 47-reaction validation slice (which selects the validation-best checkpoint and never enters a reported number), of each cell’s verbatim training reactions, and of the four barriers that appear twice under permuted-reactant names, carries 58 BH76 and 120 W4-11 reactions (name-deduplicated; the live count per cell is the n_{rxn} column of the tables below), and the density comparison runs over 189 molecular species with CCSD reference densities at the same basis and grid (atoms excluded). PBE is evaluated on exactly the same slice as each cell, so every comparison is like against like; the pooled PBE values are 7.98 (BH76), 13.74 (W4-11) and 11.86 kcal/mol (combined) reaction-energy MAE, and 8.92 kcal/mol 2-subset WTMAD-2. The functionals are evaluated with the same 3-cycle SCF they were trained with (Sec. 4.2); the PBE density is the converged reference calculation’s.

5.2 Metrics

Energy. Per-reaction absolute error $|E_{\text{rxn}}^{\text{NN}} - E_{\text{rxn}}^{\text{ref}}|$, reduced either as the plain MAE over a pool or as the two-bucket WTMAD-2 of GMTKN55 (Goerigk et al., PCCP 19, 32184 (2017)),

$$\text{WTMAD-2} = \frac{56.84}{N} \sum_{i \in \{\text{BH76}, \text{W4-11}\}} N_i \frac{\text{MAD}_i}{|\Delta E_{\text{ref}}|_i},$$

a labeled reweighting of the two present subsets, not the full-GMTKN55 quantity (the figures say so). When one pool is reported alone the same expression collapses to $56.84 \text{ MAD}_i / |\Delta E_{\text{ref}}|_i$, the pool’s mean absolute deviation rescaled by its mean absolute reference energy: the BH76 column of the 3x3 figures is a scaled relative error, which is why it reads three times the plain MAE (a mean barrier of about 19 kcal/mol) while the W4-11 column reads a fifth of it (a mean atomization energy of about 350 kcal/mol). **Density.** Per species, the grid-weighted RMSE $\sqrt{\sum_g w_g (n_g - n_g^{\text{ref}})^2} / \sum_g w_g$ (the self-calibrated figures) and the DFS Eq. 20 per-electron error

$$\varepsilon_n = \frac{1}{N_e} \int |n - n^{\text{ref}}| d^3r, \quad N_e = \int n^{\text{ref}} d^3r$$

(the DFS-units figures), each reduced to a cell mean over species with case twins collapsed. **Combined.** The DFS Eq. 21 harmonic combination

$$\mathcal{ED} = \frac{2}{1/E + 1/(\gamma D)},$$

with E the WTMAD-2 energy leg and D the density leg. In DFS units $\gamma = 1084.87$ kcal/mol, the Letter’s published slope, is dimensionally valid on ε_n and is used as is; in the RMSE units the figures self-calibrate $\gamma = E_{\text{PBE}}/D_{\text{PBE}}$ from the pooled PBE anchors, so that $\mathcal{ED}_{\text{PBE}} = E_{\text{PBE}}$ by construction and the NN against PBE ranking is independent of the density scale. A cell “beats PBE” when its $\mathcal{ED}$ is below the PBE $\mathcal{ED}$ of the same slice. The harmonic mean is dominated by its

smaller leg: with $\gamma\varepsilon_n \approx 10$ kcal/mol for every functional here and E between 6 and 14 kcal/mol, a change of 1 kcal/mol in E moves $\mathcal{ED}$ by 0.4 to 0.7 kcal/mol, while the observed spread of the density leg (0.0088 to 0.0118 per electron, 9.6 to 12.8 kcal/mol after γ) moves it by at most 1.5 kcal/mol.

5.3 Held-out energy

Fig. 5.1 Held-out energy per cell, one panel per metric, subset size on the abscissa and one line per architecture: the combined reaction-energy MAE (left) and the 2-subset WTMAD-2 (right), with PBE’s value on the union of the cells’ held-out slices as the dashed horizontal line (8.92 kcal/mol WTMAD-2; 11.57 kcal/mol MAE, `holdout_ed_combined.csv`) and the cell-slice PBE value as a black cap over each subset size (the two differ where a cell’s training reactions were removed from the slice: the WTMAD-2 caps run from 8.7 to 9.5 kcal/mol and the MAE caps from 11.4 to 11.6, as the footer states; the footer’s 11.86 kcal/mol is PBE’s MAE over the full 178-reaction test set of Sec. 5.1). The subtitle names the cell with the lowest barrier MAE (`deep_attn_3x16` at 5 molecules, 4.04 kcal/mol) and counts the cells below their slice-matched PBE on the barriers. The footer lists the training subset of every cell.

WTMAD-2: `deep_3x16` starts at 7.92 kcal/mol at one molecule (H_2O), rises to 9.61, 10.67, 11.06, 13.97 and 13.51 at two to six, drops to 6.44 at seven and stays between 6.74 and 7.41 through eighteen, then rises to 8.23 at twenty-six; `deep_attn_3x16` runs 8.26, 7.87, 7.22, 6.09, 6.03, 6.09, 5.94, 6.61 and 6.47 over its nine sizes (one to seven, twelve, fifteen), below PBE’s 8.92 at every one. `deep0_3x16`, the same network under the zero-init start: 8.18 at one, 9.16 at two, 7.99 and 8.06 at three and four, 10.48 and 10.40 at five and six, then 5.90, 7.11, 7.10, 6.82 and 8.52 from seven to twenty-six. `deep0_attn_3x16` has its first two cells, 8.11 at one and 13.64 at two. The plain MAE panel carries the same shape with PBE at 11.86 kcal/mol.

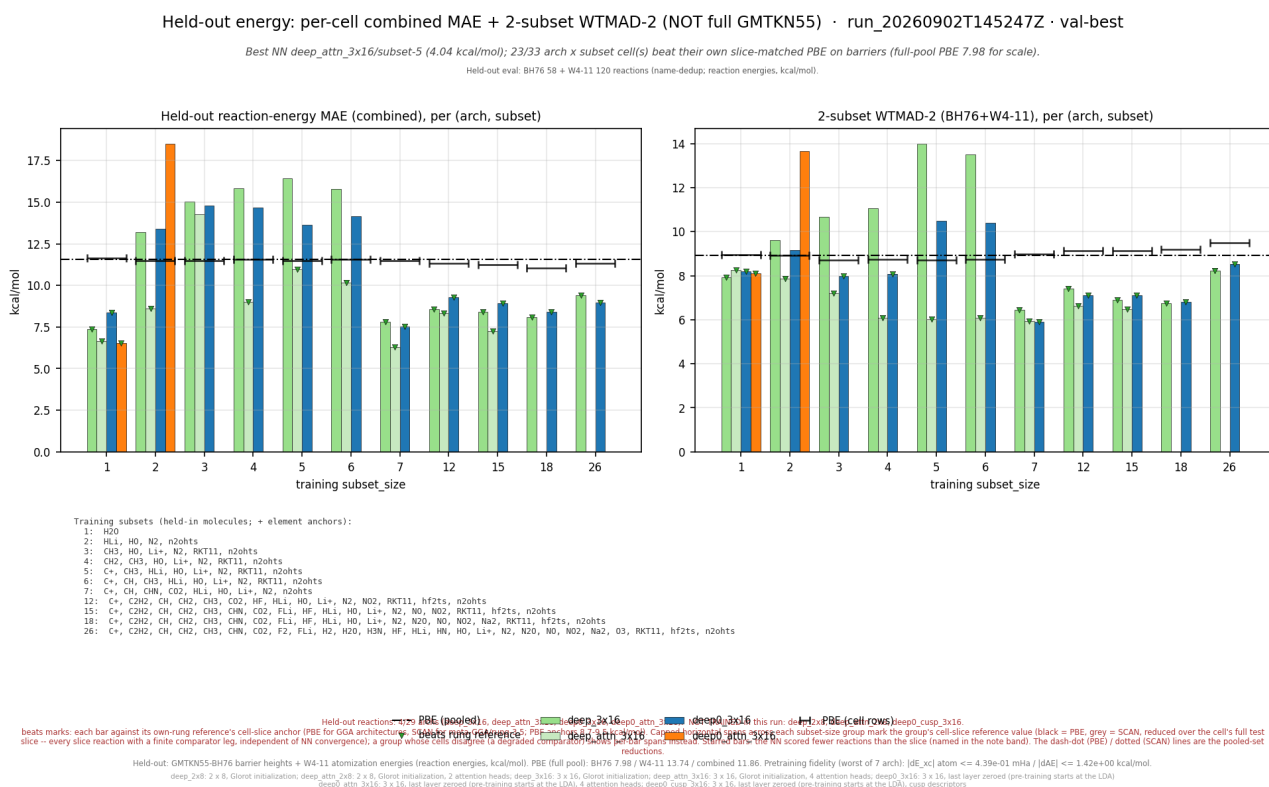


Fig. 5.2 The same energy comparison as a grid of ratios: one cell per (architecture, subset size), colored and annotated with the network’s reaction-energy MAE divided by PBE’s on the same slice, below 1 (blue) where the network is better. The `deep_3x16` row reads 0.63 at one molecule, 1.15 to 1.43 at two to six, 0.68 to 0.83 from seven to twenty-six; the `deep_attn_3x16` row is below 1 at every size except three molecules (1.24); the `deep0_3x16` row is below 1 at one, three, four and seven to eighteen and above it at two, five and six.

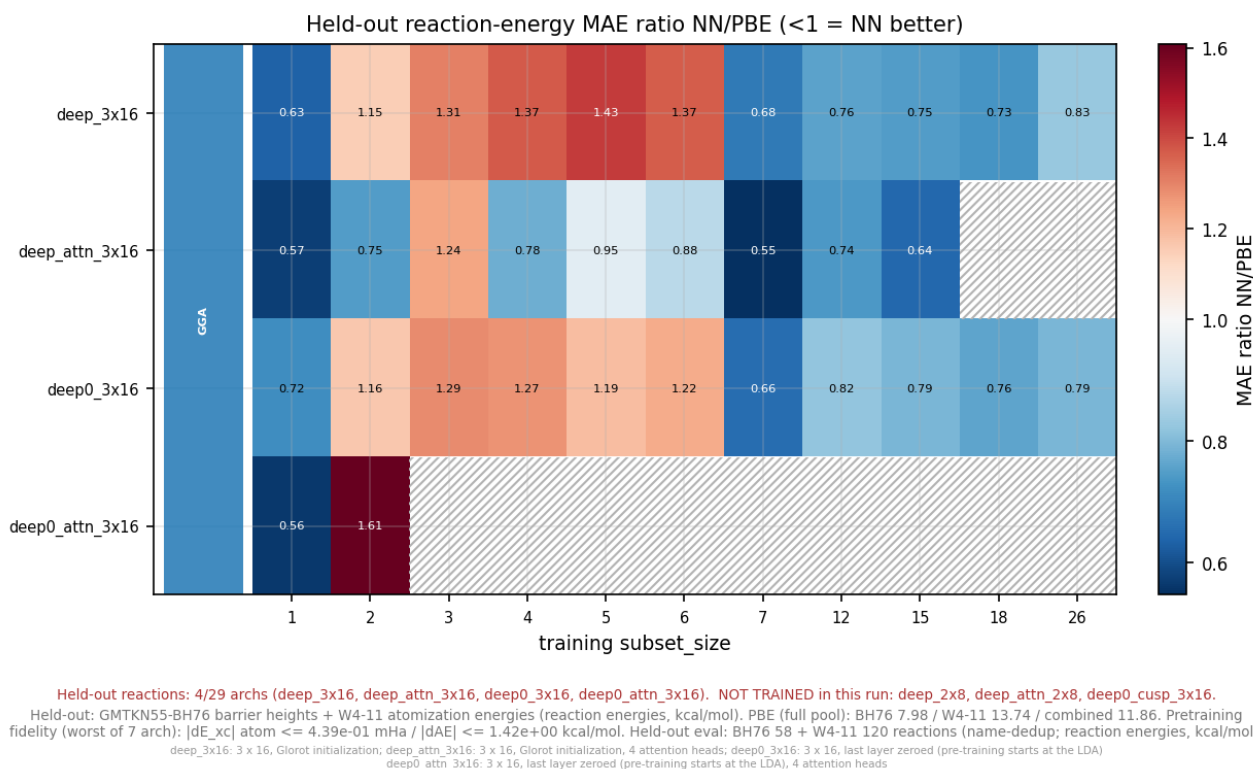
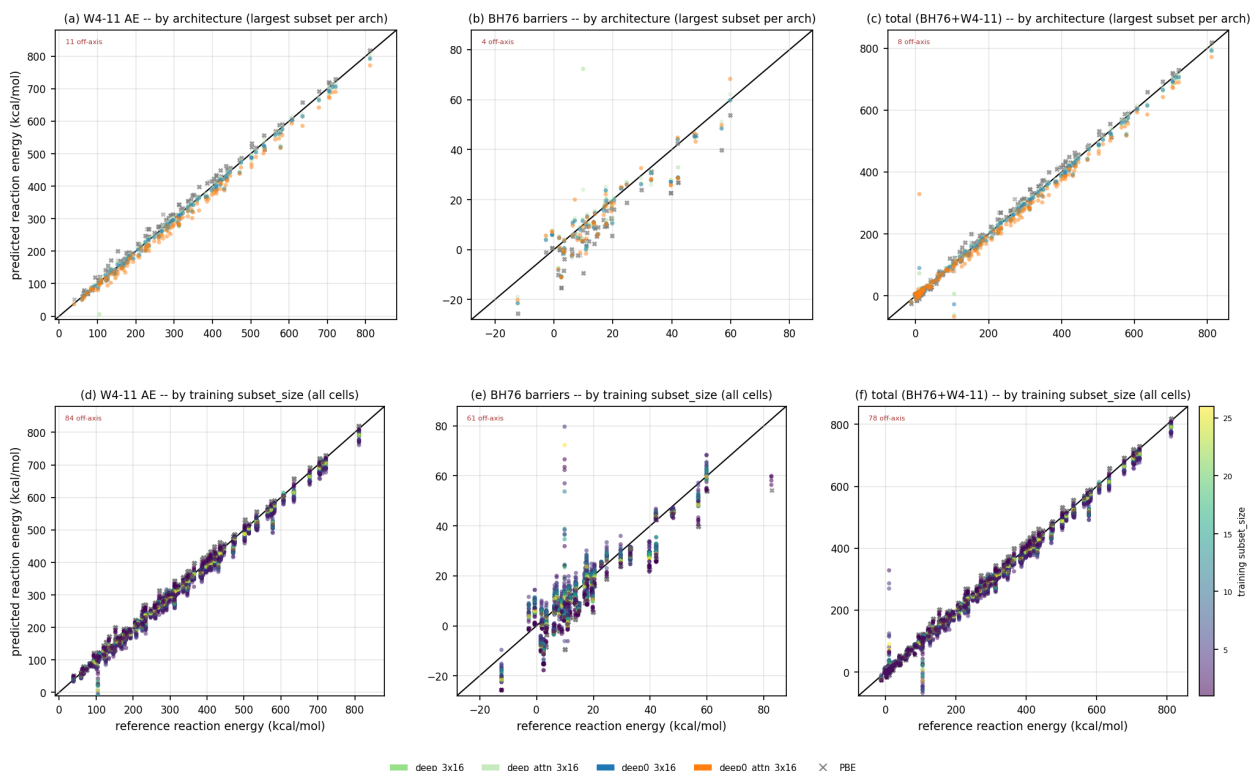


Fig. 5.3 Parity of the predicted against the reference reaction energy, one point per held-out reaction, by reaction class in three columns (W4-11 atomizations, BH76 barriers, both together). The top row draws each architecture at its largest subset; the bottom row pools every cell and colors the points by subset size; PBE on the same reactions is drawn as grey crosses in every panel and the diagonal is the exact line. Each panel prints the number of points outside its axis range. The atomization column spans 0 to 800 kcal/mol; at that scale the networks and PBE lie on the diagonal (5 points off-axis in the top panel, 44 in the bottom one, the largest of them the one-molecule cells' errors of 60 kcal/mol on the smallest molecules, Fig. 5.8), and the per-reaction errors of a few kcal/mol that the tables report are not resolved. The barrier column spans -25 to 85 kcal/mol. In the top panel the largest-subset cells (deep_3x16 at 26, deep_attn_3x16 at 7) scatter around the diagonal with residuals up to about 10 kcal/mol and one deep_attn_3x16 barrier predicted at 72 kcal/mol against a reference of 10, while the PBE crosses sit mostly below the diagonal between references of 0 and 20 kcal/mol; in the bottom panel the two-to-six-molecule cells (the darkest colors) spread widest, the reference barrier of 10 kcal/mol being predicted from -5 to 80 kcal/mol across cells, with 36 points off-axis.

Best NN deep_attn_3x16/subset-5 (4.04 kcal/mol); 23/33 arch x subset cell(s) beat their own slice-matched PBE on barriers (full-pool PBE 7.98 for scale).



Held-out reactions: 409 archs (deep_3x16, deep_attn_3x16, deep0_3x16, deep0_attn_3x16). NOT TRAINED in this run: deep_2x8, deep_attn_2x8, deep0_cup_3x16.

Held-out: GMTKX55-BH76 barrier heights + W4-11 atomization energies (reaction energies, kcal/mol). PBE (full pool): BH76 7.98 / W4-11 13.74 / combined 11.86. Pretraining fidelity (worst of 7 archs): |dE_{xc}| atom <= 4.39e-01 mHa / |dAE| <= 1.42e+00 kcal/mol. Held-out eval: BH76 58 + W4-11 120 reactions (name-depup, reaction energies, kcal/mol).

deep_3x16: 3 x 16, Glorot initialization; deep_attn_3x16: 3 x 16, Glorot initialization; 4 attention heads; deep0_3x16: 3 x 16, last layer zeroed (pre-training starts at the LDA); deep0_attn_3x16: 3 x 16, last layer zeroed (pre-training starts at the LDA); 4 attention heads.

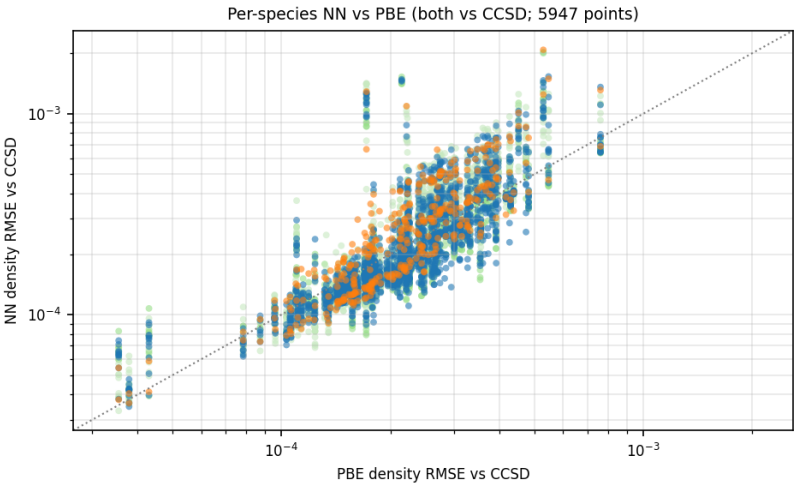
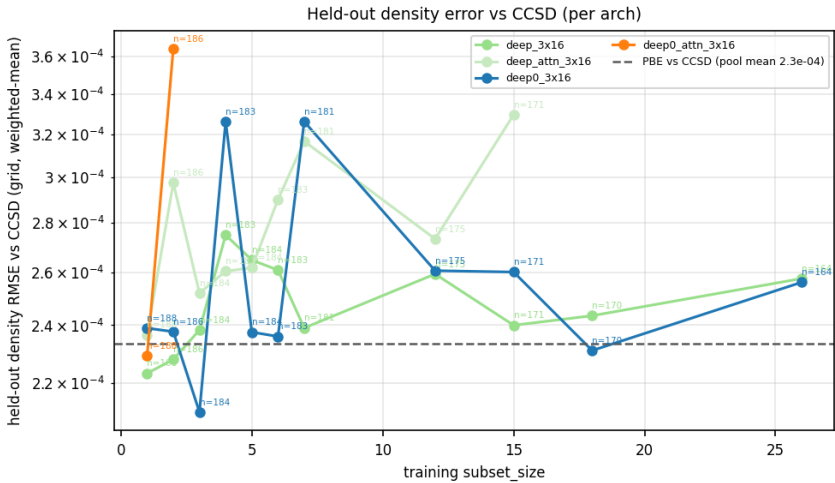
5.4 Held-out density

Fig. 5.4 Held-out density against CCSD in grid-RMSE units. Left: the cell mean of the weighted RMSE over the held-out species (the count per cell printed at each point) against the subset size, one line per architecture, PBE's pooled mean as the dashed line ($2.34\text{e-}4$). Right: every (cell, species) pair as one point, the network's RMSE against PBE's on the same species, both against CCSD, log axes, the dotted diagonal the line of equal error. The first figure of each pair is the full set; the second removes the recurring tail (the species above $1.5 \times$ PBE in at least half of the cells that carry them and in at least two; the rule, the count with its denominator and the names are in the footer; a diagnostic view selected by the errors themselves).

The values below are the D_{rmse} column of `holdout_by_pool_3x3.csv` (combined leg) of each set. Full set (189 species): deep_3x16 runs from $2.23\text{e-}4$ at one molecule (below PBE's $2.34\text{e-}4$) through 2.28, 2.38, 2.75, 2.65 and $2.61\text{e-}4$ at two to six to $2.39\text{e-}4$ at seven, $2.59\text{e-}4$ at twelve, $2.40\text{e-}4$ at fifteen, $2.43\text{e-}4$ at eighteen and $2.57\text{e-}4$ at twenty-six; deep_attn_3x16 from $2.37\text{e-}4$ at one to $2.98\text{e-}4$ at two, 2.52, 2.60, 2.62 and $2.90\text{e-}4$ at three to six, $3.17\text{e-}4$ at seven, $2.73\text{e-}4$ at twelve and $3.30\text{e-}4$ at fifteen; deep0_3x16 2.39, 2.38, 2.10, 3.26, 2.37, 2.36, 3.26, 2.61, 2.60, 2.31 and $2.56\text{e-}4$ at one to twenty-six; deep0_attn_3x16 $2.29\text{e-}4$ at one and $3.64\text{e-}4$ at two. On the right, the bulk of the points lies along the diagonal between $1\text{e-}4$ and $5\text{e-}4$, and the vertical columns of a few species reach $1\text{e-}3$ and above in every cell: those columns are the recurring tail. Over the 33 cells the recurring tail is eight species (H_2 , RKT06, RKT17, bn, bn3pi, cloo, cn and cs; the rule of the figure's footer). The tail-excluded figure (181 species) reads deep_3x16 2.14, 2.14, 2.23, 2.62, 2.45, 2.41, 2.28, 2.46, 2.29, 2.34 and $2.42\text{e-}4$ over the eleven sizes, deep_attn_3x16 2.26, 2.82, 2.46, 2.50, 2.50, 2.76, 3.04, 2.59 and $3.16\text{e-}4$, deep0_3x16 2.29, 2.25, 1.98, 3.18, 2.21, 2.20, 3.19, 2.48, 2.47, 2.21 and $2.38\text{e-}4$, and deep0_attn_3x16 2.19 and $3.49\text{e-}4$, against PBE's $2.34\text{e-}4$ on the reduced set ($2.3370\text{e-}4$; $2.3351\text{e-}4$ on the full set): deep_3x16 is below PBE at one, two, three, seven and fifteen molecules (at eighteen 2.3446 against 2.3370), deep_attn_3x16 at one only, deep0_3x16 at one, two, three, five, six and eighteen, deep0_attn_3x16 at one. The peaks at four and seven molecules in the deep0_3x16 column ($3.26\text{e-}4$) survive the tail removal, so they are not the recurring species.

Held-out density error vs CCSD (NN vs PBE) · run_20260902T145247Z · val-best

HELD-OUT density error vs CCSD reference densities on the W4-11+BH76 benchmark species (atoms excluded; weighted-mean grid RMSE, NOT N_e -normalized). PBE baseline is model-free on the same grid. Density generalization; separate from the held-out ENERGY panels by design.
Held-out eval: BH76 58 + W4-11 120 reactions (name-duped; reaction energies, kcal/mol); density: 189 species vs CCSD refs at matching basis/grid (atoms excluded).



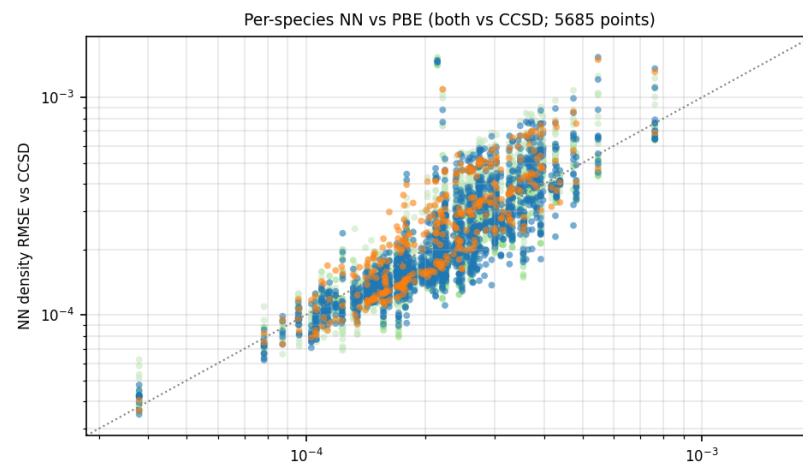
Held-out reactions: 4/29 archs (deep_3x16, deep_attn_3x16, deep0_3x16, deep0_attn_3x16). NOT TRAINED in this run: deep_2x8, deep_attn_2x8, deep0_cusp_3x16.

Held-out ~~deep_3x16~~ ~~deep_attn_3x16~~ ~~deep0_3x16~~ ~~deep0_attn_3x16~~ same grid.

deep_2x8: 2 x 8, Glorot initialization; deep_attn_2x8: 2 x 8, Glorot initialization, 2 attention heads; deep_3x16: 3 x 16, Glorot initialization; deep_attn_3x16: 3 x 16, Glorot initialization, 4 attention heads; deep0_3x16: 3 x 16, last layer zeroed (pre-training starts at the LDA); deep0_attn_3x16: 3 x 16, last layer zeroed (pre-training starts at the LDA), 4 attention heads; deep0_cusp_3x16: 3 x 16, last layer zeroed (pre-training starts at the LDA), cusp descriptors

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deep_2x8_2x8_Glorot initialization; deep_attn_2x8_2x8_Glorot initialization; deep_3x16_3x16_Glorot initialization; deep_attn_3x16_3x16_Glorot initialization; deep_4x32_4x32_Glorot initialization; deep_attn_4x32_4x32_Glorot initialization; deep_8x64_8x64_Glorot initialization; deep_attn_8x64_8x64_Glorot initialization; deep_16x128_16x128_Glorot initialization; deep_attn_16x128_16x128_Glorot initialization; deep_32x256_32x256_Glorot initialization; deep_attn_32x256_32x256_Glorot initialization; deep_64x512_64x512_Glorot initialization; deep_attn_64x512_64x512_Glorot initialization; deep_128x1024_128x1024_Glorot initialization; deep_attn_128x1024_128x1024_Glorot initialization; deep_256x2048_256x2048_Glorot initialization; deep_attn_256x2048_256x2048_Glorot initialization; deep_512x4096_512x4096_Glorot initialization; deep_attn_512x4096_512x4096_Glorot initialization; deep_1024x8192_1024x8192_Glorot initialization; deep_attn_1024x8192_1024x8192_Glorot initialization; deep_2048x16384_2048x16384_Glorot initialization; deep_attn_2048x16384_2048x16384_Glorot initialization; deep_4096x32768_4096x32768_Glorot initialization; deep_attn_4096x32768_4096x32768_Glorot initialization; deep_8192x65536_8192x65536_Glorot initialization; deep_attn_8192x65536_8192x65536_Glorot initialization; deep_16384x131072_16384x131072_Glorot initialization; deep_attn_16384x131072_16384x131072_Glorot initialization; deep_32768x262144_32768x262144_Glorot initialization; deep_attn_32768x262144_32768x262144_Glorot initialization; deep_65536x524288_65536x524288_Glorot initialization; deep_attn_65536x524288_65536x524288_Glorot initialization; deep_131072x1048576_131072x1048576_Glorot initialization; deep_attn_131072x1048576_131072x1048576_Glorot initialization; deep_262144x2097152_262144x2097152_Glorot initialization; deep_attn_262144x2097152_262144x2097152_Glorot initialization; deep_524288x4194304_524288x4194304_Glorot initialization; deep_attn_524288x4194304_524288x4194304_Glorot initialization; deep_1048576x8388608_1048576x8388608_Glorot initialization; 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PBE density RMSE vs CCSD

5.5 The per-pool 3x3 in DFS units

Fig. 5.5 The held-out result in the Letter’s units, nine panels: the rows are the energy leg (WTMAD-2), the density leg (ε_n , Eq. 20) and the combination ($\mathcal{ED}$ under the one shared $\gamma = 1084.87$ kcal/mol); the columns are the BH76 barriers, the W4-11 atomizations and both pooled. In every panel the abscissa is the subset size, the bars are the architectures (the greens deep_3x16 and deep_attn_3x16, the blue deep0_3x16, the orange deep0_attn_3x16, in the display order), the dashed line is PBE on the pooled slice and the black cap over each subset size is PBE on that cell’s own slice; a triangle on a bar marks a cell below its cap. The first figure of each pair is the full set, the second the tail-excluded one. The panel values are in the four tables of Sec. 5.7.

Row 1 (energy). Panel A, BH76: PBE’s scaled relative error is 22.1 kcal/mol; deep_3x16 reads 21.3 at one molecule, 23.6 to 36.6 at two to six (the peak at five), 16.3 at seven, 16.7 to 20.0 from twelve to twenty-six; deep_attn_3x16 22.8 at one, then 20.6, 15.3, 14.6, 13.3, 14.0 and 15.5. Panel B, W4-11: PBE 2.54; deep_3x16 1.41 at one, 2.95 to 3.54 at two to six, 1.70 to 2.01 from seven; deep_attn_3x16 1.17, 1.80, 3.46, 2.06, 2.63, 2.38, 1.31. Panel C, the two-subset WTMAD-2 of Fig. 5.1.

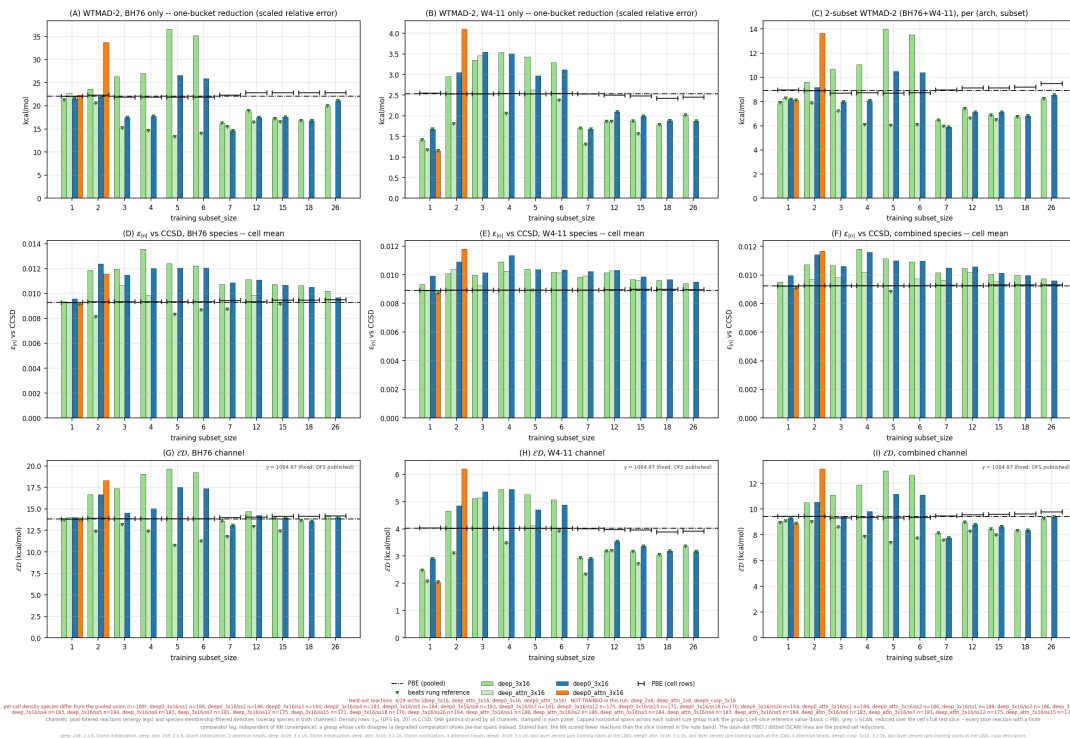
Row 2 (density). PBE’s ε_n is 0.00928 (BH76 species), 0.00891 (W4-11) and 0.00923 (combined) per electron. Panel D: deep_3x16 0.0093 at one, 0.0119 to 0.0136 at two to six, 0.0102 to 0.0111 from seven; deep_attn_3x16 0.0093, 0.0081, 0.0107, 0.0099, 0.0083, 0.0087, 0.0087, so the attention twin is below PBE at two, five, six and seven. Panel E: every deep_3x16 cell is above PBE (0.0093 to 0.0109), and so is every deep_attn_3x16 cell, at one and five molecules by less than 0.4 percent (0.00894 and 0.00892 against 0.00891). Panel F, the combination of both.

Row 3 (the combination). Panel G, BH76: PBE 13.7; deep_3x16 13.7 at one, 16.7 to 19.7 at two to six, 13.6 to 14.7 from seven; deep_attn_3x16 14.0 at one and 10.8 to 13.2 from two to seven. Panel H, W4-11: PBE 4.02; deep_3x16 2.48 at one, 4.65 to 5.45 at two to six, 2.93 to 3.36 from seven; deep_attn_3x16 2.09, 3.10, 5.14, 3.48, 4.13, 3.92, 2.33. Panel I, combined, the verdict column of the tables.

deep0_3x16, the same panels: BH76 energy 21.5, 22.0, 17.5, 17.8, 26.6, 25.9, 14.6, 17.5, 17.6, 16.8, 21.1 at one to twenty-six; W4-11 energy 1.68, 3.05, 3.54, 3.50, 2.97, 3.12, 1.67, 2.10, 1.99, 1.88, 1.87; combined ε_n 0.0096 to 0.0116, above PBE in every cell; combined $\mathcal{ED}$ 9.32, 10.54, 9.44, 9.83, 11.16, 11.10, 7.77, 8.78, 8.63, 8.37, 9.38 against the cell caps 9.46, 9.44, 9.32, 9.34, 9.32, 9.34, 9.49, 9.56, 9.60, 9.62, 9.80. The two new cells of deep_attn_3x16 (twelve and fifteen molecules): BH76 energy 16.45 and 16.56, W4-11 1.87 and 1.56, combined ε_n 0.0102 and 0.0096, combined $\mathcal{ED}$ 8.27 and 7.97 against 9.56 and 9.60. deep0_attn_3x16 at one and two molecules: BH76 energy 22.38 and 33.73, W4-11 1.15 and 4.10, combined ε_n 0.0091 and 0.0117, combined $\mathcal{ED}$ 8.89 and 13.14 against 9.46 and 9.44.

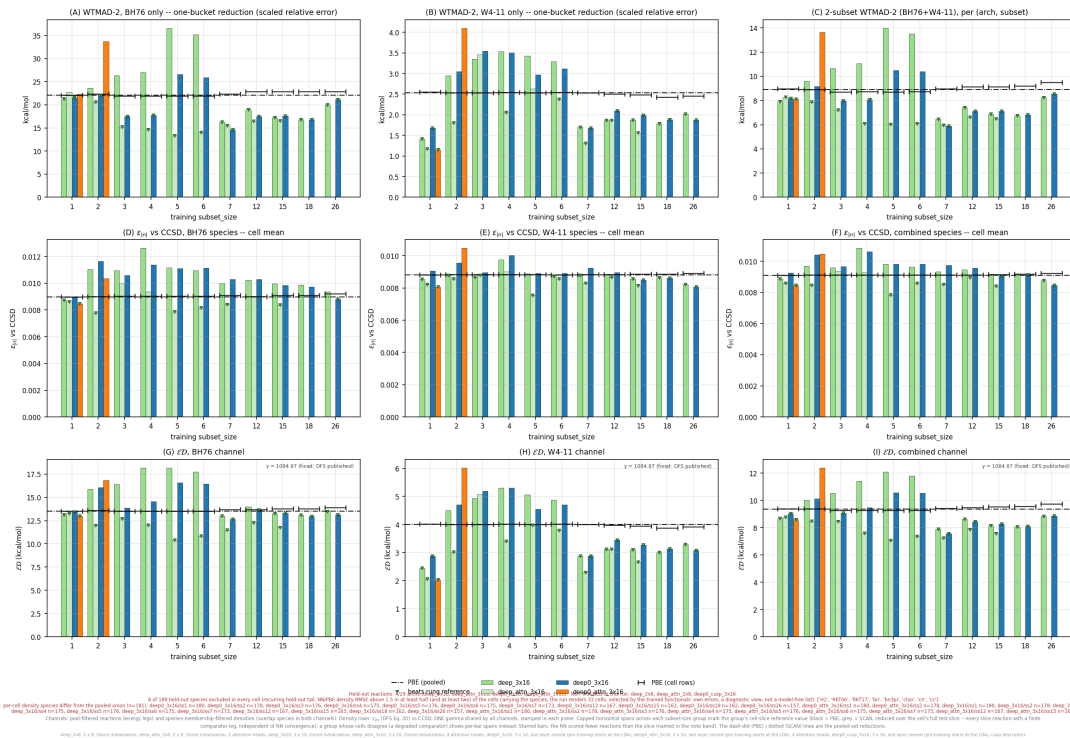
Per-channel held-out story (DFS units): WTMA2-2 | ϵ_{ref} | ϵ_{D} (BH76, W4-11, combined) · run_20260902T145247Z · val-best

Columns are channels: BH76 | W4-11 | combined, AB and the GH ED energy legs use the SINGLE-POOL, WTMA2-2, which collapses to 56.84MAD_pooledmean[ϵ_{ref} , pool] – the pool's mean absolute deviation rescaled by its mean absolute reference energy; a scaled relative error, NOT a weighting (only the combined column weights); NOT full GUTKOS).
 $\epsilon_{\text{D}} = 2 \times 10^{-6} + 1/(\text{gamma } \epsilon_{\text{D}})$; $\epsilon_{\text{D}} = \frac{1}{n} \sum_{i=1}^n (n - n_{\text{ref}})$ (Eq. 20 per species); ONE GEMMA SHARED by all channels, value = source stamped in each panel (on-axis six functional fit when the calibration cache resolves, the Letter's published 1084.87 otherwise) – ϵ_{D} compares across columns; ϵ_{D} of PBE = ϵ_{PBE} . Density row = cell mean ϵ_{D} ; per-species parity in holdout_density_parity_by_pool_esp.png.
 Held out eval: BH76 50 × W4-11 120 reactions (name delay, reaction energies, kcal/mol, density 180 species vs CCSD refs at matching biological labels excluded).



Per-channel held-out story (DFS units): WTMA2-2 | ϵ_{ref} | ϵ_{D} (BH76, W4-11, combined) · run_20260902T145247Z · val-best

Columns are channels: BH76 | W4-11 | combined, AB and the GH ED energy legs use the SINGLE-POOL, WTMA2-2, which collapses to 56.84MAD_pooledmean[ϵ_{ref} , pool] – the pool's mean absolute deviation rescaled by its mean absolute reference energy; a scaled relative error, NOT a weighting (only the combined column weights); NOT full GUTKOS).
 $\epsilon_{\text{D}} = 2 \times 10^{-6} + 1/(\text{gamma } \epsilon_{\text{D}})$; $\epsilon_{\text{D}} = \frac{1}{n} \sum_{i=1}^n (n - n_{\text{ref}})$ (Eq. 20 per species); ONE GEMMA SHARED by all channels, value = source stamped in each panel (on-axis six functional fit when the calibration cache resolves, the Letter's published 1084.87 otherwise) – ϵ_{D} compares across columns; ϵ_{D} of PBE = ϵ_{PBE} . Density row = cell mean ϵ_{D} ; per-species parity in holdout_density_parity_by_pool_esp.png.
 Held out eval: BH76 50 × W4-11 120 reactions (name delay, reaction energies, kcal/mol, density 180 species vs CCSD refs at matching biological labels excluded).



5.6 The combined view in self-calibrated units

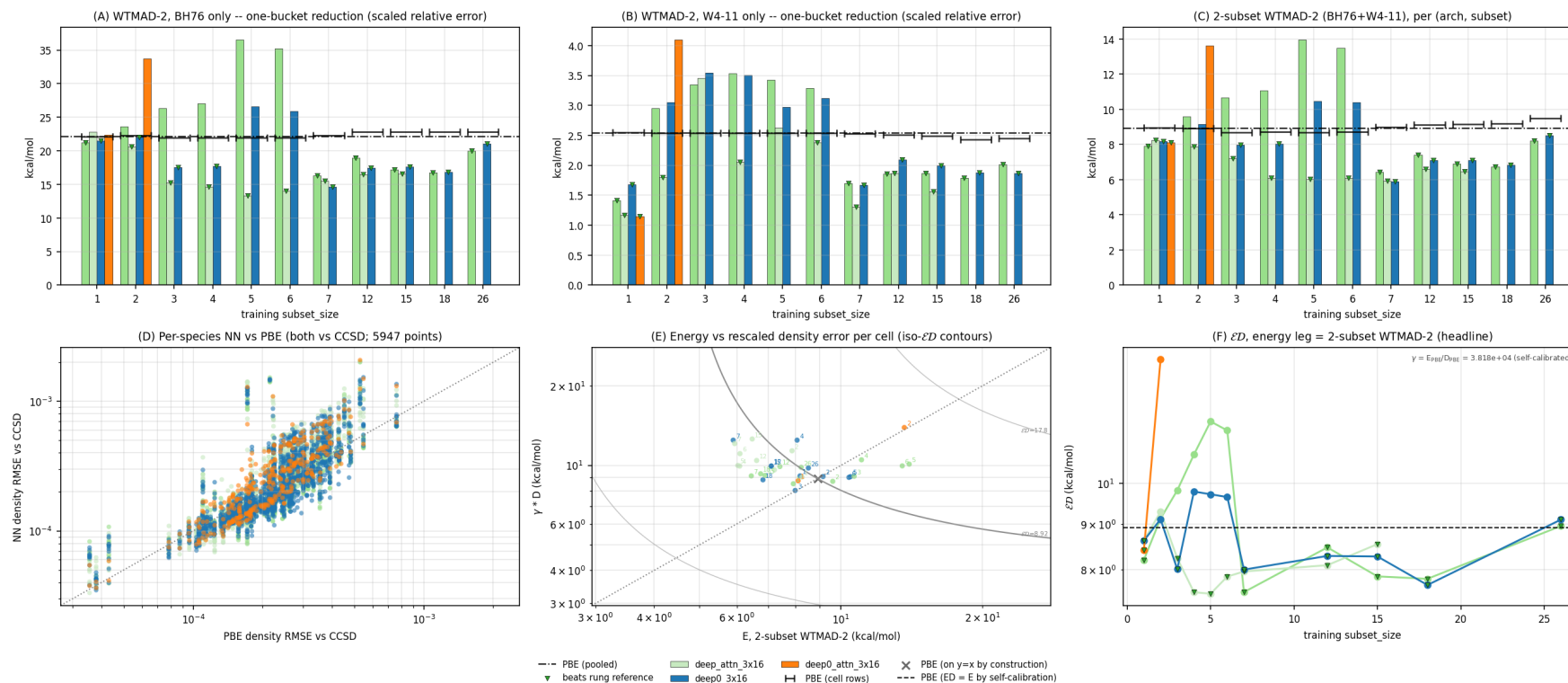
Fig. 5.6 The held-out overview on one canvas, in grid-RMSE units with the self-calibrated γ . Top left: WTMA2-2 by pool (BH76, W4-11, combined) per cell as grouped bars with the PBE

lines. Top right: the per-species density parity of Fig. 5.4. Bottom left: every cell as a point in the $(E, \gamma D)$ plane, log axes, with two iso- $\mathcal{ED}$ contours drawn as curves and PBE as the black cross at $(E_{\text{PBE}}, E_{\text{PBE}}) = (8.92, 8.92)$ on the dotted diagonal $\gamma D = E$; a point above the diagonal has its density leg larger than its energy leg. Bottom right: the $\mathcal{ED}$ headline per cell against PBE's 8.92 (dashed). In the plane every cell but two lies above the diagonal: with the self-calibrated γ the rescaled density leg is 8.5 to 12.1 kcal/mol (deep_3x16 at one and two molecules, 8.52 and 8.70, are the two below PBE's 8.92) while the energy leg runs from 5.9 to 14.0. The deep_3x16 cells from seven molecules upward (E 6.4 to 8.2, γD 9.1 to 9.9) and every deep_attn_3x16 cell (E 5.9 to 8.3, γD 9.0 to 12.1) lie to the left of the cross and, deep_attn_3x16 at two, six and seven molecules apart, near or below its contour; the two-to-six-molecule deep_3x16 cells (E 9.6 to 14.0) lie to its right. In the headline panel deep_attn_3x16 at two molecules is above PBE (9.30) under these units, whereas in DFS units (Fig. 5.5) it is below: the two density scales weight the tail species differently.

Held-out overview: WTMAD-2 by pool + density vs CCSD + ED · run 20260902T145247Z · val-best

Single-pool "WTMAD-2" (panels A, B) reduces to 56.84 * MAD pool / mean[ref] pool -- a scaled relative error, NOT a reweighting; only panel C (2-subset) reweights BH76 vs W4-11, and it is NOT full GMTKN55. E/f: $\varepsilon_D = 2/(1/E + 1/(\text{gamma}^*D))$, $\text{gamma} = E \text{ PBE}/D \text{ PBE}$ self-calibrated from the pooled PBE anchors (value printed in the panels), so $\varepsilon_D \text{ of PBE} = E \text{ PBE}$ (dashed).

Held-out eval: BH76 58 + W4-11 120 reactions (name-dedup; reaction energies, kcal/mol); density: 189 species vs CCSD refs at matching basis/grid (atoms excluded)



Hold over: **deep_x16**, **deep_x16_deep**, **deep_x16_deep_x16**, **deep_x16_deep_x16_deep**, **deep_x16_deep_x16_deep_x16**
 Held out overview: All one-bucket WTMAD-2 reduction 56.84Madd[ref] per pool. C-2 subset WTMAD-2; no SCAN lines (no SCAN WTMAD-2 cache). DE: grid-weight-averaged dense RMSE vs CSDS (not CSDST) refs at matching basing. PBE model-run on the same grid. F-ED: full diagnostics on holdout; ed_combined.gpr: beats marks; each against its own-running reference's cell-slice anchor (PBE for GGA architectures, SCAN for meta-GGA/rtg-3.5; PBE anchors 8-7.9-5 kcal/mol). Capped horizontal spans across each subset-size group mark the group's cell-slice-reference value (black = PBE). gray = SCAN, reduced over the cell's full test suite – every slice reaction with a finite comparator (ref. independent of NN convergence); a group whose cells disagree (a degraded comparator) shows per-bar spans instead. Starred bars: the NN scored fewer reactions than the slice (named in the next column). The dash-dot (PBE) / dotted (SCAN) lines are the pooled-set reductions.

Fig. 5.7 The combined $\mathcal{ED}$ per cell with the self-calibrated γ (left), the decomposition plane of Fig. 5.6 enlarged (middle), and the leg-independence check (right): the same $\mathcal{ED}$ recomputed with the plain reaction-energy MAE as the energy leg in place of WTMAD-2, against PBE’s MAE of 11.57 kcal/mol. The two legs agree on every deep_3x16 and deep0_3x16 cell and disagree on four deep_attn_3x16 cells: at two molecules the WTMAD-2 leg puts the cell above PBE (9.30 against 8.92) and the MAE leg below (10.86 against 11.57); at three, five and six molecules the WTMAD-2 leg puts it below (8.24, 7.52, 7.86) and the MAE leg above (13.31, 11.88, 11.90). The footer reports a Spearman rank correlation of 0.57 between the two legs over the 18 shared cells: the ranking of cells within about 10 percent of each other is not metric-independent.

Combined energy-density $\mathcal{E}D$ (DFS Eq. 21, harmonic mean) -- held-out, NN vs PBE · run_20260902T145247Z · val-best

$\mathcal{E}D = 2/(1/E + 1/(\gamma D))$ (Dick & Fernandez-Serra, PRB 104, L161109 (2021), Eq. 21); γ self-calibrated per leg from pooled PBE anchors ($\gamma = E_{PBE}/D_{PBE}$ -- the Letter's regression slope, calibrated here on PBE alone), so $\mathcal{E}D$ of PBE == E_{PBE} by construction (dashed).
Held-out eval: BH76 58 + W4-11 120 reactions (name-dupup: reaction energies, kcal/mol); density: 189 species vs CCSD refs at matching basis/grid (atoms excluded).

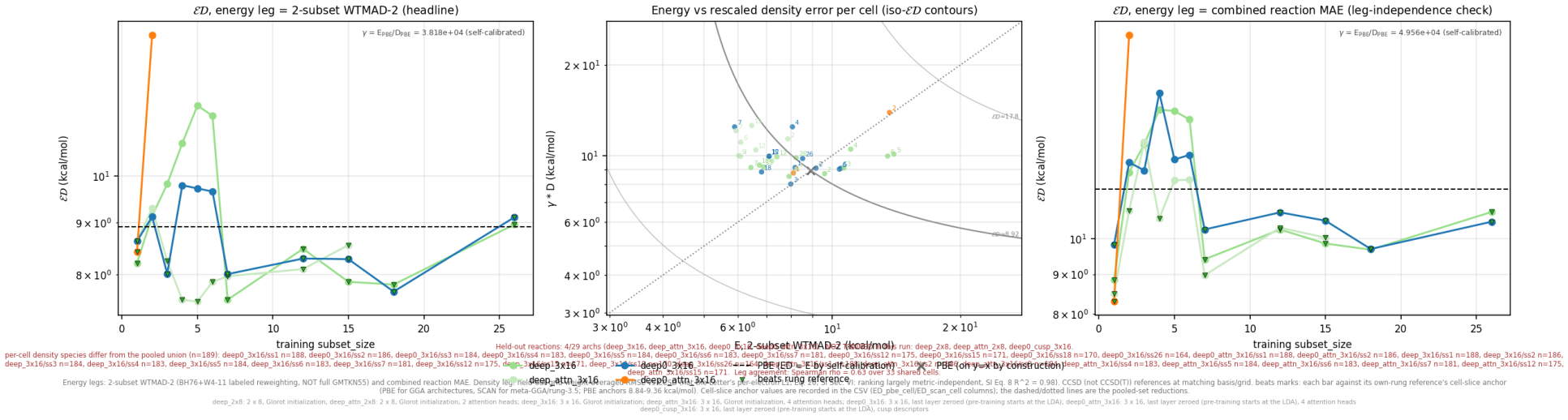
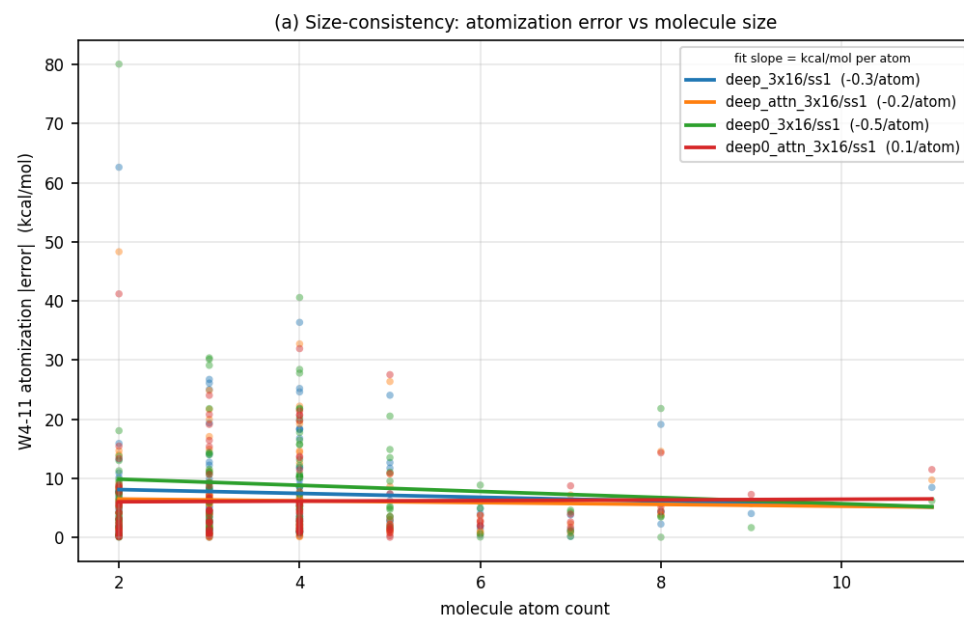


Fig. 5.8 Size-consistency diagnostic of the one-molecule cells (trained on H₂O alone). Left: the held-out absolute atomization-energy error against the number of atoms of the molecule, one point per W4-11 reaction and per cell, with a linear fit per cell whose slope is printed in kcal/mol per atom; a functional whose error grows with the size of the molecule carries a per-atom energy offset, which is what the diagnostic is for. Right: the BH76 and W4-11 MAE of the same cells side by side. The fitted slopes are -0.3 (deep_3x16) and -0.2 kcal/mol per atom (deep_attn_3x16): the error does not grow with molecule size, and the largest errors (62 and 48 kcal/mol) are on two-atom molecules. The right panel reads 7.4 and 7.5 kcal/mol (deep_3x16, barriers and atomizations) and 8.0 and 6.2 kcal/mol (deep_attn_3x16).

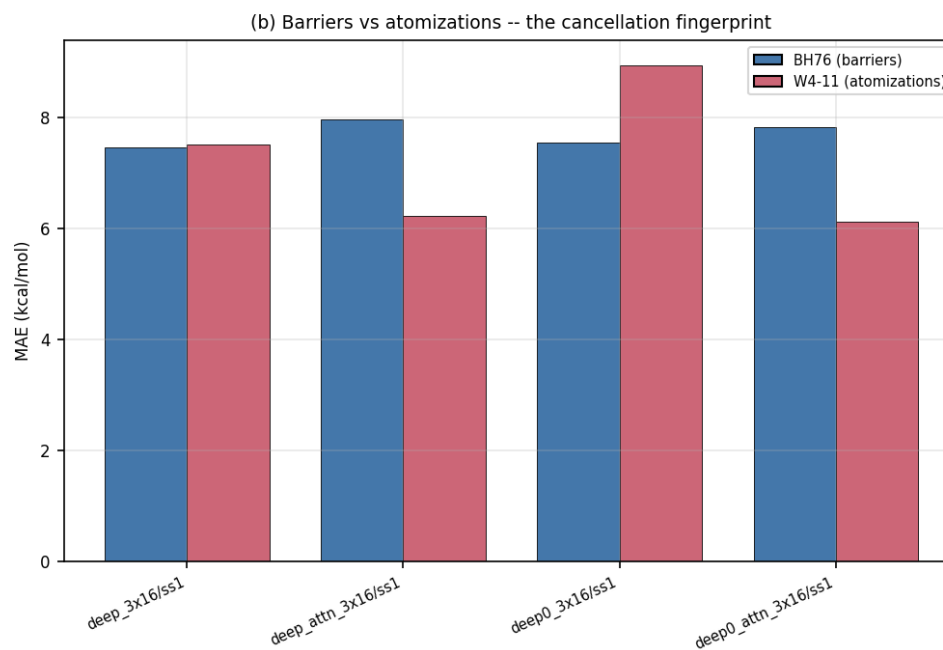
Size-consistency (additivity) diagnostic -- deep_3x16/ss1, deep_attn_3x16/ss1, deep0_3x16/ss1, deep0_attn_3x16/ss1 · run_20260902T145247Z · val-best

Held-out eval: BH76 58 + W4-11 120 reactions (name-dedup; reaction energies, kcal/mol).

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Held-out reactions: 4/29 archs (deep_3x16, deep_attn_3x16, deep0_3x16, deep0_attn_3x16). NOT TRAINED in this run: deep_2x8, deep_attn_2x8, deep0_cusp_3x16.
Held-out: GMTKN55-BH76 barrier heights + W4-11 atomization energies (reaction energies, kcal/mol).



5.7 Results per cell

The four tables below are read from `holdout_by_pool_3x3_eps.csv` of the `family_val_best` set (the tail-excluded one from the `_excl_tail` sibling): `E` is the WTMAD-2 of the leg in kcal/mol (the one-pool legs collapse to the scaled relative error of Sec. 5.2), `eps` the per-electron density error of the leg's species, `ED` the combination under $\gamma = 1084.87$ kcal/mol; the `E PBE` and `eps PBE` columns are the leg's pooled PBE values (the same in every row), `ED PBE (cell)` is PBE's ED on the cell's own slice, and "beats" whether the network's ED is below it. `n rxn` and `n species` are the reactions and species of the cell's slice. Every table carries the 33 finished cells in the display order of the architectures and then the subset size; the run of a cell is that of its architecture in the scope table (the tables of Sec. 4.6 carry it as a column). The pooled PBE columns of the tail-excluded table are its own file's, computed without the excluded species (`eps PBE` 0.00909 against 0.00923).

BH76 leg (the barriers and their species).

Architecture	Subset	n rxn	n species	E NN	E PBE (pool)	eps NN	eps PBE (pool)	ED NN	ED PBE (cell)	dED	beats
deep_3x16	1	58	66	21.27	22.11	0.00935	0.00928	13.73	13.85	-0.12	yes
deep_3x16	2	57	64	23.61	22.11	0.01187	0.00928	16.66	13.94	+2.72	no
deep_3x16	3	56	62	26.36	22.11	0.01195	0.00928	17.38	13.87	+3.50	no
deep_3x16	4	56	62	27.04	22.11	0.01357	0.00928	19.07	13.87	+5.19	no
deep_3x16	5	56	62	36.57	22.11	0.01242	0.00928	19.69	13.87	+5.82	no
deep_3x16	6	56	62	35.22	22.11	0.01220	0.00928	19.24	13.87	+5.37	no
deep_3x16	7	57	61	16.27	22.11	0.01075	0.00928	13.58	14.03	-0.45	yes
deep_3x16	12	55	59	18.92	22.11	0.01111	0.00928	14.72	14.06	+0.67	no
deep_3x16	15	55	56	17.21	22.11	0.01075	0.00928	13.90	14.16	-0.26	yes
deep_3x16	18	55	56	16.75	22.11	0.01064	0.00928	13.66	14.16	-0.50	yes
deep_3x16	26	55	51	19.98	22.11	0.01020	0.00928	14.24	14.19	+0.05	no
deep_attn_3x16	1	58	66	22.81	22.11	0.00933	0.00928	14.02	13.85	+0.17	no
deep_attn_3x16	2	57	64	20.64	22.11	0.00813	0.00928	12.36	13.94	-1.58	yes
deep_attn_3x16	3	56	62	15.27	22.11	0.01066	0.00928	13.16	13.87	-0.71	yes
deep_attn_3x16	4	56	62	14.65	22.11	0.00987	0.00928	12.37	13.87	-1.50	yes
deep_attn_3x16	5	56	62	13.32	22.11	0.00833	0.00928	10.77	13.87	-3.11	yes
deep_attn_3x16	6	56	62	13.98	22.11	0.00867	0.00928	11.24	13.87	-2.63	yes
deep_attn_3x16	7	57	61	15.52	22.11	0.00873	0.00928	11.77	14.03	-2.27	yes
deep_attn_3x16	12	55	59	16.45	22.11	0.00986	0.00928	12.96	14.06	-1.09	yes
deep_attn_3x16	15	55	56	16.56	22.11	0.00916	0.00928	12.42	14.16	-1.74	yes
deep0_3x16	1	58	66	21.52	22.11	0.00956	0.00928	14.00	13.85	+0.15	no
deep0_3x16	2	57	64	22.03	22.11	0.01238	0.00928	16.69	13.94	+2.75	no
deep0_3x16	3	56	62	17.52	22.11	0.01147	0.00928	14.55	13.87	+0.67	no
deep0_3x16	4	56	62	17.75	22.11	0.01202	0.00928	15.03	13.87	+1.16	no
deep0_3x16	5	56	62	26.57	22.11	0.01204	0.00928	17.52	13.87	+3.64	no
deep0_3x16	6	56	62	25.88	22.11	0.01207	0.00928	17.39	13.87	+3.51	no
deep0_3x16	7	57	61	14.64	22.11	0.01088	0.00928	13.07	14.03	-0.97	yes
deep0_3x16	12	55	59	17.50	22.11	0.01109	0.00928	14.26	14.06	+0.20	no
deep0_3x16	15	55	56	17.60	22.11	0.01068	0.00928	13.97	14.16	-0.19	yes
deep0_3x16	18	55	56	16.80	22.11	0.01050	0.00928	13.57	14.16	-0.59	yes
deep0_3x16	26	55	51	21.11	22.11	0.00965	0.00928	14.00	14.19	-0.19	yes
deep0_attn_3x16	1	58	66	22.38	22.11	0.00917	0.00928	13.77	13.85	-0.07	yes
deep0_attn_3x16	2	57	64	33.73	22.11	0.01157	0.00928	18.29	13.94	+4.36	no

W4-11 leg (the atomizations and their species).

Architecture	Subset	n rxn	n species	E NN	E PBE (pool)	eps NN	eps PBE (pool)	ED NN	ED PBE (cell)	dED	beats
deep_3x16	1	119	119	1.41	2.54	0.00933	0.00891	2.48	4.04	-1.56	yes
deep_3x16	2	120	118	2.95	2.54	0.01007	0.00891	4.65	4.02	+0.63	no
deep_3x16	3	120	117	3.35	2.54	0.00998	0.00891	5.12	4.02	+1.10	no
deep_3x16	4	119	116	3.54	2.54	0.01091	0.00891	5.45	4.03	+1.42	no
deep_3x16	5	120	117	3.43	2.54	0.01040	0.00891	5.26	4.02	+1.24	no
deep_3x16	6	119	116	3.29	2.54	0.01018	0.00891	5.07	4.03	+1.04	no
deep_3x16	7	118	115	1.70	2.54	0.00982	0.00891	2.93	4.01	-1.09	yes
deep_3x16	12	114	109	1.86	2.54	0.01013	0.00891	3.18	3.98	-0.80	yes
deep_3x16	15	113	107	1.87	2.54	0.00966	0.00891	3.17	3.96	-0.79	yes
deep_3x16	18	111	106	1.79	2.54	0.00962	0.00891	3.05	3.89	-0.83	yes
deep_3x16	26	104	101	2.01	2.54	0.00939	0.00891	3.36	3.92	-0.56	yes
deep_attn_3x16	1	119	119	1.17	2.54	0.00894	0.00891	2.09	4.04	-1.95	yes
deep_attn_3x16	2	120	118	1.80	2.54	0.01037	0.00891	3.10	4.02	-0.92	yes
deep_attn_3x16	3	120	117	3.46	2.54	0.00925	0.00891	5.14	4.02	+1.12	no
deep_attn_3x16	4	119	116	2.06	2.54	0.01026	0.00891	3.48	4.03	-0.55	yes
deep_attn_3x16	5	120	117	2.63	2.54	0.00892	0.00891	4.13	4.02	+0.11	no
deep_attn_3x16	6	119	116	2.38	2.54	0.01013	0.00891	3.92	4.03	-0.11	yes
deep_attn_3x16	7	118	115	1.31	2.54	0.00990	0.00891	2.33	4.01	-1.68	yes
deep_attn_3x16	12	114	109	1.87	2.54	0.01029	0.00891	3.20	3.98	-0.78	yes
deep_attn_3x16	15	113	107	1.56	2.54	0.00961	0.00891	2.72	3.96	-1.24	yes
deep0_3x16	1	119	119	1.68	2.54	0.00992	0.00891	2.91	4.04	-1.13	yes
deep0_3x16	2	120	118	3.05	2.54	0.01088	0.00891	4.85	4.02	+0.83	no
deep0_3x16	3	120	117	3.54	2.54	0.01014	0.00891	5.36	4.02	+1.34	no
deep0_3x16	4	119	116	3.50	2.54	0.01134	0.00891	5.46	4.03	+1.43	no
deep0_3x16	5	120	117	2.97	2.54	0.01036	0.00891	4.70	4.02	+0.68	no
deep0_3x16	6	119	116	3.12	2.54	0.01033	0.00891	4.88	4.03	+0.85	no
deep0_3x16	7	118	115	1.67	2.54	0.01023	0.00891	2.91	4.01	-1.10	yes
deep0_3x16	12	114	109	2.10	2.54	0.01031	0.00891	3.53	3.98	-0.45	yes
deep0_3x16	15	113	107	1.99	2.54	0.00985	0.00891	3.36	3.96	-0.60	yes
deep0_3x16	18	111	106	1.88	2.54	0.00966	0.00891	3.18	3.89	-0.70	yes
deep0_3x16	26	104	101	1.87	2.54	0.00950	0.00891	3.17	3.92	-0.75	yes
deep0_attn_3x16	1	119	119	1.15	2.54	0.00872	0.00891	2.05	4.04	-1.99	yes
deep0_attn_3x16	2	120	118	4.10	2.54	0.01180	0.00891	6.21	4.02	+2.19	no

Combined leg (both pools).

Architecture	Subset	n rxn	n species	E NN	E PBE (pool)	eps NN	eps PBE (pool)	ED NN	ED PBE (cell)	dED	beats
deep_3x16	1	177	188	7.92	8.92	0.00951	0.00923	8.96	9.46	-0.50	yes
deep_3x16	2	177	186	9.61	8.92	0.01071	0.00923	10.52	9.44	+1.08	no
deep_3x16	3	176	184	10.67	8.92	0.01068	0.00923	11.11	9.32	+1.79	no
deep_3x16	4	175	183	11.06	8.92	0.01182	0.00923	11.88	9.34	+2.53	no
deep_3x16	5	176	184	13.97	8.92	0.01113	0.00923	12.96	9.32	+3.64	no
deep_3x16	6	175	183	13.51	8.92	0.01091	0.00923	12.62	9.34	+3.28	no
deep_3x16	7	175	181	6.44	8.92	0.01017	0.00923	8.13	9.49	-1.35	yes
deep_3x16	12	169	175	7.41	8.92	0.01047	0.00923	8.97	9.56	-0.59	yes
deep_3x16	15	168	171	6.89	8.92	0.01004	0.00923	8.44	9.60	-1.16	yes
deep_3x16	18	166	170	6.74	8.92	0.00999	0.00923	8.31	9.62	-1.31	yes
deep_3x16	26	159	164	8.23	8.92	0.00973	0.00923	9.25	9.80	-0.55	yes
deep_attn_3x16	1	177	188	8.26	8.92	0.00925	0.00923	9.06	9.46	-0.40	yes
deep_attn_3x16	2	177	186	7.87	8.92	0.00972	0.00923	9.01	9.44	-0.43	yes
deep_attn_3x16	3	176	184	7.22	8.92	0.00981	0.00923	8.60	9.32	-0.72	yes
deep_attn_3x16	4	175	183	6.09	8.92	0.01020	0.00923	7.85	9.34	-1.49	yes
deep_attn_3x16	5	176	184	6.03	8.92	0.00883	0.00923	7.40	9.32	-1.92	yes
deep_attn_3x16	6	175	183	6.09	8.92	0.00973	0.00923	7.73	9.34	-1.61	yes
deep_attn_3x16	7	175	181	5.94	8.92	0.00963	0.00923	7.57	9.49	-1.91	yes
deep_attn_3x16	12	169	175	6.61	8.92	0.01019	0.00923	8.27	9.56	-1.29	yes
deep_attn_3x16	15	168	171	6.47	8.92	0.00956	0.00923	7.97	9.60	-1.63	yes
deep0_3x16	1	177	188	8.18	8.92	0.00997	0.00923	9.32	9.46	-0.14	yes
deep0_3x16	2	177	186	9.16	8.92	0.01143	0.00923	10.54	9.44	+1.10	no
deep0_3x16	3	176	184	7.99	8.92	0.01062	0.00923	9.44	9.32	+0.12	no
deep0_3x16	4	175	183	8.06	8.92	0.01160	0.00923	9.83	9.34	+0.48	no
deep0_3x16	5	176	184	10.48	8.92	0.01099	0.00923	11.16	9.32	+1.83	no
deep0_3x16	6	175	183	10.40	8.92	0.01097	0.00923	11.10	9.34	+1.76	no
deep0_3x16	7	175	181	5.90	8.92	0.01051	0.00923	7.77	9.49	-1.72	yes
deep0_3x16	12	169	175	7.11	8.92	0.01058	0.00923	8.78	9.56	-0.78	yes
deep0_3x16	15	168	171	7.10	8.92	0.01013	0.00923	8.63	9.60	-0.97	yes
deep0_3x16	18	166	170	6.82	8.92	0.00996	0.00923	8.37	9.62	-1.26	yes
deep0_3x16	26	159	164	8.52	8.92	0.00961	0.00923	9.38	9.80	-0.42	yes
deep0_attn_3x16	1	177	188	8.11	8.92	0.00906	0.00923	8.89	9.46	-0.57	yes
deep0_attn_3x16	2	177	186	13.64	8.92	0.01168	0.00923	13.14	9.44	+3.70	no

Combined leg without the recurring tail (the species listed in Sec. 5.4 removed from the density leg; the energy leg is unchanged and is not repeated).

Architecture	Subset	n rxn	n species	E NN	E PBE (pool)	eps NN	eps PBE (pool)	ED NN	ED PBE (cell)	dED	beats
deep_3x16	1	177	180	7.92	8.92	0.00886	0.00909	8.68	9.39	-0.71	yes
deep_3x16	2	177	178	9.61	8.92	0.00969	0.00909	10.04	9.37	+0.67	no
deep_3x16	3	176	176	10.67	8.92	0.00959	0.00909	10.53	9.26	+1.28	no
deep_3x16	4	175	175	11.06	8.92	0.01086	0.00909	11.41	9.28	+2.13	no
deep_3x16	5	176	176	13.97	8.92	0.00981	0.00909	12.08	9.26	+2.83	no
deep_3x16	6	175	175	13.51	8.92	0.00964	0.00909	11.79	9.28	+2.51	no
deep_3x16	7	175	173	6.44	8.92	0.00933	0.00909	7.87	9.42	-1.55	yes
deep_3x16	12	169	167	7.41	8.92	0.00947	0.00909	8.61	9.49	-0.88	yes
deep_3x16	15	168	163	6.89	8.92	0.00917	0.00909	8.14	9.53	-1.38	yes
deep_3x16	18	166	162	6.74	8.92	0.00921	0.00909	8.05	9.55	-1.50	yes
deep_3x16	26	159	157	8.23	8.92	0.00876	0.00909	8.82	9.74	-0.92	yes
deep_attn_3x16	1	177	180	8.26	8.92	0.00858	0.00909	8.75	9.39	-0.64	yes
deep_attn_3x16	2	177	178	7.87	8.92	0.00846	0.00909	8.47	9.37	-0.90	yes
deep_attn_3x16	3	176	176	7.22	8.92	0.00936	0.00909	8.43	9.26	-0.82	yes
deep_attn_3x16	4	175	175	6.09	8.92	0.00929	0.00909	7.59	9.28	-1.69	yes
deep_attn_3x16	5	176	176	6.03	8.92	0.00785	0.00909	7.06	9.26	-2.19	yes
deep_attn_3x16	6	175	175	6.09	8.92	0.00859	0.00909	7.37	9.28	-1.91	yes
deep_attn_3x16	7	175	173	5.94	8.92	0.00852	0.00909	7.23	9.42	-2.19	yes
deep_attn_3x16	12	169	167	6.61	8.92	0.00895	0.00909	7.87	9.49	-1.62	yes
deep_attn_3x16	15	168	163	6.47	8.92	0.00841	0.00909	7.57	9.53	-1.95	yes
deep0_3x16	1	177	180	8.18	8.92	0.00927	0.00909	9.02	9.39	-0.37	yes
deep0_3x16	2	177	178	9.16	8.92	0.01041	0.00909	10.12	9.37	+0.75	no
deep0_3x16	3	176	176	7.99	8.92	0.00966	0.00909	9.07	9.26	-0.19	yes
deep0_3x16	4	175	175	8.06	8.92	0.01062	0.00909	9.49	9.28	+0.21	no
deep0_3x16	5	176	176	10.48	8.92	0.00982	0.00909	10.57	9.26	+1.31	no
deep0_3x16	6	175	175	10.40	8.92	0.00983	0.00909	10.53	9.28	+1.26	no
deep0_3x16	7	175	173	5.90	8.92	0.00974	0.00909	7.57	9.42	-1.85	yes
deep0_3x16	12	169	167	7.11	8.92	0.00956	0.00909	8.44	9.49	-1.05	yes
deep0_3x16	15	168	163	7.10	8.92	0.00908	0.00909	8.26	9.53	-1.27	yes
deep0_3x16	18	166	162	6.82	8.92	0.00913	0.00909	8.08	9.55	-1.47	yes
deep0_3x16	26	159	157	8.52	8.92	0.00847	0.00909	8.84	9.74	-0.90	yes
deep0_attn_3x16	1	177	180	8.11	8.92	0.00844	0.00909	8.60	9.39	-0.79	yes
deep0_attn_3x16	2	177	178	13.64	8.92	0.01047	0.00909	12.40	9.37	+3.03	no

5.8 Reading

Energy. On the 2-subset WTMAD-2 the deep_3x16 architecture is below PBE (8.92 kcal/mol) at one molecule and at every subset from 7 upward (6.44 to 8.23), and above it at 2 to 6 molecules (9.61 to 13.97); on the plain reaction-energy MAE (Fig. 5.2) the same cells read 0.64 and 0.67 to 0.81 of PBE against 1.14 to 1.42. The attention twin is below PBE at every one of its nine cells on WTMAD-2 (5.94 to 8.26), and on the MAE ratio at eight of them (3 molecules: 1.23). deep0_3x16, the same network under the zero-init start, is below PBE on WTMAD-2 at 1, 3, 4 and 7 to 26 (5.90 to 8.52) and above it at 2, 5 and 6 (9.16, 10.48, 10.40); its verdicts agree with deep_3x16's at every size but 3 and 4, where it sits just below PBE (7.99, 8.06) and deep_3x16 above (10.67, 11.06). deep0_attn_3x16 is below PBE at one molecule (8.11) and above it at two (13.64). The 2-to-6-molecule rise is common to the architectures and to both starts. The subsets at those sizes are not nested and are not small-molecule subsets (Sec. 2.4): the 2-point subset is HLi with the OH + N₂ barrier, the 3-point subset is two barriers with the Li ionization, and the 4-, 5- and 6-point subsets add CH₂, or HLi with the C ionization, or HLi, CH and the C ionization, to the two barriers and the Li ionization; from seven points on, the subsets hold four or more atomization energies. Which property of these subsets produces the rise is not established. From 7 molecules the correction transfers, and the largest subsets do not improve on 7 to 18 (at 26: deep_3x16 8.23, deep0_3x16 8.52).

Density. In DFS units the held-out ε_n of the networks sits at or above PBE's (0.00923 per electron on the full slice) in every cell but two (deep_attn_3x16 at 5 molecules, 0.00883; deep0_attn_3x16 at 1, 0.00906), by 0.2 to 28 percent; the per-species parity (Figs. 5.4 and 5.6) shows the bulk of species near the diagonal and a tail well above it. The tail is recurring: over the 33 cells, eight species (H₂, RKT06, RKT17, bn, bn3pi, cloo, cn and cs) are above 1.5 x PBE in at least half of the cells carrying them, computed from eval_holdout_val_best/per_molecule.json by the rule of Fig. 5.4; bn is the largest at every size. Without them the networks' ε_n moves to 0.0079 to 0.0109 against PBE's 0.0091 on the same reduced set, below PBE in deep_attn_3x16 at 1, 2, 5, 6, 7, 12 and 15, in deep_3x16 at 1 and 26, in deep0_3x16 at 15 and 26 and in deep0_attn_3x16 at 1, within 1.5 percent of it in deep_3x16 at 15 and 18 and in deep0_3x16 at 18, and above it elsewhere; in the deep_3x16 column from 7 molecules upward the ratio is 0.96 to 1.04. The tail species are the multireference and open-shell cases (BN and its 3-Pi state, CN, CS, CLOO, the OH + Cl and the H + CH₄ transition states, and H₂), and they are the species whose three-cycle SCF from the PBE seed is farthest from settled (Sec. 4.2). The 2026-09-07 density record found the comparison asymmetric by protocol (an unconverged network density against the converged PBE one) and could not decide from the pulled data whether the tail belongs to the functionals or to the three-cycle evaluation; the density channel itself trains (Sec. 4.5), and the converged-SCF re-evaluation in flight (Sec. 6) is the direct test.

Combined. On $\mathcal{ED}$ in DFS units, against the cell-slice PBE value: deep_3x16 is below PBE at 1, 7, 12, 15, 18 and 26 and above it at 2 to 6; deep_attn_3x16 is below it at every one of its nine cells; deep0_3x16 is below it at 1, 7, 12, 15, 18 and 26 and above it at 2 to 6, where at 3 and 4 molecules the small energy margin (7.99 and 8.06 against 8.92) is outweighed by the density leg (0.0106 and 0.0116 against 0.0092), so that the two starts of the 3x16 network agree on every verdict; deep0_attn_3x16 is below it at 1 (8.89 against 9.46) and above at 2 (13.14 against 9.44). The verdicts are unchanged by removing the tail except deep0_3x16 at 3 molecules, which then moves below PBE (9.07 against 9.26). In every cell below PBE the margin comes from the energy leg: on the decomposition plane (Fig. 5.6) each such cell lies to the left of the PBE point, and its density leg is below PBE's only in deep_attn_3x16 at 5 molecules and deep0_attn_3x16 at 1. The lowest $\mathcal{ED}$ values are deep_attn_3x16 at 5 to 7 molecules (7.40 to 7.73 against 9.32 to 9.49) and deep0_3x16 at 7 (7.77 against 9.49). The tables of Sec. 5.7 print the difference $\mathcal{ED}_{\text{NN}} - \mathcal{ED}_{\text{PBE}}$ per cell, the negative values (the network below PBE) in bold.

Units and legs. The verdicts above are the DFS-units ones (the ε_n leg under the Letter's γ). Under the self-calibrated grid-RMSE units of Figs. 5.6 and 5.7 (holdout_ed_combined.csv, WTMAD-2 leg) the verdicts differ from the DFS-units ones at deep_3x16 at 2 molecules (below PBE there), deep_attn_3x16 at 2 (above) and deep0_3x16 at 2 and 3 (below); under the plain-MAE energy leg the verdicts of deep_attn_3x16 at 3, 5 and 6 differ from the WTMAD-2 ones (Sec. 5.6), and the deep_3x16, deep0_3x16 and deep0_attn_3x16 verdicts are the DFS-units ones. The cells whose verdict depends on the choice of leg or units are the ones within about 5 percent of PBE.

6. Pending work and open items

- Cells: `deep_attn_3x16` at 18 and 26, `deep0_attn_3x16` from 3 to 26, `deep0_cusp_3x16` at every subset, the `deep_2x8` pair at every subset, are queued on the cluster; the figures and tables are regenerated from the same suite as they land.
- Meta-GGA group: four certificates pending; the `deep0_cusp_mgga_3x16` FAIL holds the training array (Sec. 3.6). The decision between dropping the architecture, refitting with more capacity or a higher energy-term weight, and the anchored class follows the four results.
- Converged-SCF re-evaluation of the 33 finished cells (the converged channel, DIIS at 10^{-8} Ha) and the CCSD T1 diagnostics for the model-free multireference list: two cluster jobs pending; they produce the `_excl_t1` and `_excl_unconverged` siblings of every density figure, the model-free counterparts of the `_excl_tail` view used here.
- The two training arms of the density program on the `deep_3x16` architecture at 7, 12, 15, 18 and 26 molecules: 25 SCF cycles with the current protocol (arm A) and the `dpyscf`-parity protocol (arm B: no convergence freeze, the executed first-mix index, the randomized atomic/PBE seed mixture, Adam with the plateau schedule and the Letter's L2 and channel weights). Submitted 2026-09-08, no results yet.
- In-sample: the suite's in-sample atomization panels stay excluded until that metric uses the network's own atoms; Figs. 4.1 and 4.5 carry the trained quantities instead.